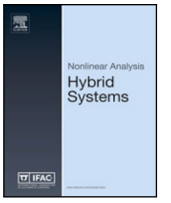
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A conditional mean-field type stochastic differential game of hybrid switching diffusions systems with delay and its applications☆

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ABSTRACT

This paper focuses on a conditional mean-field type game of hybrid switching diffusions systems with delay where both the state dynamics and costs are dependent on the conditional expectations of state given switching process. Firstly, we prove that a nonlinear conditional mean-field anticipated backward stochastic differential equation with regime switching admits a unique solution under mild conditions, which is necessary to guarantee the well-posedness of adjoint equations that arise in the optimality condition for Nash equilibrium point. Then, using established results on conditional mean-field anticipated backward stochastic differential equation, we develop a Pontryagin type maximum principle that provides necessary condition for open-loop Nash equilibrium points. Additionally, we establish two verification theorems under different assumptions, which provide sufficient conditions for Nash equilibrium points. Finally, we present three financial applications. Employing the theoretical results derived, we obtain explicit solutions of all the financial applications and provide some numerical examples with sound economic interpretations for demonstration.

1. Introduction

Stochastic differential game concerns with optimization on stochastic dynamics involving multi-players decision-making. (Conditional) mean-field type game refers to stochastic differential game where dynamics and costs are subject to not only state-strategy pair but also their (conditional) distributions. This kind of problems have been studied extensively over the last decade, and the (conditional) mean-field systems have been applied in many areas, such as mathematical finance, smart grid, engineering and so on, see e.g., [1–5] and the references therein. This paper investigates a conditional mean-field type game of hybrid switching diffusions systems with delay, where the instant and delayed conditional distributions of state appear in the coefficients of dynamics and payoffs through conditional expectations given the history of Markov chain.

In ecological population modeling, evolution of economic quantity, production planning, and complicated industrial systems, it is common that the parameters of dynamics are affected by random environments [6]. For example, epidemic models may be perturbed by telegraph noise causing the system to switch from one environmental regime to another [7], the appreciate return rate and volatility rate of stocks may be very different in bull market and bear market [8]. In such cases, the continuous processes represented merely by stochastic differential equations (SDEs for short) are inadequate to describe those features. Hybrid switching diffusions system (also known as switching diffusions), which refers to stochastic systems involving both continuous dynamics and

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discrete events, exhibit the ability to depict the influence of random environments. Therefore, the investigations about stochastic optimal control and differential games based on hybrid switching diffusions have received intensive attention. Here, we list some related works about this topic. Several maximum principles for optimal control problems of hybrid switching diffusions system were established in various frameworks [9–12]. Linear quadratic (LQ for short) optimal control problems with regime switching mechanism were considered in [13–15], where the optimal strategies were characterized by different kinds of Riccati equations. Refs. [2,16–18] studied LQ stochastic differential games of hybrid switching diffusions systems. Partially observable setting for switching diffusions were investigated in [19]. Recently, motivated by the need to reduce computational complexity, [20] considered the particles systems with mean-field interaction and a common switching process. A law of large numbers for this system was obtained in [20], showing that the limit of the empirical measure is a random measure given the history of Markov chain. In particular, consider the following interacting particles system:

$$dX_t^\ell = b(t,\alpha_t,X_t^\ell,\frac{1}{N}\sum_{n=1}^N X_t^n)dt+\sigma(t,\alpha_t,X_t^\ell,\frac{1}{N}\sum_{n=1}^N X_t^n)dW_t^\ell, \tag{1}$$

where $W_\cdot^\ell$ ($\ell\in\{1,\dots,N\}$) is a collection of $\mathbb{R}^d$-valued independent standard Brownian motions, $\alpha_\cdot$ is a continuous time Markov chain with finite states, and $W_\cdot^\ell$ is independent of $\alpha_\cdot$ for any $\ell\in\{1,\dots,N\}$. In particles system (1), Brownian motion $W_\cdot^\ell$ is an idiosyncratic noise which is felt by agent ℓ only. Markov chain $\alpha_\cdot$ serves as a common noise which brings random switching among different environments that can be felt by all agents simultaneously. The appearance of common Markov chain $\alpha_\cdot$ induces dependence of agents on the history of this switching process. As a result, empirical mean term $\frac{1}{N}\sum_{n=1}^N X_t^n$ and its limits are dependent on the history of Markov chain α_t as well. Different with works [1,12,21] where expectation terms of state were considered, this dependence on Markov chain makes $\frac{1}{N}\sum_{n=1}^N X_t^n$ converges to conditional expectation $\mathbb{E}[X_t|\mathcal{F}_t^\alpha]$ in L^2 sense as $N\to\infty$, where X_t satisfies the following conditional mean-field SDE:

$$dX_t = b(t,\alpha_t,X_t,\mathbb{E}[X_t|\mathcal{F}_t^\alpha])dt+\sigma(t,\alpha_t,X_t,\mathbb{E}[X_t|\mathcal{F}_t^\alpha])dW_t.$$

Hence, it is of great theoretical significance to investigate the optimization problems of the systems where the coefficients depend on conditional distribution regarding switching process. Inspired by this, works [22,23] explored the conditional mean-field type optimal control problems of regime-switching diffusions and obtained maximum principles in local and global forms, respectively. The exact convergence rate with respect to (w.r.t. for short) 2-Wasserstein distance of an N-player LQ game with common Markov chain noise towards mean-field game was obtained in [24]. Incorporating conditional mean-field term w.r.t. Markov chain into dynamics and cost functional, nonzero-sum LQ games and leader–follower differential games were studied in [25,26], both of which obtained state feedback representations for equilibrium points.

However, all the above works do not consider the time delay structure. In fact, stochastic system with delay has been a recurrent theme in the realm of control theory. It is common that certain systems show past-independence feature, i.e., current system state depends on its previous value, especially in ecological population modeling and financial market. Usually, optimization problems with delay structure are hard to treat for the infinite-dimensional property. To the best of our knowledge, there are mainly two approaches to derive maximum principles for delayed systems. The first one is to formulate adjoint equations separately for instant state and delay terms. However, due to the lack of Itô's formula for point delay, an assumption is necessary which requires that the adjoint equation for point delay process admits zero solution only. This assumption essentially makes the problem become finite-dimensional and thus solvable. Using this method, a verification theorem of variational inequality was obtained and an optimal harvesting problem was solved by [27]. [28,29] established maximum principles of singular control and infinite horizon case for delayed systems. The second approach is based on anticipated BSDE (ABSDE for short), which was firstly proposed by Peng and Yang [30]. In [30], they presented the well-posedness results of nonlinear ABSDEs and illustrated ingenious duality relation between ABSDEs and stochastic differential delay equations (SDDEs for short). By virtue of ABSDEs, stochastic optimal control problems driven by SDDEs systems and backward delayed systems were studied in [31,32]. [33] obtained a maximum principle for delayed nonzero-sum games.

From the theoretical point of view, as we alluded to earlier, the conditional mean-field term $\mathbb{E}[X_t|\mathcal{F}_t^\alpha]$ in SDE occurs as the L^2 limit of interacting particles system with common Markov chain noise. Heuristically, $\int_{t-\delta}^t e^{-\rho(t-s)}\mathbb{E}\left[X_s|\mathcal{F}_s^\alpha\right]ds$ and $\mathbb{E}[X_{t-\delta}|\mathcal{F}_{t-\delta}^\alpha]$, which are average (distributed) delay and point delay of conditional mean-field terms, will appear if the interacting particles system depend on the delayed empirical mean terms $\frac{1}{N}\sum_{n=1}^N\int_{t-\delta}^t e^{-\rho(t-s)}X_s^n ds$ and $\frac{1}{N}\sum_{n=1}^N X_{t-\delta}^n$. Such delay structures in interacting particles system is very common, see [34–36]. Therefore, it is of great theoretical significance to explore the optimization problems driven by conditional mean-field hybrid switching systems with delay. From the practical point of view, the conditional mean-field hybrid switching systems with delay can describe complex phenomena in real-world and have great practical potentiality. Therefore, this paper investigates a conditional mean-field type game of regime-switching diffusions with delay and its applications. The major contributions of our work could be summarized as follows:

(1) We prove that a new kind of nonlinear conditional mean-field ABSDE with regime switching admits a unique solution under mild conditions. The adjoint equations in works [22,23,25,26,37] are conditional mean-field type BSDEs or forward and backward stochastic differential equations with regime switching without anticipated terms. However, since we consider the average delay and point delay of state and associated conditional mean-field terms in systems and costs, one of the adjoint equations in our paper (see (16)) turns out to be a conditional mean-field ABSDE with regime switching, where the generator at time t depends on the value of solution at a future time $t+\delta$ and the integral on the future path of solution during $[t,t+\delta]$ and related conditional mean-field terms. To the best of our knowledge, this is a new kind of equation that has never been

studied in existing literature. Under mild conditions, we prove that a nonlinear conditional mean-field ABSDE with regime switching equipped with above features admits a unique solution, which is the prerequisite in the establishment of necessary condition and verification theorems for open-loop Nash equilibrium point.

(2) We study a conditional mean-field type game of hybrid switching diffusions systems with delay. In our paper, both the average delay and point delay of state and associated conditional mean-field terms enter into dynamics and cost functionals, which endows the model with the ability to describe complex phenomena. However, the appearance of time-delay factors and conditional mean-field terms makes the problem become time-inconsistent. Therefore, we aim to establish the optimality condition for open-loop Nash equilibrium point in the form of maximum principle. Firstly, by introducing a conditional mean-field ABSDE as adjoint equation, we provide a necessary condition and a verification theorem for open-loop Nash equilibrium point. Then, by virtue of three coupled conditional mean-field BSDEs, another verification theorem is established. It should be emphasized that these two verification theorems, which hold true under different conditions, are exploited in different application scenarios.

(3) We present three financial applications of the above theoretical results. The first one is an N-agent conditional mean–variance portfolio selection problem with capital adjustment. We impose a new condition different from that in [21,28,29,38–40] to ensure the solvability of this problem (see (46) and Remark 5.2 for details), which is instructive to investigate the solvability for general stochastic differential games with delay. Under this condition, efficient portfolio and optimal cost for each agent are derived. The second one is an advertising and dividend choice game problem. Similar problems were investigated by virtue of abstract evolution equations in [41–43], where only some existence results about optimal control were given. We obtain the explicit optimal intensity of advertising expenditure and optimal dividend via regime-switching ABSDE, which becomes a backward ordinary differential equation (ODE for short) in some special cases. The third one is an N-agent inter-bank borrowing and lending problem with regime switching. Distinguished from [44] where exact Nash equilibrium was concerned, we solve its cooperative equilibrium through a conditional mean-field type control problem of hybrid switching diffusions systems. Optimal borrowing and lending rates of the representative bank are derived and the probability of systemic event is illustrated. Moreover, we find that, by introducing certain condition on the coefficients, the optimal control of representative bank constitutes an ϵ-Nash equilibrium of the N-agent inter-bank borrowing and lending problem. For each application, we present a number of numerical examples with sound economic interpretations, which verify the effectiveness of our results.

The rest content of the paper is structured as follows. In Section 2, we introduce some preliminaries and give the problem formulation. In Section 3, the existence and uniqueness of solution to a nonlinear conditional mean-field ABSDE with regime switching is proved. In Section 4, we establish one necessary condition and two verification theorems in terms of maximum principle for open-loop Nash equilibrium point. In Section 5, three financial applications are provided with several numerical examples. In Section 6, we give some concluding remarks.

2. Preliminaries and problem formulation

In this section, we present some notations and preliminaries, and give the formulation of the conditional mean-field type game of hybrid switching diffusions systems with delay.

2.1. Notations and preliminaries

Throughout this paper, we denote by $\mathbb{R}^n$ the n-dimensional vectors space with Euclidean norm $|\cdot|$, by $\mathbb{R}^{n\times m}$ the $n\times m$-dimensional matrices space. $A^\top$ denotes the transpose of matrix or vector A, $\|A\| = \sqrt{\mathrm{tr}\left[A^\top A\right]}$ denotes the norm of matrix A, and $\mathrm{tr}[A]$ denotes the trace of square matrix A. Hereinafter, we work with a finite terminal time T and a delay parameter $0 \le \delta \le T$. For $-\delta \le a \le b \le T$, let $C([a,b];\mathbb{R}^n)$ represent the set containing all continuous functions f_t from $[a,b]$ to $\mathbb{R}^n$ and $C^{k,l,p}([a,b]\times\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^m)$ represent the set of continuous functions $h(t,x,y)$ from $[a,b]\times\mathbb{R}^n\times\mathbb{R}^n$ to $\mathbb{R}^m$ which have k (resp. l,p)-order continuously differentiable partial derivatives w.r.t. variables t (resp. x,y) at most. Consider a complete filtered probability space $(\Omega,\mathcal{F},(\mathcal{F}_t)_{0\le t\le T},\mathbb{P})$ satisfying the usual conditions, where an $\mathbb{R}^d$-valued standard Brownian motion $W_\cdot$ and a continuous-time Markov chain $\alpha_\cdot$ are defined. Assume that the Markov chain $\alpha_\cdot$ is independent of $W_\cdot$ and takes values in a finite regime space $\mathcal{S} = \{1,\dots,S\}$. Let $Q = \left(Q^{i,j}\right)_{i,j=1}^{S\times S} \in \mathbb{R}^{S\times S}$ be the generator of $\alpha_\cdot$ such that $Q^{i,j} \ge 0$ for any $i \ne j \in \mathcal{S}$ and $\sum_{j\in\mathcal{S}} Q^{i,j} = 0$ for each $i \in \mathcal{S}$. Set $\mathcal{F}_t^\alpha = \sigma\left\{\alpha_r, 0 \le r \le t\right\}$, $\mathcal{F}_t = \sigma\left\{\alpha_r, W_r, 0 \le r \le t\right\}$ for $t \in [0,T]$ and $\mathcal{F}_t^\alpha = \mathcal{F}_0^\alpha$ for $t \in [-\delta,0]$.

Let $\mathbb{P}^\alpha(t) = \left(\mathbb{P}_{i,j}(t)\right)_{i,j}^{S\times S}$ be the transition matrix of homogeneous Markov chain $\alpha_\cdot$, where

$$\mathbb{P}_{i,j}(t) = \mathbb{P}\left\{\alpha(t+\tau) = j \,|\, \alpha(\tau) = i\right\}, \qquad i,j \in \mathcal{S}.$$

Denote by $\mathcal{I}_{S\times S}$ the $S\times S$-dimensional identity matrix. Then $\mathbb{P}^\alpha(t)$ satisfies the backward Kolmogorov differential equation

$$\begin{cases} \dfrac{d\mathbb{P}^\alpha(t)}{dt} = Q\mathbb{P}^\alpha(t), & t \in [0,T], \\ \mathbb{P}^\alpha(0) = \mathcal{I}_{S\times S}, \end{cases}$$

which has a unique solution $\mathbb{P}^\alpha(t) = e^{Qt}$. Suppose that the initial state of $\alpha_\cdot$ is i_0, i.e., $\alpha_0 = i_0$.

Next, we provide some preliminary results about the martingale associated with Markov chain $\alpha_\cdot$. For any $t \in [0,T]$ and $i \neq j \in \mathcal{S}$, let $\left[M^{i,j}\right]_t = \sum_{0\le s\le t} \mathbb{1}_{\{\alpha_{s-}=i\}} \mathbb{1}_{\{\alpha_s=j\}}$ and $\left\langle M^{i,j}\right\rangle_t = \int_0^t Q^{i,j} \mathbb{1}_{\{\alpha_{s-}=i\}} ds$, where $\mathbb{1}_{\mathcal{A}}$ denotes the indicator function of set $\mathcal{A}$. Then $M_t^{i,j} = \left[M^{i,j}\right]_t - \left\langle M^{i,j}\right\rangle_t$ is a purely discontinuous and square-integrable martingale w.r.t. $\mathcal{F}_t^\alpha$ and is null at the origin [45]. In fact, $\left[M^{i,j}\right]_t$ and $\left\langle M^{i,j}\right\rangle_t$ are the optional and predictable variation processes of $M_t^{i,j}$, respectively. Moreover, we define that $M_t^{i,i} = \left[M^{i,i}\right]_t = \left\langle M^{i,i}\right\rangle_t = 0$ for any $i \in \mathcal{S}$, $t \in [0,T]$.

Then, we introduce the following spaces.

$$\mathcal{L}^2_{\mathcal{F}_T}(\mathbb{R}^n) = \left\{ \xi : \Omega \to \mathbb{R}^n \middle| \xi \text{ is } \mathcal{F}_T\text{-measurable random variable and } \mathbb{E}|\xi|^2 < \infty \right\},$$

$$S^2_{\mathcal{F}}([0,T];\mathbb{R}^n) = \left\{ \mathcal{X}_\cdot : [0,T]\times\Omega \to \mathbb{R}^n \middle| \mathcal{X}_t \text{ is } \mathcal{F}_t\text{-adapted, càdlàg process and } \mathbb{E}\sup_{t\in[0,T]}|\mathcal{X}_t|^2 < \infty \right\},$$

$$\mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^n) = \left\{ \mathcal{X}_\cdot : [0,T]\times\Omega \to \mathbb{R}^n \middle| \mathcal{X}_t \text{ is } \mathcal{F}_t\text{-progressively measurable process and } \mathbb{E}\left[\int_0^T |\mathcal{X}_t|^2 dt\right] < \infty \right\},$$

$$\mathcal{K}^2_{\mathcal{F}}([0,T];\mathbb{R}^n) = \left\{ \mathcal{R}_\cdot = \left\{ \left(\mathcal{R}^{1,i,j}_\cdot\right)_{i,j=1}^{S\times S}, \dots, \left(\mathcal{R}^{n,i,j}_\cdot\right)_{i,j=1}^{S\times S} \right\} \middle| \mathcal{R}^{\ell,i,j}_t \text{ is } \mathcal{F}_t\text{-progressively measurable process} \right.$$
$$\left. \text{for any } i \in \mathcal{S}, \ell \in \{1,\dots,n\} \text{ and } \sum_{\ell=1}^n \sum_{i,j\in\mathcal{S}} \mathbb{E}\left[\int_0^T |\mathcal{R}^{\ell,i,j}_t|^2 d\left[M^{i,j}\right]_t\right] < \infty \right\}.$$

Given that $M_t^{i,j}$ $(i,j \in \mathcal{S})$ is a martingale w.r.t. $\mathcal{F}_t^\alpha$, for a process $\mathcal{R}_\cdot \in \mathcal{K}^2_{\mathcal{F}}([0,T];\mathbb{R}^n)$, we denote

$$\mathcal{R}_t \bullet dM_t = \left(\sum_{i,j\in\mathcal{S}} \mathcal{R}^{1,i,j}_t dM^{i,j}_t, \dots, \sum_{i,j\in\mathcal{S}} \mathcal{R}^{n,i,j}_t dM^{i,j}_t \right)^\top.$$

Finally, we present a useful result that will be employed in the subsequent derivation. Its proof is similar to Lemma 5.4 in [46]. Therefore, we omit it here to save space.

Lemma 2.1. *If $\mathcal{X}_\cdot$ is a square-integrable and $\mathcal{F}_t$-adapted process, then, for any $t \in [0,T]$, we have*

1. $\mathbb{E}\left[\int_0^t \mathcal{X}_s ds \middle| \mathcal{F}_t^\alpha\right] = \int_0^t \mathbb{E}\left[\mathcal{X}_s | \mathcal{F}_s^\alpha\right] ds$;
2. $\mathbb{E}\left[\int_0^t \mathcal{X}_s dW_s \middle| \mathcal{F}_t^\alpha\right] = 0$.

2.2. Problem formulation

Suppose that there are N decision makers (players) denoted by $\mathcal{N} = \{1,\dots,N\}$. Consider the following controlled nonlinear conditional mean-field hybrid switching diffusions systems with delay:

$$\begin{cases} dX_t^v = b(t,\alpha_t,X_t^v,Y_t^v,Z_t^v,\hat{X}_t^v,\hat{Y}_t^v,\hat{Z}_t^v,v_t^1,\dots,v_t^N)dt + \sigma(t,\alpha_t,X_t^v,Y_t^v,Z_t^v,\hat{X}_t^v,\hat{Y}_t^v,\hat{Z}_t^v,v_t^1,\dots,v_t^N)dW_t, & t\in[0,T], \\ X_t^v = f_t, \quad t\in[-\delta,0], \quad f_\cdot \in C([-\delta,0],\mathbb{R}^n). \end{cases} \tag{2}$$

In (2), we let the $\mathbb{R}^{d_k}$-valued process $v_\cdot^k$ denote the control of player k ($k \in \mathcal{N}$), the $\mathbb{R}^n$-valued process $X_\cdot^v$ denote the controlled state where the superscript v highlights its dependence on control $v_\cdot = (v_\cdot^1,\dots,v_\cdot^N)$. Y_t^v and Z_t^v are given linear functionals of previous path X_{t+s} for $s \in [-\delta,0]$, which are defined by

$$Y_t^v = \int_{t-\delta}^t e^{-\rho(t-s)} X_s^v ds, \qquad Z_t^v = X_{t-\delta}^v, \qquad t\in[0,T], \tag{3}$$

where $\rho > 0$. More precisely, $Y_\cdot^v$ and $Z_\cdot^v$ are referred to as average delay and point delay of $X_\cdot^v$, respectively. $\hat{X}_\cdot^v, \hat{Y}_\cdot^v, \hat{Z}_\cdot^v$ are conditional expectations of $X_\cdot^v, Y_\cdot^v, Z_\cdot^v$ w.r.t. the history of $\alpha_\cdot$, i.e.,

$$\hat{X}_t^v = \mathbb{E}[X_t^v|\mathcal{F}_t^\alpha], \qquad \hat{Y}_t^v = \mathbb{E}\left[Y_t^v|\mathcal{F}_t^\alpha\right], \qquad \hat{Z}_t^v = \mathbb{E}[X_{t-\delta}^v|\mathcal{F}_{t-\delta}^\alpha] = \hat{X}_{t-\delta}^v.$$

Actually, from Lemma 2.1, we derive that

$$\hat{Y}_t^v = \mathbb{E}\left[Y_t^v|\mathcal{F}_t^\alpha\right] = \mathbb{E}\left[\int_{t-\delta}^t e^{-\rho(t-s)} X_s^v ds \middle| \mathcal{F}_t^\alpha\right] = \int_{t-\delta}^t e^{-\rho(t-s)} \mathbb{E}\left[X_s^v|\mathcal{F}_s^\alpha\right] ds = \int_{t-\delta}^t e^{-\rho(t-s)} \hat{X}_s ds.$$

Therefore, $\hat{Y}_\cdot^v$ and $\hat{Z}_\cdot^v$ can be regarded as average delay and point delay of conditional expectation $\hat{X}_\cdot^v$.

For player k ($k \in \mathcal{N}$), we consider a cost functional defined as

$$J^k[v_\cdot] = \mathbb{E}\left[\int_0^T l^k\left(t,\alpha_t,X_t^v,Y_t^v,Z_t^v,\hat{X}_t^v,\hat{Y}_t^v,\hat{Z}_t^v,v_t\right)dt + g^k\left(\alpha_T,X_T^v,Y_T^v,\hat{X}_T^v,\hat{Y}_T^v\right)\right]. \tag{4}$$

Letting $\mathbb{U} = \prod_{k\in\mathcal{N}} \mathbb{U}^k$ where $\mathbb{U}^k \subseteq \mathbb{R}^{d_k}$ is a non-empty convex set, the coefficients in (2) and (4) are suitable mappings

$$b,\sigma,l^k : [0,T]\times\mathcal{S}\times(\mathbb{R}^n)^6\times\mathbb{U} \to \mathbb{R}^n, \mathbb{R}^{n\times d}, \mathbb{R},$$
$$g^k : \mathcal{S}\times(\mathbb{R}^n)^4 \to \mathbb{R},$$

and satisfy the following condition.

Assumption (A1).

(i) For any $t \in [0,T]$, $i \in \mathcal{S}$, $b(t,i,\cdot,\cdot,\cdot,\cdot,\cdot,\cdot,\cdot)$ and $\sigma(t,i,\cdot,\cdot,\cdot,\cdot,\cdot,\cdot,\cdot,\cdot)$ are continuously differentiable w.r.t. $(x,y,z,\hat{x},\hat{y},\hat{z},v)$. All the partial derivatives of b and σ are uniformly bounded.

(ii) For any $t \in [0,T]$, $i \in \mathcal{S}$, $k \in \mathcal{N}$, $l^k(t,i,\cdot,\cdot,\cdot,\cdot,\cdot,\cdot,\cdot,\cdot)$ and $g^k(i,\cdot,\cdot,\cdot,\cdot)$ are continuously differentiable w.r.t. $(x,y,z,\hat{x},\hat{y},\hat{z},v)$ and $(x,y,\hat{x},\hat{y})$, respectively. All the partial derivatives of l^k and g^k are continuous and bounded by $C\left(1+|x|+|y|+|z|+|\hat{x}|+|\hat{y}|+|\hat{z}|+|v|\right)$ and $C\left(1+|x|+|y|+|\hat{x}|+|\hat{y}|\right)$.

Consider the set $\mathcal{U}^k = \left\{v_\cdot^k | v_\cdot^k \in \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{U}^k)\right\}$ where $k \in \mathcal{N}$. Each $v_\cdot^k \in \mathcal{U}^k$ is called an open-loop admissible control for player k ($k \in \mathcal{N}$). We let $\mathcal{U} = \prod_{k\in\mathcal{N}} \mathcal{U}^k$ represent the set which contains all the open-loop admissible controls of N players.

Under Assumption (A1), by utilizing similar arguments of Lemma 2.1 in [21], we can prove that Eq. (2) admits a unique solution $X_\cdot^v \in \mathcal{S}^2_{\mathcal{F}}([0,T];\mathbb{R}^n)$ for any admissible control $v_\cdot \in \mathcal{U}$. Additionally, if Assumption (A1) holds, then cost functional (4) is well-defined for any $v_\cdot \in \mathcal{U}$. The problem we concern is summarized as follows.

Problem (CMFG): For system (2) and cost (4), find an admissible control $u_\cdot = (u_\cdot^1,\dots,u_\cdot^N) \in \mathcal{U}$ under which

$$J^k[u_\cdot] \le J^k[(u_\cdot^{-k}, v_\cdot^k)], \qquad \forall k \in \mathcal{N}, v_\cdot^k \in \mathcal{U}^k \tag{5}$$

holds, where $u_\cdot^{-k} = (u_\cdot^1,\dots,u_\cdot^{k-1},u_\cdot^{k+1},\dots,u_\cdot^N)$ is the control except for player k.

The admissible control $u_\cdot$ satisfying (5) is referred to as an open-loop Nash equilibrium point of Problem (CMFG). It should be noted that both dynamics (2) and cost functional (4) depend on the conditional distributions of state through conditional expectation terms. Moreover, the coefficients in (2) and (4) are Markov modulated processes and involve delay terms. Therefore, Problem (CMFG) is a conditional mean-field type game of hybrid switching diffusions systems with delay. The main result of this paper is devoted to establishing the necessary condition and verification theorems for open-loop Nash equilibrium point of Problem (CMFG).

3. Existence and uniqueness of solution to a nonlinear conditional mean-field ABSDE with regime switching

In this section, we provide a well-posedness result to a nonlinear conditional mean-field ABSDE with regime switching, which covers adjoint equations (16) and (26)–(28) in the optimality conditions for Nash equilibrium as special cases. Consider

$$\begin{cases} d\mathcal{Y}_t = -\mathbb{E}\Big[F\Big(t,\alpha_t,\mathcal{Y}_t,\mathcal{Z}_t,\mathcal{Y}_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t),\mathcal{Z}_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t),\int_t^{t+\delta} e^{\rho(t-s)}\mathcal{Y}_s\mathbb{1}_{[0,T]}(s)ds, \\ \qquad \int_t^{t+\delta} e^{\rho(t-s)}\mathcal{Z}_s\mathbb{1}_{[0,T]}(s)ds, \mathbb{E}[\mathcal{Y}_t|\mathcal{F}_t^\alpha],\mathbb{E}[\mathcal{Z}_t|\mathcal{F}_t^\alpha],\mathbb{E}[\mathcal{Y}_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t^\alpha],\mathbb{E}[\mathcal{Z}_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t^\alpha], \\ \qquad \mathbb{E}\left[\int_t^{t+\delta} e^{\rho(t-s)}\mathcal{Y}_s\mathbb{1}_{[0,T]}(s)ds\Big|\mathcal{F}_t^\alpha\right],\mathbb{E}\left[\int_t^{t+\delta} e^{\rho(t-s)}\mathcal{Z}_s\mathbb{1}_{[0,T]}(s)ds\Big|\mathcal{F}_t^\alpha\right]\Big)\Big|\mathcal{F}_t\Big]dt + \mathcal{Z}_t dW_t + \mathcal{R}_t \bullet dM_t, \\ \mathcal{Y}_T = \xi, \end{cases} \tag{6}$$

where F is a mapping from $\Omega\times[0,T]\times\mathcal{S}\times(\mathbb{R}^n\times\mathbb{R}^{n\times d})^6$ to $\mathbb{R}^n$ and ξ is a random variable that is $\mathcal{F}_T$-measurable. Noteworthily, the generator F at time t depends on the value of solution at a future time $t+\delta$ and the integral on the future path of solution during $[t,t+\delta]$ and corresponding conditional mean-field terms.

To unburden the complexity of notations, for a given $(\mathcal{Y}_\cdot,\mathcal{Z}_\cdot,\mathcal{R}_\cdot) \in \mathcal{S}^2_{\mathcal{F}}([0,T];\mathbb{R}^n)\times\mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^{n\times d})\times\mathcal{K}^2_{\mathcal{F}}([0,T];\mathbb{R}^n)$, we set

$$\begin{aligned} \Gamma_t &= \left(\mathcal{Y}_t,\mathcal{Z}_t\right), \\ \Gamma_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t) &= (\mathcal{Y}_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t),\mathcal{Z}_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t)), \\ \Gamma_{t,\delta}\mathbb{1}_{[0,T-\delta]}(t) &= \left(\int_t^{t+\delta} e^{\rho(t-s)}\mathcal{Y}_s ds,\int_t^{t+\delta} e^{\rho(t-s)}\mathcal{Z}_s ds\right)\mathbb{1}_{[0,T-\delta]}(t) = \left(\int_t^{t+\delta} e^{\rho(t-s)}\mathcal{Y}_s\mathbb{1}_{[0,T]}(s)ds,\int_t^{t+\delta} e^{\rho(t-s)}\mathcal{Z}_s\mathbb{1}_{[0,T]}(s)ds\right), \end{aligned}$$

and introduce the norm $|\Gamma_\cdot|_* = \sqrt{|\mathcal{Y}_\cdot|^2 + \|\mathcal{Z}_\cdot\|^2}$. Then regime-switching conditional mean-field ABSDE (6) is reformulated as

$$\begin{cases} d\mathcal{Y}_t = -\mathbb{E}\left[F\left(t,\alpha_t,\Gamma_t,\Gamma_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t),\Gamma_{t,\delta}\mathbb{1}_{[0,T-\delta]}(t),\mathbb{E}[\Gamma_t|\mathcal{F}_t^\alpha],\mathbb{E}[\Gamma_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t^\alpha],\right.\right. \\ \qquad \left.\left.\mathbb{E}[\Gamma_{t,\delta}\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t^\alpha]\right)|\mathcal{F}_t\right]dt + \mathcal{Z}_t dW_t + \mathcal{R}_t \bullet dM_t, \\ \mathcal{Y}_T = \xi. \end{cases}$$

Furthermore, we introduce the following assumption.

Assumption (A2).

(i) $\xi \in \mathcal{L}^2_{\mathcal{F}_T}(\mathbb{R}^n)$;

(ii) For any $i \in \mathcal{S}$, $F(\cdot,i,0,0,0,0,0,0) \in \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^n)$;

(iii) There exists a constant $L>0$ such that, for any $t\in[0,T]$, $i\in\mathcal{S}$, $(\phi^\ell,\varphi^\ell,\psi^\ell,\hat{\phi}^\ell,\hat{\varphi}^\ell,\hat{\psi}^\ell) \in (\mathbb{R}^n\times\mathbb{R}^{n\times d})^6$, $\ell=1,2$,

$$\begin{aligned} &|F(t,i,\phi^1,\varphi^1,\psi^1,\hat{\phi}^1,\hat{\varphi}^1,\hat{\psi}^1) - F(t,i,\phi^2,\varphi^2,\psi^2,\hat{\phi}^2,\hat{\varphi}^2,\hat{\psi}^2)| \\ &\le L\left(|\phi^1-\phi^2|_* + |\varphi^1-\varphi^2|_* + |\psi^1-\psi^2|_* + |\hat{\phi}^1-\hat{\phi}^2|_* + |\hat{\varphi}^1-\hat{\varphi}^2|_* + |\hat{\psi}^1-\hat{\psi}^2|_*\right), \qquad \mathbb{P}-a.s. \end{aligned}$$

Then we present the following well-posedness result about (6).

Theorem 3.1. *If Assumption (A2) is satisfied, then nonlinear regime-switching conditional mean-field ABSDE (6) admits a unique solution* $(\mathcal{Y}_\cdot, \mathcal{Z}_\cdot, \mathcal{R}_\cdot) \in \mathcal{S}^2_{\mathcal{F}}([0,T];\mathbb{R}^n) \times \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^{n\times d}) \times \mathcal{K}^2_{\mathcal{F}}([0,T];\mathbb{R}^n)$.

Proof. For a given $(y_\cdot, z_\cdot, r_\cdot) \in \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^n) \times \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^{n\times d}) \times \mathcal{K}^2_{\mathcal{F}}([0,T];\mathbb{R}^n)$, we denote

$$\begin{aligned}
\gamma_t &= \left(y_t, z_t\right),\\
\gamma_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t) &= (y_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t), z_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t)),\\
\gamma_{t,\delta}\mathbb{1}_{[0,T-\delta]}(t) &= \left(\int_t^{t+\delta} e^{\rho(t-s)} y_s ds, \int_t^{t+\delta} e^{\rho(t-s)} z_s ds\right)\mathbb{1}_{[0,T-\delta]}(t) = \left(\int_t^{t+\delta} e^{\rho(t-s)} y_s \mathbb{1}_{[0,T]}(s) ds, \int_t^{t+\delta} e^{\rho(t-s)} z_s \mathbb{1}_{[0,T]}(s) ds\right),
\end{aligned}$$

and introduce the following BSDE with regime switching:

$$\begin{cases}
d\mathcal{Y}_t = -\mathbb{E}\Big[F\Big(t, \alpha_t, \gamma_t, \gamma_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t), \gamma_{t,\delta}\mathbb{1}_{[0,T-\delta]}(t), \mathbb{E}[\gamma_t|\mathcal{F}^\alpha_t], \mathbb{E}[\gamma_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}^\alpha_t],\\
\qquad \mathbb{E}[\gamma_{t,\delta}\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}^\alpha_t]\Big)|\mathcal{F}_t\Big]dt + \mathcal{Z}_t dW_t + \mathcal{R}_t \bullet dM_t,\\
\mathcal{Y}_T = \xi.
\end{cases} \tag{7}$$

It follows from Propositions 5.1 and 5.2 in [47] that regime-switching BSDE (7) admits a unique solution $(\mathcal{Y}_\cdot, \mathcal{Z}_\cdot, \mathcal{R}_\cdot) \in \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^n) \times \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^{n\times d}) \times \mathcal{K}^2_{\mathcal{F}}([0,T];\mathbb{R}^n)$. Consequently, we are able to construct a mapping Φ from space $\mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^n) \times \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^{n\times d}) \times \mathcal{K}^2_{\mathcal{F}}([0,T];\mathbb{R}^n)$ to itself by letting $(\mathcal{Y}_\cdot, \mathcal{Z}_\cdot, \mathcal{R}_\cdot) = \Phi((y_\cdot, z_\cdot, r_\cdot))$, where $(\mathcal{Y}_\cdot, \mathcal{Z}_\cdot, \mathcal{R}_\cdot)$ is the unique solution of (7) corresponding to $(y_\cdot, z_\cdot, r_\cdot)$. For $(y^\ell_\cdot, z^\ell_\cdot, r^\ell_\cdot) \in \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^n) \times \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}^{n\times d}) \times \mathcal{K}^2_{\mathcal{F}}([0,T];\mathbb{R}^n)$ where $\ell = 1,2$, let $(\mathcal{Y}^\ell_\cdot, \mathcal{Z}^\ell_\cdot, \mathcal{R}^\ell_\cdot)$ be the associated solution of (7), i.e., $(\mathcal{Y}^\ell_\cdot, \mathcal{Z}^\ell_\cdot, \mathcal{R}^\ell_\cdot) = \Phi((y^\ell_\cdot, z^\ell_\cdot, r^\ell_\cdot))$. For simplicity, we use the short-hand notations

$$\begin{aligned}
\Delta\varrho_t &= \varrho^1_t - \varrho^2_t, \qquad \varrho = \mathcal{Y}, \mathcal{Z}, \mathcal{R}, y, z, r,\\
\Delta F(t) &= \mathbb{E}\Big[F\Big(t, \alpha_t, \gamma^1_t, \gamma^1_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t), \gamma^1_{t,\delta}\mathbb{1}_{[0,T-\delta]}(t), \mathbb{E}[\gamma^1_t|\mathcal{F}^\alpha_t], \mathbb{E}[\gamma^1_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}^\alpha_t], \mathbb{E}[\gamma^1_{t,\delta}\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}^\alpha_t]\Big)\\
&\quad - F\Big(t, \alpha_t, \gamma^2_t, \gamma^2_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t), \gamma^2_{t,\delta}\mathbb{1}_{[0,T-\delta]}(t), \mathbb{E}[\gamma^2_t|\mathcal{F}^\alpha_t], \mathbb{E}[\gamma^2_{t+\delta}\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}^\alpha_t], \mathbb{E}[\gamma^2_{t,\delta}\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}^\alpha_t]\Big)|\mathcal{F}_t\Big].
\end{aligned}$$

It yields from (7) that $(\Delta\mathcal{Y}_\cdot, \Delta\mathcal{Z}_\cdot, \Delta\mathcal{R}_\cdot)$ satisfies

$$\begin{cases}
d\Delta\mathcal{Y}_t = -\Delta F(t)dt + \Delta\mathcal{Z}_t dW_t + \Delta\mathcal{R}_t \bullet dM_t,\\
\Delta\mathcal{Y}_T = 0.
\end{cases}$$

Applying Itô's formula for semi-martingale to $e^{\beta t}|\Delta\mathcal{Y}_t|^2$ $(\beta \geq 0)$ yields

$$\begin{aligned}
&\mathbb{E}\left[e^{\beta t}|\Delta\mathcal{Y}_t|^2 + \int_t^T e^{\beta s}\|\Delta\mathcal{Z}_s\|^2 ds + \sum_{\ell=1}^n \sum_{i,j\in\mathcal{S}} \int_t^T e^{\beta s}|\Delta\mathcal{R}^{\ell,i,j}_s|^2 d\left[M^{i,j}\right]_s\right]\\
&= \mathbb{E}\left[\int_t^T -e^{\beta s}\beta|\Delta\mathcal{Y}_s|^2 ds + \int_t^T 2e^{\beta s}\Delta\mathcal{Y}_s^\top \Delta F(s) ds\right]\\
&\leq \frac{6L^2}{\beta}\mathbb{E}\left[\int_t^T e^{\beta s}\mathbb{E}\left[\left(|\Delta\gamma_s|^2_* + |\mathbb{E}[\Delta\gamma_s|\mathcal{F}^\alpha_s]|^2_* + |\Delta\gamma_{s+\delta}\mathbb{1}_{[0,T-\delta]}(s)|^2_* + |\mathbb{E}[\Delta\gamma_{s+\delta}\mathbb{1}_{[0,T-\delta]}(s)|\mathcal{F}^\alpha_s]|^2_*\right.\right.\right.\\
&\quad \left.\left.\left. + |\Delta\gamma_{s,\delta}\mathbb{1}_{[0,T-\delta]}(s)|^2_* + |\mathbb{E}[\Delta\gamma_{s,\delta}\mathbb{1}_{[0,T-\delta]}(s)|\mathcal{F}^\alpha_s]|^2_*\right)|\mathcal{F}_s\right] ds\right].
\end{aligned} \tag{8}$$

By Jensen's inequality, we derive

$$\begin{aligned}
&\mathbb{E}\left[\int_t^T e^{\beta s}\mathbb{E}\left[\left(|\Delta\gamma_s|^2_* + |\mathbb{E}[\Delta\gamma_s|\mathcal{F}^\alpha_s]|^2_*\right)|\mathcal{F}_s\right] ds\right]\\
&\leq 2\mathbb{E}\left[\int_t^T e^{\beta s}|\Delta\gamma_s|^2_* ds\right]\\
&= 2\mathbb{E}\left[\int_t^T e^{\beta s}\left(|\Delta y_s|^2 + \|\Delta z_s\|^2\right) ds\right],
\end{aligned} \tag{9}$$

and

$$\begin{aligned}
&\mathbb{E}\left[\int_t^T e^{\beta s}\mathbb{E}\left[\left(|\Delta\gamma_{s+\delta}\mathbb{1}_{[0,T-\delta]}(s)|^2_* + |\mathbb{E}[\Delta\gamma_{s+\delta}\mathbb{1}_{[0,T-\delta]}(s)|\mathcal{F}^\alpha_s]|^2_*\right)|\mathcal{F}_s\right] ds\right]\\
&\leq 2\mathbb{E}\left[\int_t^T e^{\beta s}|\Delta\gamma_{s+\delta}\mathbb{1}_{[0,T-\delta]}(s)|^2_* ds\right]\\
&\leq 2e^{-\beta\delta}\mathbb{E}\left[\int_t^T e^{\beta s}|\Delta\gamma_s|^2_* ds\right]\\
&\leq 2\mathbb{E}\left[\int_t^T e^{\beta s}\left(|\Delta y_s|^2 + \|\Delta z_s\|^2\right) ds\right].
\end{aligned} \tag{10}$$

Considering $\rho > 0$ and $0 \le \delta \le T$, it follows that

$$\int_s^{s+\delta} e^{2\rho(s-\tau)} d\tau \le \delta \le T.$$

Then

$$\begin{aligned}
&\mathbb{E}\left[\int_t^T e^{\beta s}\left(|\Delta\gamma_{s,\delta}\mathbb{1}_{[0,T-\delta]}(s)|_*^2 + |\mathbb{E}[\Delta\gamma_{s,\delta}\mathbb{1}_{[0,T-\delta]}(s)|\mathcal{F}_s^\alpha]|_*^2\right) ds\right] \\
&\le 2\mathbb{E}\left[\int_t^T e^{\beta s}\left(\left|\int_s^{s+\delta} e^{\rho(s-\tau)}\Delta y_\tau \mathbb{1}_{[0,T]}(\tau) d\tau\right|^2 + \left\|\int_s^{s+\delta} e^{\rho(s-\tau)}\Delta z_\tau \mathbb{1}_{[0,T]}(\tau) d\tau\right\|^2\right) ds\right] \\
&\le 2\mathbb{E}\left[\int_t^T \int_s^{s+\delta} e^{2\rho(s-\tau)} d\tau e^{\beta s} \int_s^{s+\delta} \left(|\Delta y_\tau|^2 + \|\Delta z_\tau\|^2\right) \mathbb{1}_{[0,T]}(\tau) d\tau ds\right] \\
&\le 2T\mathbb{E}\left[\int_t^T \int_s^{s+\delta} e^{\beta s}\left(|\Delta y_\tau|^2 + \|\Delta z_\tau\|^2\right) \mathbb{1}_{[0,T]}(\tau) d\tau ds\right] \\
&\le 2T^2\mathbb{E}\left[\int_t^T e^{\beta s}\left(|\Delta y_s|^2 + \|\Delta z_s\|^2\right) ds\right].
\end{aligned} \tag{11}$$

Inserting (9)–(11) into (8), we get

$$\begin{aligned}
&\mathbb{E}\left[e^{\beta t}|\Delta\mathcal{Y}_t|^2 + \int_t^T e^{\beta s}\|\Delta\mathcal{Z}_s\|^2 ds + \sum_{\ell=1}^n \sum_{i,j\in\mathcal{S}} \int_t^T e^{\beta s}|\Delta\mathcal{R}_s^{\ell,i,j}|^2 d\left[M^{i,j}\right]_s\right] \\
&\le \frac{12L^2\left(2+T^2\right)}{\beta}\mathbb{E}\left[\int_t^T e^{\beta s}\left(|\Delta y_s|^2 + \|\Delta z_s\|^2\right) ds\right].
\end{aligned}$$

Then, one obtains

$$\mathbb{E}\left[\int_0^T e^{\beta s}|\Delta\mathcal{Y}_s|^2 ds\right] \le \frac{12L^2\left(2+T^2\right)T}{\beta}\mathbb{E}\left[\int_0^T e^{\beta s}\left(|\Delta y_s|^2 + \|\Delta z_s\|^2\right) ds\right],$$

and

$$\begin{aligned}
&\mathbb{E}\left[\int_0^T e^{\beta s}\left(|\Delta\mathcal{Y}_s|^2 + \|\Delta\mathcal{Z}_s\|^2\right) ds + \sum_{\ell=1}^n \sum_{i,j\in\mathcal{S}} \int_0^T e^{\beta s}|\Delta\mathcal{R}_s^{\ell,i,j}|^2 d\left[M^{i,j}\right]_s\right] \\
&\le \frac{12L^2\left(2+T^2\right)(T+1)}{\beta}\mathbb{E}\left[\int_0^T e^{\beta s}\left(|\Delta y_s|^2 + \|\Delta z_s\|^2\right) ds + \sum_{\ell=1}^n \sum_{i,j\in\mathcal{S}} \int_0^T e^{\beta s}|\Delta r_s^{\ell,i,j}|^2 d\left[M^{i,j}\right]_s\right].
\end{aligned} \tag{12}$$

Since $\mathcal{L}_{\mathcal{F}}^2([0,T];\mathbb{R}^n) \times \mathcal{L}_{\mathcal{F}}^2([0,T];\mathbb{R}^{n\times d}) \times \mathcal{K}_{\mathcal{F}}^2([0,T];\mathbb{R}^n)$ is a Banach space equipped with the norm

$$|(\mathcal{Y}_\cdot,\mathcal{Z}_\cdot,\mathcal{R}_\cdot)|_\beta = \left\{\mathbb{E}\left[\int_0^T e^{\beta s}\left(|\mathcal{Y}_s|^2 + \|\mathcal{Z}_s\|^2\right) ds + \sum_{\ell=1}^n \sum_{i,j\in\mathcal{S}} \int_0^T e^{\beta s}|\mathcal{R}_s^{\ell,i,j}|^2 d\left[M^{i,j}\right]_s\right]\right\}^{\frac{1}{2}}, \tag{13}$$

taking $\beta = 48L^2\left(2+T^2\right)(T+1)$ in (12) obtains

$$|(\Delta\mathcal{Y}_\cdot,\Delta\mathcal{Z}_\cdot,\Delta\mathcal{R}_\cdot)|_\beta \le \frac{1}{2}|(\Delta y_\cdot,\Delta z_\cdot,\Delta r_\cdot)|_\beta,$$

which shows that Φ is a contraction mapping. It yields from Banach fixed point theorem that there exists a unique triplet $(\mathcal{Y}_\cdot,\mathcal{Z}_\cdot,\mathcal{R}_\cdot)$ satisfying $(\mathcal{Y}_\cdot,\mathcal{Z}_\cdot,\mathcal{R}_\cdot) = \Phi((\mathcal{Y}_\cdot,\mathcal{Z}_\cdot,\mathcal{R}_\cdot))$. Therefore, Eq. (6) has a unique solution $(\mathcal{Y}_\cdot,\mathcal{Z}_\cdot,\mathcal{R}_\cdot) \in \mathcal{L}_{\mathcal{F}}^2([0,T];\mathbb{R}^n) \times \mathcal{L}_{\mathcal{F}}^2([0,T];$

$\mathbb{R}^{n\times d})\times\mathcal{K}_{\mathcal{F}}^2([0,T];\mathbb{R}^n)$. In addition, from B-D-G inequality, Jensen's inequality, and Hölder's inequality, we derive

$$\begin{aligned}
&\mathbb{E}\sup_{t\in[0,T]}|\mathcal{Y}_t|^2\\
&\le 4\Bigg\{\mathbb{E}|\xi|^2+\mathbb{E}\Bigg[\Big|\int_0^T\mathbb{E}\Big[F\left(s,\alpha_s,\Gamma_s,\Gamma_{s+\delta}\mathbb{1}_{[0,T-\delta]}(s),\Gamma_{s,\delta}\mathbb{1}_{[0,T-\delta]}(s),\mathbb{E}[\Gamma_s|\mathcal{F}_s^\alpha],\mathbb{E}[\Gamma_{s+\delta}\mathbb{1}_{[0,T-\delta]}(s)|\mathcal{F}_s^\alpha],\right.\\
&\qquad\left.\mathbb{E}[\Gamma_{s,\delta}\mathbb{1}_{[0,T-\delta]}(s)|\mathcal{F}_s^\alpha]\right)|\mathcal{F}_s\Big]ds\Big|^2\Bigg]+\mathbb{E}\sup_{t\in[0,T]}\Big|\int_t^T\mathcal{Z}_s dW_s\Big|^2+\mathbb{E}\sup_{t\in[0,T]}\Big|\int_t^T\mathcal{R}_s\bullet dM_s\Big|^2\Bigg\}\\
&\le 4\Bigg\{\mathbb{E}|\xi|^2+T\mathbb{E}\Bigg[\int_0^T\Big|F\left(s,\alpha_s,\Gamma_s,\Gamma_{s+\delta}\mathbb{1}_{[0,T-\delta]}(s),\Gamma_{s,\delta}\mathbb{1}_{[0,T-\delta]}(s),\mathbb{E}[\Gamma_s|\mathcal{F}_s^\alpha],\mathbb{E}[\Gamma_{s+\delta}\mathbb{1}_{[0,T-\delta]}(s)|\mathcal{F}_s^\alpha],\right.\\
&\qquad\left.\mathbb{E}[\Gamma_{s,\delta}\mathbb{1}_{[0,T-\delta]}(s)|\mathcal{F}_s^\alpha]\right)\Big|^2ds\Bigg]+4\mathbb{E}\sup_{t\in[0,T]}\Big|\int_0^t\mathcal{Z}_s dW_s\Big|^2+4\mathbb{E}\sup_{t\in[0,T]}\Big|\int_0^t\mathcal{R}_s\bullet dM_s\Big|^2\Bigg\}\\
&\le 4\Bigg\{\mathbb{E}|\xi|^2+TC\mathbb{E}\Bigg[\int_0^T\left(|\mathcal{Y}_s|^2+\|\mathcal{Z}_s\|^2\right)ds\Bigg]+2T\mathbb{E}\Bigg[\int_0^T|F(s,\alpha_s,0,0,0,0,0,0)|^2ds\Bigg]+4C\mathbb{E}\Bigg[\int_0^T\|\mathcal{Z}_s\|^2ds\Bigg]\\
&\qquad+4C\sum_{\ell=1}^n\sum_{i,j\in\mathcal{S}}\mathbb{E}\int_0^T|\mathcal{R}_s^{\ell,i,j}|^2d\left[M^{i,j}\right]_s\Bigg\}\\
&<\infty,
\end{aligned}$$

which implies that $\mathcal{Y}_\cdot\in\mathcal{S}_{\mathcal{F}}^2([0,T];\mathbb{R}^n)$. This completes the proof. □

4. Necessary condition and verification theorems for open-loop Nash equilibrium point

In this section, we establish one necessary condition and two verification theorems for open-loop Nash equilibrium point of Problem (CMFG) under different assumptions.

4.1. Necessary condition

In this part, we give a necessary condition employing conditional mean-field ABSDE. It should be pointed out that the introduction of ABSDE makes the maximum principle for delayed system concise. However, it cannot handle the situation where the average delay of state enters into terminal cost function since it leads to some difficulty in formulating adjoint equation of ABSDE type for variation inequality. Therefore, we require the following assumption holds.

Assumption (A3). For each $k\in\mathcal{N}, i\in\mathcal{S}$, terminal cost function g^k is independent of y and $\hat{y}$, i.e., there exists a function $\widetilde{g}^k(i,x,\hat{x}):\mathcal{S}\times(\mathbb{R}^n)^2\to\mathbb{R}$ such that $g^k(i,x,\hat{x},y,\hat{y})=\widetilde{g}^k(i,x,\hat{x})$.

Suppose that $u_\cdot=(u_\cdot^1,\dots,u_\cdot^N)\in\mathcal{U}$ is a Nash equilibrium point and, for $k\in\mathcal{N}$, $v_\cdot^k\in\mathcal{U}^k$ is an admissible control of player k. For notational convenience, we denote by $(X_\cdot,Y_\cdot,Z_\cdot)$ the optimal state corresponding to Nash equilibrium point $u_\cdot$. In addition, we use the notations $\varphi(t)$ and $\widetilde{g}^k(T)$ to denote the functions φ and $\widetilde{g}^k$ evaluated at $(t,\alpha_t,X_t,Y_t,Z_t,\hat{X}_t,\hat{Y}_t,\hat{Z}_t,u_t)$ and $(\alpha_T,X_T,\hat{X}_T)$ where $\varphi=b,\sigma,l^k$ with $k\in\mathcal{N}$. Letting $\Delta v_\cdot^k=v_\cdot^k-u_\cdot^k$ $(k\in\mathcal{N})$, it follows from the convexity of $\mathcal{U}^k$ that, for any $\varepsilon\in[0,1]$, $u_\cdot^{k,\varepsilon}=u_\cdot^k+\varepsilon\Delta v_\cdot^k=(1-\varepsilon)u_\cdot^k+\varepsilon v_\cdot^k\in\mathcal{U}^k$ and $(u_\cdot^{-k},u_\cdot^{k,\varepsilon})\in\mathcal{U}$ is an admissible control. Denote by $(X_\cdot^{(u^{-k},u^{k,\varepsilon})},Y_\cdot^{(u^{-k},u^{k,\varepsilon})},Z_\cdot^{(u^{-k},u^{k,\varepsilon})})$ the state corresponding to $(u_\cdot^{-k},u_\cdot^{k,\varepsilon})$.

Here we introduce the following first-order variational equation

$$\begin{cases}
d\mathbb{X}_t^k=\Big[b_x(t)\mathbb{X}_t^k+b_y(t)\mathbb{Y}_t^k+b_z(t)\mathbb{Z}_t^k+b_{\hat{x}}(t)\hat{\mathbb{X}}_t^k+b_{\hat{y}}(t)\hat{\mathbb{Y}}_t^k+b_{\hat{z}}(t)\hat{\mathbb{Z}}_t^k+b_{v^k}(t)\Delta v_t^k\Big]dt\\
\qquad\quad+\sum_{\ell=1}^d\Big[\sigma_x^\ell(t)\mathbb{X}_t^k+\sigma_y^\ell(t)\mathbb{Y}_t^k+\sigma_z^\ell(t)\mathbb{Z}_t^k+\sigma_{\hat{x}}^\ell(t)\hat{\mathbb{X}}_t^k+\sigma_{\hat{y}}^\ell(t)\hat{\mathbb{Y}}_t^k+\sigma_{\hat{z}}^\ell(t)\hat{\mathbb{Z}}_t^k+\sigma_{v^k}^\ell(t)\Delta v_t^k\Big]dW_t^\ell,\qquad t\in[0,T],\\
\mathbb{X}_t^k=0,\qquad t\in[-\delta,0],
\end{cases}\tag{14}$$

where

$$\mathbb{Y}_t^k=\int_{t-\delta}^t e^{-\rho(t-s)}\mathbb{X}_s^k ds,\qquad \mathbb{Z}_t^k=\mathbb{X}_{t-\delta}^k,$$

$$\hat{\mathbb{X}}_t^k=\mathbb{E}[\mathbb{X}_t^k|\mathcal{F}_t^\alpha],\qquad \hat{\mathbb{Y}}_t^k=\mathbb{E}\left[\mathbb{Y}_t^k|\mathcal{F}_t^\alpha\right],\qquad \hat{\mathbb{Z}}_t^k=\mathbb{E}[\mathbb{X}_{t-\delta}^k|\mathcal{F}_{t-\delta}^\alpha]=\hat{\mathbb{X}}_{t-\delta}^k,$$

and $\sigma^\ell(t)$ is the ℓth column of $\sigma(t)$ for $\ell\in\{1,\dots,d\}$.

Next, we present the following estimate, which implies that $\mathbb{X}_\cdot^k$ is the derivative process $\frac{d}{d\varepsilon}X_\cdot^{(u^{-k},u^{k,\varepsilon})}\Big|_{\epsilon=0}$ belonging to L^2 space.

Lemma 4.1. *Suppose Assumption (A1) holds. Then, for any $k\in\mathcal{N}$,*

$$\lim_{\varepsilon\to0^+}\mathbb{E}\sup_{0\le s\le t}\left|\frac{X_s^{(u^{-k},u^{k,\varepsilon})}-X_s}{\varepsilon}-\mathbb{X}_s^k\right|^2=0.$$

Proof. To ensure a more fluent exposition, its proof is aggregated in Appendix. □

To proceed, we formulate the Hamiltonian function $H^k : [0,T]\times S\times(\mathbb{R}^n)^6\times\mathbb{U}\times\mathbb{R}^n\times\mathbb{R}^{n\times d}\to\mathbb{R}$ for player k ($k\in\mathcal{N}$) by

$$\begin{aligned}&H^k(t,i,x,y,z,\hat{x},\hat{y},\hat{z},(v^{-k},v^k),P^k,Q^k)\\&=l^k(t,i,x,y,z,\hat{x},\hat{y},\hat{z},(v^{-k},v^k))+b(t,i,x,y,z,\hat{x},\hat{y},\hat{z},(v^{-k},v^k))^\top P^k+\operatorname{tr}\left[\sigma(t,i,x,y,z,\hat{x},\hat{y},\hat{z},(v^{-k},v^k))^\top Q^k\right].\end{aligned}\tag{15}$$

By virtue of (15), we introduce the following conditional mean-field ABSDE with regime switching for player k ($k\in\mathcal{N}$):

$$\begin{cases}dP_t^k=-\Big\{H_x^k(t)+\mathbb{E}\left[\int_t^{t+\delta}H_y^k(s)e^{\rho(t-s)}\mathbb{1}_{[0,T]}(s)ds\Big|\mathcal{F}_t\right]+\mathbb{E}[H_z^k(t+\delta)\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t]+\mathbb{E}[H_{\hat{x}}^k(t)|\mathcal{F}_t^\alpha]\\\qquad+\mathbb{E}\left[\int_t^{t+\delta}H_{\hat{y}}^k(s)e^{\rho(t-s)}\mathbb{1}_{[0,T]}(s)ds\Big|\mathcal{F}_t^\alpha\right]+\mathbb{E}[H_{\hat{z}}^k(t+\delta)\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t^\alpha]\Big\}dt+Q_t^k dW_t+R_t^k\bullet dM_t,\\P_T^k=\widetilde{g}_x^k(T)+\mathbb{E}[\widetilde{g}_{\hat{x}}^k(T)|\mathcal{F}_T^\alpha],\end{cases}\tag{16}$$

where $H^k(t)=H^k(t,\alpha_t,X_t,Y_t,Z_t,\hat{X}_t,\hat{Y}_t,\hat{Z}_t,(u_t^{-k},u_t^k),P_t^k,Q_t^k)$. It can be derived from Theorem 3.1 that, for any $k\in\mathcal{N}$, Eq. (16) has a unique solution $(P_\cdot^k,Q_\cdot^k,R_\cdot^k)\in\mathcal{S}_{\mathcal{F}}^2([0,T];\mathbb{R}^n)\times\mathcal{L}_{\mathcal{F}}^2([0,T];\mathbb{R}^{n\times d})\times\mathcal{K}_{\mathcal{F}}^2([0,T];\mathbb{R}^n)$ under Assumption (A1).

Then, we have the following Pontryagin's type maximum principle for open-loop Nash equilibrium point of Problem (CMFG).

Theorem 4.1. *Let Assumptions (A1) and (A3) hold. If $u_\cdot=(u_\cdot^1,\dots,u_\cdot^N)\in\mathcal{U}$ is a Nash equilibrium point of Problem (CMFG) and $(X_\cdot,Y_\cdot,Z_\cdot)$ is the optimal state, then it holds that,*

$$H_{v^k}^k(t,\alpha_t,X_t,Y_t,Z_t,\hat{X}_t,\hat{Y}_t,\hat{Z}_t,(u_t^{-k},u_t^k),P_t^k,Q_t^k)^\top(\mu^k-u_t^k)\geq 0,\qquad\forall k\in\mathcal{N},\mu^k\in\mathbb{U}^k,\tag{17}$$

$dt\times d\mathbb{P}-a.e.$ on $[0,T]\times\Omega$, where Hamiltonian function H^k is defined by (15) and $(P_\cdot^k,Q_\cdot^k,R_\cdot^k)$ is the solution of (16).

Proof. For a fixed $k\in\mathcal{N}$, in light of (5) and Lemma 4.1, we have

$$\begin{aligned}0\leq&\lim_{\varepsilon\to0^+}\frac{J^k[(u_\cdot^{-k},u_\cdot^{k,\varepsilon})]-J^k[u_\cdot]}{\varepsilon}\\&=\frac{d}{d\varepsilon}J^k[(u_\cdot^{-k},u_\cdot^{k,\varepsilon})]\Big|_{\varepsilon=0}\\&=\mathbb{E}\left[\int_0^T\left(l_x^k(t)^\top\mathbb{X}_t^k+l_y^k(t)^\top\int_{t-\delta}^t e^{-\rho(t-s)}\mathbb{X}_s^k ds+l_z^k(t)^\top\mathbb{X}_{t-\delta}^k+l_{\hat{x}}^k(t)^\top\mathbb{E}[\mathbb{X}_t^k|\mathcal{F}_t^\alpha]\right.\right.\\&\quad\left.\left.+l_{\hat{y}}^k(t)^\top\mathbb{E}\left[\int_{t-\delta}^t e^{-\rho(t-s)}\mathbb{X}_s^k ds\Big|\mathcal{F}_t^\alpha\right]+l_{\hat{z}}^k(t)^\top\mathbb{E}[\mathbb{X}_{t-\delta}^k|\mathcal{F}_{t-\delta}^\alpha]+l_{v^k}^k(t)^\top\Delta v_t^k\right)dt+\widetilde{g}_x^k(T)^\top\mathbb{X}_T^k+\widetilde{g}_{\hat{x}}^k(T)^\top\mathbb{E}[\mathbb{X}_T^k|\mathcal{F}_T^\alpha]\right].\end{aligned}\tag{18}$$

Let $Q_t^{k,\ell}$ denote the ℓth column of Q_t^k for $k\in\mathcal{N},\ell\in\{1,\dots,d\}$. Applying Itô's formula for semi-martingale to $(P_t^k)^\top\mathbb{X}_t^k$ and taking expectation, it yields that

$$\begin{aligned}&\mathbb{E}[(P_T^k)^\top\mathbb{X}_T^k]\\&=\mathbb{E}[\widetilde{g}_x^k(T)^\top\mathbb{X}_T^k+\widetilde{g}_{\hat{x}}^k(T)^\top\mathbb{E}[\mathbb{X}_T^k|\mathcal{F}_T^\alpha]]\\&=\mathbb{E}\left\{\int_0^T\left[-\left(H_x^k(t)+\mathbb{E}\left[\int_t^{t+\delta}H_y^k(s)e^{\rho(t-s)}\mathbb{1}_{[0,T]}(s)ds\Big|\mathcal{F}_t\right]+\mathbb{E}[H_z^k(t+\delta)\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t]+\mathbb{E}[H_{\hat{x}}^k(t)|\mathcal{F}_t^\alpha]\right.\right.\right.\\&\quad\left.+\mathbb{E}\left[\int_t^{t+\delta}H_{\hat{y}}^k(s)e^{\rho(t-s)}\mathbb{1}_{[0,T]}(s)ds\Big|\mathcal{F}_t^\alpha\right]+\mathbb{E}[H_{\hat{z}}^k(t+\delta)\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t^\alpha]\right)^\top\mathbb{X}_t^k\\&\quad+(P_t^k)^\top\left(b_x(t)\mathbb{X}_t^k+b_y(t)\int_{t-\delta}^t e^{-\rho(t-s)}\mathbb{X}_s^k ds+b_z(t)\mathbb{X}_{t-\delta}^k+b_{\hat{x}}(t)\mathbb{E}[\mathbb{X}_t^k|\mathcal{F}_t^\alpha]\right.\\&\quad\left.+b_{\hat{y}}(t)\mathbb{E}\left[\int_{t-\delta}^t e^{-\rho(t-s)}\mathbb{X}_s^k ds\Big|\mathcal{F}_t^\alpha\right]+b_{\hat{z}}(t)\mathbb{E}[\mathbb{X}_{t-\delta}^k|\mathcal{F}_{t-\delta}^\alpha]+b_{v^k}(t)\Delta v_t^k\right)\\&\quad+\sum_{\ell=1}^d(Q_t^{k,\ell})^\top\left(\sigma_x^\ell(t)\mathbb{X}_t^k+\sigma_y^\ell(t)\int_{t-\delta}^t e^{-\rho(t-s)}\mathbb{X}_s^k ds+\sigma_z^\ell(t)\mathbb{X}_{t-\delta}^k+\sigma_{\hat{x}}^\ell(t)\mathbb{E}[\mathbb{X}_t^k|\mathcal{F}_t^\alpha]\right.\\&\quad\left.\left.\left.+\sigma_{\hat{y}}^\ell(t)\mathbb{E}\left[\int_{t-\delta}^t e^{-\rho(t-s)}\mathbb{X}_s^k ds\Big|\mathcal{F}_t^\alpha\right]+\sigma_{\hat{z}}^\ell(t)\mathbb{E}[\mathbb{X}_{t-\delta}^k|\mathcal{F}_{t-\delta}^\alpha]+\sigma_{v^k}^\ell(t)\Delta v_t^k\right)\right]dt\right\}\\&=\mathbb{E}\left\{\int_0^T\left[-\left(l_x^k(t)^\top+\mathbb{E}\left[\int_t^{t+\delta}e^{\rho(t-s)}l_y^k(s)^\top\mathbb{1}_{[0,T]}(s)ds\Big|\mathcal{F}_t\right]+\mathbb{E}[l_z^k(t+\delta)^\top\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t]\right.\right.\right.\\&\quad+\mathbb{E}[l_{\hat{x}}^k(t)^\top|\mathcal{F}_t^\alpha]+\mathbb{E}\left[\int_t^{t+\delta}e^{\rho(t-s)}l_{\hat{y}}^k(s)^\top\mathbb{1}_{[0,T]}(s)ds\Big|\mathcal{F}_t^\alpha\right]+\mathbb{E}[l_{\hat{z}}^k(t+\delta)^\top\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t^\alpha]\Big)\mathbb{X}_t^k\\&\quad\left.\left.+\left((P_t^k)^\top b_{v^k}(t)+\sum_{\ell=1}^d(Q_t^{k,\ell})^\top\sigma_{v^k}^\ell(t)\right)\Delta v_t^k\right]dt\right\}.\end{aligned}\tag{19}$$

Combining (18) with (19) and employing integration by parts, we arrive at

$$
\begin{aligned}
0 \le\ & \mathbb{E}\left[\int_0^T \left(l_{\hat{x}}^k(t)^\top \mathbb{E}[\mathbb{X}_t^k|\mathcal{F}_t^\alpha] - \mathbb{E}[l_{\hat{x}}^k(t)^\top|\mathcal{F}_t^\alpha]\mathbb{X}_t^k\right) dt\right] \\
& + \mathbb{E}\left[\int_0^T \left(l_y^k(t)^\top \int_{t-\delta}^t e^{-\rho(t-s)}\mathbb{X}_s^k ds - \mathbb{E}\left[\int_t^{t+\delta} e^{\rho(t-s)} l_y^k(s)^\top \mathbb{1}_{[0,T]}(s) ds \middle| \mathcal{F}_t\right]\mathbb{X}_t^k\right) dt\right] \\
& + \mathbb{E}\left[\int_0^T \left(l_z^k(t)^\top \mathbb{X}_{t-\delta}^k - \left(\mathbb{E}[l_z^k(t+\delta)\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t]\right)^\top \mathbb{X}_t^k\right) dt\right] \\
& + \mathbb{E}\left[\int_0^T \left(l_{\hat{y}}^k(t)^\top \mathbb{E}\left[\int_{t-\delta}^t e^{-\rho(t-s)}\mathbb{X}_s^k ds \middle| \mathcal{F}_t^\alpha\right] - \mathbb{E}\left[\int_t^{t+\delta} e^{\rho(t-s)} l_{\hat{y}}^k(s)^\top \mathbb{1}_{[0,T]}(s) ds \middle| \mathcal{F}_t^\alpha\right]\mathbb{X}_t^k\right) dt\right] \\
& + \mathbb{E}\left[\int_0^T \left(l_{\hat{z}}^k(t)^\top \mathbb{E}[\mathbb{X}_{t-\delta}^k|\mathcal{F}_{t-\delta}^\alpha] - \mathbb{E}[l_{\hat{z}}^k(t+\delta)^\top \mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t^\alpha]\mathbb{X}_t^k\right) dt\right] \\
& + \mathbb{E}\left[\int_0^T \left(l_{v^k}^k(t)^\top + (P_t^k)^\top b_{v^k}(t) + \sum_{\ell=1}^d (Q_t^{k,\ell})^\top \sigma_{v^k}^\ell(t)\right) \Delta v_t^k dt\right] \\
=\ & \mathbb{E}\left[\int_0^T H_{v^k}^k(t)^\top \Delta v_t^k dt\right].
\end{aligned}
$$

Utilizing similar technique in [48], the desired result follows. □

4.2. Verification theorems

The necessary condition provides us all the candidate open-loop Nash equilibrium points. Furthermore, if we want to determine that whether the candidate strategies are optimal or not, a verification theorem is needed. In this subsection, we would establish two different verification theorems for Nash equilibrium point of Problem (CMFG).

4.2.1. A first verification theorem

To begin with, we impose the following assumption.

Assumption (A4).

(i) For any $k \in \mathcal{N}$, $i \in S$, the terminal cost function $\widetilde{g}^k(i,x,\hat{x})$ is convex on $(x,\hat{x})$;
(ii) For any $k \in \mathcal{N}$, $i \in S$, $t \in [0,T]$, the Hamiltonian function $H^k(t,i,x,y,z,\hat{x},\hat{y},\hat{z},(v^{-k},v^k),P^k,Q^k)$ is convex on $(x,y,z,\hat{x},\hat{y},\hat{z},v^k)$.

Theorem 4.2. *Suppose that Assumptions (A1), (A3) and (A4) hold. Let $u_\cdot = (u_\cdot^1,\dots,u_\cdot^N) \in \mathcal{U}$ be an admissible control, $(X_\cdot,Y_\cdot,Z_\cdot)$ and $(P_\cdot^k,Q_\cdot^k)$ $(k \in \mathcal{N})$ be the corresponding solutions of (2) and (16), respectively. If*

$$
H^k(t,\alpha_t,X_t,Y_t,Z_t,\hat{X}_t,\hat{Y}_t,\hat{Z}_t,(u_t^{-k},u_t^k),P_t^k,Q_t^k) = \inf_{\mu^k \in \mathbb{U}^k} H^k(t,\alpha_t,X_t,Y_t,Z_t,\hat{X}_t,\hat{Y}_t,\hat{Z}_t,(u_t^{-k},\mu^k),P_t^k,Q_t^k), \qquad \forall k \in \mathcal{N}, \tag{20}
$$

holds $dt \times d\mathbb{P} - a.e.$ on $[0,T] \times \Omega$, then $u_\cdot$ is a Nash equilibrium point of Problem (CMFG).

Proof. It suffices to prove that, for a given $k \in \mathcal{N}$, $J^k[u_\cdot] - J^k[(u_\cdot^{-k},v_\cdot^k)] \le 0$ holds for any $v_\cdot^k \in \mathcal{U}^k$. The cases for other players can be proved similarly.

For any $v_\cdot^k \in \mathcal{U}^k$ $(k \in \mathcal{N})$, we let $(X^{v^k},Y^{v^k},Z^{v^k})$ denote the state corresponding to $(u_\cdot^{-k},v_\cdot^k)$. Moreover, for any $k \in \mathcal{N}$, we adopt the notations $\varpi^{v^k}(t) = \varpi(t,\alpha_t,X_t^{v^k},Y_t^{v^k},Z_t^{v^k},\hat{X}_t^{v^k},\hat{Y}_t^{v^k},\hat{Z}_t^{v^k},(u_t^{-k},v_t^k))$ where $\varpi = b,\sigma,l^k$, $\widetilde{g}^{k,v^k}(T) = \widetilde{g}^k(\alpha_T,X_T^{v^k},\hat{X}_T^{v^k})$, and $H^{k,v^k}(t) = H^k(t,\alpha_t,X_t^{v^k},Y_t^{v^k},Z_t^{v^k},\hat{X}_t^{v^k},\hat{Y}_t^{v^k},\hat{Z}_t^{v^k},(u_t^{-k},v_t^k),P_t^k,Q_t^k)$.

For a fixed $k \in \mathcal{N}$, defining

$$
\mathcal{J}_1^k = \mathbb{E}\left[\int_0^T \left(l^k(t) - l^{k,v^k}(t)\right) dt\right],
$$

and

$$
\mathcal{J}_2^k = \mathbb{E}\left[\widetilde{g}^k(T) - \widetilde{g}^{k,v^k}(T)\right],
$$

then it holds that

$$
J^k[u_\cdot] - J^k[(u_\cdot^{-k},v_\cdot^k)] = \mathcal{J}_1^k + \mathcal{J}_2^k. \tag{21}
$$

Utilizing the convexity assumption of Hamiltonian function H^k ($k \in \mathcal{N}$), we derive that

$$\begin{aligned}\mathcal{J}_1^k &= \mathbb{E}\left\{\int_0^T \left[H^k(t) - H^{k,v^k}(t) - \left(b(t) - b^{v^k}(t)\right)^\top P_t^k - \mathrm{tr}\left[\left(\sigma(t) - \sigma^{v^k}(t)\right)^\top Q_t^k\right]\right] dt\right\} \\ &\le \mathbb{E}\left\{\int_0^T \left[-\left(b(t) - b^{v^k}(t)\right)^\top P_t^k - \mathrm{tr}\left[\left(\sigma(t) - \sigma^{v^k}(t)\right)^\top Q_t^k\right] + H_x^k(t)^\top \left(X_t - X_t^{v^k}\right)\right.\right. \\ &\quad + H_y^k(t)^\top \left(Y_t - Y_t^{v^k}\right) + H_z^k(t)^\top \left(Z_t - Z_t^{v^k}\right) + H_{\hat{x}}^k(t)^\top \left(\hat{X}_t - \hat{X}_t^{v^k}\right) + H_{\hat{y}}^k(t)^\top \left(\hat{Y}_t - \hat{Y}_t^{v^k}\right) \\ &\quad \left.\left. + H_{\hat{z}}^k(t)^\top \left(\hat{Z}_t - \hat{Z}_t^{v^k}\right) + H_{v^k}^k(t)^\top \left(u_t^k - v_t^k\right)\right] dt\right\}.\end{aligned} \tag{22}$$

Similarly, for $k \in \mathcal{N}$, employing the convexity of $\tilde{g}^k$ and applying Itô's formula to $(P_t^k)^\top \left(X_t - X_t^{v^k}\right)$, it yields that

$$\begin{aligned}\mathcal{J}_2^k &\le \mathbb{E}\left[\tilde{g}_x^k(T)^\top \left(X_T - X_T^{v^k}\right) + \tilde{g}_{\hat{x}}^k(T)^\top \left(\hat{X}_T - \hat{X}_T^{v^k}\right)\right] \\ &= \mathbb{E}\left\{\int_0^T \left[-\left(H_x^k(t) + \mathbb{E}\left[\int_t^{t+\delta} H_y^k(s) e^{\rho(t-s)} \mathbb{1}_{[0,T]}(s) ds \middle| \mathcal{F}_t\right] + \mathbb{E}[H_z^k(t+\delta)\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t]\right.\right.\right. \\ &\quad + \mathbb{E}[H_{\hat{x}}^k(t)|\mathcal{F}_t^\alpha] + \mathbb{E}\left[\int_t^{t+\delta} H_{\hat{y}}^k(s) e^{\rho(t-s)} \mathbb{1}_{[0,T]}(s) ds \middle| \mathcal{F}_t^\alpha\right] + \mathbb{E}[H_{\hat{z}}^k(t+\delta)\mathbb{1}_{[0,T-\delta]}(t)|\mathcal{F}_t^\alpha]\Big)^\top \left(X_t - X_t^{v^k}\right) \\ &\quad \left.\left.\left. + (P_t^k)^\top \left(b(t) - b^{v^k}(t)\right) + \mathrm{tr}\left[\left(\sigma(t) - \sigma^{v^k}(t)\right)^\top Q_t^k\right]\right] dt\right\}.\end{aligned} \tag{23}$$

Combining (21), (22), and (23), we arrive at

$$J^k[u_\cdot] - J^k[(u_\cdot^{-k}, v_\cdot^k)] \le \mathbb{E}\left[\int_0^T H_{v^k}^k(t)^\top \left(u_t^k - v_t^k\right) dt\right] \le 0, \qquad \forall v_\cdot^k \in \mathcal{U}^k, \tag{24}$$

where the last inequality holds for the reason that condition (17) is equivalent with (20) under Assumption (A4) (see Proposition 2.1 in [49]). Considering the arbitrariness of $v_\cdot^k$, identity (24) indicates that $u_\cdot$ is a Nash equilibrium point of Problem (CMFG). This completes the proof. □

4.2.2. A second verification theorem

Recall that both Theorems 4.1 and 4.2 rely heavily on Assumption (A3) that the terminal cost function is independent of average delay terms, which may bring some limit on its application scenarios. Therefore, we try to establish another verification theorem allowing that average delay terms enter into terminal cost function.

To this end, we define a new Hamiltonian function $\mathcal{H}^k : [0,T] \times \mathcal{S} \times (\mathbb{R}^n)^6 \times \mathbb{U} \times (\mathbb{R}^n)^3 \times (\mathbb{R}^{n\times d})^3 \to \mathbb{R}$ for player k ($k \in \mathcal{N}$) by

$$\begin{aligned}\mathcal{H}^k(t,i,x,y,z,\hat{x},\hat{y},\hat{z},(v^{-k},v^k),p^k,q^k) &= l^k\left(t,i,x,y,z,\hat{x},\hat{y},\hat{z},(v^{-k},v^k)\right) + b\left(t,i,x,y,z,\hat{x},\hat{y},\hat{z},(v^{-k},v^k)\right)^\top p^{k,1} \\ &\quad + \mathrm{tr}[\sigma\left(t,i,x,y,z,\hat{x},\hat{y},\hat{z},(v^{-k},v^k)\right)^\top q^{k,1}] + \left(x - \rho y - e^{-\rho\delta} z\right)^\top p^{k,2},\end{aligned} \tag{25}$$

where $p^k = (p^{k,1}, p^{k,2}, p^{k,3}) \in (\mathbb{R}^n)^3$ and $q^k = (q^{k,1}, q^{k,2}, q^{k,3}) \in (\mathbb{R}^{n\times d})^3$.

Next, we present a result about Itô's formula for average state delay, whose proof can be found in [27].

Lemma 4.2. *If $(X_\cdot^v, Y_\cdot^v, Z_\cdot^v)$ satisfies (2), then, for a given function $L(t,x,y) \in C^{1,2,1}([0,T]\times\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^m)$, we have*

$$dL(t, X_t^v, Y_t^v) = \left[\mathcal{D}L(\Xi_t^v) + L_y(t, X_t^v, Y_t^v)\left(X_t^v - \rho Y_t^v - e^{-\rho\delta} Z_t^v\right)\right] dt + L_x(t, X_t^v, Y_t^v)\sigma(\Xi_t^v) dW_t,$$

where

$$\Xi_t^v = \left(t, \alpha_t, X_t^v, Y_t^v, Z_t^v, \hat{X}_t^v, \hat{Y}_t^v, \hat{Z}_t^v, v_t\right),$$
$$\mathcal{D}L(\Xi_t^v) = \left(\mathcal{D}L^1(\Xi_t^v), \ldots, \mathcal{D}L^m(\Xi_t^v)\right)^\top$$

with operator $\mathcal{D}$ given by

$$\begin{aligned}\mathcal{D}L^\ell(t,i,x,y,z,\hat{x},\hat{y},\hat{z},v) &= L_t^\ell(t,x,y) + L_x^\ell(t,x,y)^\top b(t,i,x,y,z,\hat{x},\hat{y},\hat{z},v) \\ &\quad + \frac{1}{2} tr\left[L_{xx}^\ell(t,x,y)\,\sigma\sigma(t,i,x,y,z,\hat{x},\hat{y},\hat{z},v)^\top\right], \qquad \ell \in \{1,\ldots,m\}.\end{aligned}$$

In particular, it holds that

$$dY_t^v = \left(X_t^v - \rho Y_t^v - e^{-\rho\delta} Z_t^v\right) dt.$$

To proceed, we formulate three adjoint equations for player k ($k \in \mathcal{N}$) by

$$\begin{cases} dp_t^{k,1} = -\left[\mathcal{H}_x^k(t) + \mathbb{E}[\mathcal{H}_{\hat{x}}^k(t)|\mathcal{F}_t^\alpha]\right] dt + q_t^{k,1} dW_t + r_t^{k,1} \bullet dM_t, \\ p_T^{k,1} = g_x^k(T) + \mathbb{E}[g_{\hat{x}}^k(T)|\mathcal{F}_T^\alpha], \end{cases} \tag{26}$$

$$\begin{cases} dp_t^{k,2} = -\left[\mathcal{H}_y^k(t) + \mathbb{E}[\mathcal{H}_{\hat{y}}^k(t)|\mathcal{F}_t^\alpha]\right] dt + q_t^{k,2} dW_t + r_t^{k,2} \bullet dM_t, \\ p_T^{k,2} = g_y^k(T) + \mathbb{E}[g_{\hat{y}}^k(T)|\mathcal{F}_T^\alpha], \end{cases} \tag{27}$$

$$\begin{cases} dp_t^{k,3} = -\left[\mathcal{H}_z^k(t) + \mathbb{E}[\mathcal{H}_{\hat{z}}^k(t)|\mathcal{F}_t^\alpha]\right] dt + q_t^{k,3} dW_t + r_t^{k,3} \bullet dM_t, \\ p_T^{k,3} = 0, \end{cases} \tag{28}$$

where $g^k(T) = g^k(\alpha_T, X_T, \hat{X}_T, Y_T, \hat{Y}_T)$, $\mathcal{H}^k(t) = \mathcal{H}^k(t, \alpha_t, X_t, Y_t, Z_t, \hat{X}_t, \hat{Y}_t, \hat{Z}_t, (u_t^{-k}, u_t^k), p_t^k, q_t^k)$ with $p_t^k = (p_t^{k,1}, p_t^{k,2}, p_t^{k,3})$, $q_t^k = (q_t^{k,1}, q_t^{k,2}, q_t^{k,3})$.

Different from (16) which is a conditional mean-field ABSDE, adjoint equations (26)–(28) are three coupled conditional mean-field BSDEs without anticipated terms. Under Assumption (A1), it yields from Theorem 3.1 that (26)–(28) admit unique solution $(p_\cdot^k, q_\cdot^k, r_\cdot^k) \in (\mathcal{S}_{\mathcal{F}}^2([0,T];\mathbb{R}^n))^3 \times (\mathcal{L}_{\mathcal{F}}^2([0,T];\mathbb{R}^{n\times d}))^3 \times (\mathcal{K}_{\mathcal{F}}^2([0,T];\mathbb{R}^n))^3$ for each $k \in \mathcal{N}$.

Next, we provide a verification theorem by virtue of (25) and (26)–(28). The following assumption is essential.

Assumption (A5).

(i) For any $k \in \mathcal{N}$, $i \in \mathcal{S}$, the terminal cost function $g^k(i, x, \hat{x}, y, \hat{y})$ is convex on $(x, \hat{x}, y, \hat{y})$;
(ii) For any $t \in [0,T]$, $k \in \mathcal{N}$, $i \in \mathcal{S}$, the Hamiltonian function $\mathcal{H}^k(t, i, x, y, z, \hat{x}, \hat{y}, \hat{z}, (v^{-k}, v^k), p^k, q^k)$ is convex on $(x, y, z, \hat{x}, \hat{y}, \hat{z}, v^k)$;
(iii) For each $k \in \mathcal{N}$, $(p_\cdot^{k,3}, q_\cdot^{k,3}, r_\cdot^{k,3}) = (0, 0, 0)$.

Theorem 4.3. *Under Assumptions (A1) and (A5), let $u_\cdot = (u_\cdot^1, \dots, u_\cdot^N) \in \mathcal{U}$ be an admissible control, $(X_\cdot, Y_\cdot, Z_\cdot)$ and $(p_\cdot^k, q_\cdot^k, r_\cdot^k)$ $(k \in \mathcal{N})$ be the related solutions of (2) and (26)–(28), respectively. If*

$$\mathcal{H}^k(t, \alpha_t, X_t, Y_t, Z_t, \hat{X}_t, \hat{Y}_t, \hat{Z}_t, (u_t^{-k}, u_t^k), p_t^k, q_t^k) = \inf_{\mu^k \in \mathbb{U}^k} \mathcal{H}^k(t, \alpha_t, X_t, Y_t, Z_t, \hat{X}_t, \hat{Y}_t, \hat{Z}_t, (u_t^{-k}, \mu^k), p_t^k, q_t^k), \quad \forall k \in \mathcal{N}, \tag{29}$$

holds for $dt \times d\mathbb{P} - a.e.$ on $[0,T] \times \Omega$, then $u_\cdot$ is a Nash equilibrium point of Problem (CMFG).

Proof. Unless otherwise stated, we still use the notations in previous sections. Moreover, for any $v_\cdot^k \in \mathcal{U}^k$ $(k \in \mathcal{N})$, we adopt the notations $\zeta_t = X_t - \rho Y_t - e^{-\rho\delta} Z_t$, $\zeta_t^{v^k} = X_t^{v^k} - \rho Y_t^{v^k} - e^{-\rho\delta} Z_t^{v^k}$, $g^{k,v^k}(T) = g^k(\alpha_T, X_T^{v^k}, \hat{X}_T^{v^k}, Y_T^{v^k}, \hat{Y}_T^{v^k})$, $\mathcal{H}^{k,v^k}(t) = \mathcal{H}^k(t, \alpha_t, X_t^{v^k}, Y_t^{v^k}, Z_t^{v^k}, \hat{X}_t^{v^k}, \hat{Y}_t^{v^k}, \hat{Z}_t^{v^k}, (u_t^{-k}, v_t^k), p_t^k, q_t^k)$. For a fixed $k \in \mathcal{N}$, defining

$$\mathcal{I}_1^k = \mathbb{E}\left[\int_0^T \left(l^k(t) - l^{k,v^k}(t)\right) dt\right],$$

and

$$\mathcal{I}_2^k = \mathbb{E}\left[g^k(T) - g^{k,v^k}(T)\right],$$

then we have

$$J^k[u_\cdot] - J^k[(u_\cdot^{-k}, v_\cdot^k)] = \mathcal{I}_1^k + \mathcal{I}_2^k. \tag{30}$$

It can be derived from the convexity of Hamiltonian function $\mathcal{H}^k$ $(k \in \mathcal{N})$ that

$$\begin{aligned} \mathcal{I}_1^k &= \mathbb{E}\left\{\int_0^T \left[\mathcal{H}^k(t) - \mathcal{H}^{k,v^k}(t) - \left(b(t) - b^{v^k}(t)\right)^\top p_t^{k,1} - \mathrm{tr}\left[\left(\sigma(t) - \sigma^{v^k}(t)\right)^\top q_t^{k,1}\right] - \left(\zeta_t - \zeta_t^{v^k}\right)^\top p_t^{k,2}\right] dt\right\} \\ &\le \mathbb{E}\left\{\int_0^T \left[\mathcal{H}_x^k(t)^\top \left(X_t - X_t^{v^k}\right) + \mathcal{H}_{\hat{x}}^k(t)^\top \left(\hat{X}_t - \hat{X}_t^{v^k}\right) + \mathcal{H}_y^k(t)^\top \left(Y_t - Y_t^{v^k}\right) + \mathcal{H}_{\hat{y}}^k(t)^\top \left(\hat{Y}_t - \hat{Y}_t^{v^k}\right)\right. \right. \\ &\quad + \mathcal{H}_z^k(t)^\top \left(Z_t - Z_t^{v^k}\right) + \mathcal{H}_{\hat{z}}^k(t)^\top \left(\hat{Z}_t - \hat{Z}_t^{v^k}\right) + \mathcal{H}_{v^k}^k(t)^\top \left(u_t^k - v_t^k\right) - \left(b(t) - b^{v^k}(t)\right)^\top p_t^{k,1} \\ &\quad \left.\left. - \mathrm{tr}\left[\left(\sigma(t) - \sigma^{v^k}(t)\right)^\top q_t^{k,1}\right] - \left(\zeta_t - \zeta_t^{v^k}\right)^\top p_t^{k,2}\right] dt\right\}. \end{aligned} \tag{31}$$

Employing the convexity of terminal cost function g^k and the property of conditional expectation, we have

$$\mathcal{I}_2^k \le \mathbb{E}\left[\left(g_x^k(T) + \mathbb{E}[g_{\hat{x}}^k(T)|\mathcal{F}_T^\alpha]\right)^\top \left(X_T - X_T^{v^k}\right) + \left(g_y^k(T) + \mathbb{E}[g_{\hat{y}}^k(T)|\mathcal{F}_T^\alpha]\right)^\top \left(Y_T - Y_T^{v^k}\right)\right]. \tag{32}$$

Applying Itô's formula to $(p_t^{k,1})^\top \left(X_t - X_t^{v^k}\right) + (p_t^{k,2})^\top \left(Y_t - Y_t^{v^k}\right) + (p_t^{k,3})^\top \left(Z_t - Z_t^{v^k}\right)$ and utilizing the assumption that $(p_\cdot^{k,3}, q_\cdot^{k,3}, r_\cdot^{k,3}) = (0,0,0)$, one gets

$$\begin{aligned} \mathcal{I}_2^k \le \mathbb{E}\left\{\int_0^T \left[-\left(\mathcal{H}_x^k(t) + \mathbb{E}[\mathcal{H}_{\hat{x}}^k(t)|\mathcal{F}_t^\alpha]\right)^\top \left(X_t - X_t^{v^k}\right) + \left(b(t) - b^{v^k}(t)\right)^\top p_t^{k,1} + \mathrm{tr}\left[\left(\sigma(t) - \sigma^{v^k}(t)\right)^\top q_t^{k,1}\right]\right. \right. \\ \left.\left. - \left(\mathcal{H}_y^k(t) + \mathbb{E}[\mathcal{H}_{\hat{y}}^k(t)|\mathcal{F}_t^\alpha]\right)^\top \left(Y_t - Y_t^{v^k}\right) - \left(\mathcal{H}_z^k(t) + \mathbb{E}[\mathcal{H}_{\hat{z}}^k(t)|\mathcal{F}_t^\alpha]\right)^\top \left(Z_t - Z_t^{v^k}\right) + \left(\zeta_t - \zeta_t^{v^k}\right)^\top p_t^{k,2}\right] dt\right\}. \end{aligned} \tag{33}$$

By analogy with (24), it yields from (30)–(33) that

$$J^k[u_\cdot] - J^k[(u_\cdot^{-k}, v_\cdot^k)] \le \mathbb{E}\left[\int_0^T \mathcal{H}_{v^k}^k(t)^\top \left(u_t^k - v_t^k\right) dt\right] \le 0, \qquad \forall v_\cdot^k \in \mathcal{U}^k,$$

which implies $u_\cdot$ is a Nash equilibrium point of Problem (CMFG). The proof is complete. □

Regarding Theorems 4.2 and 4.3, we replenish the following two remarks.

Remark 4.1. By Theorem 4.2, we can find Nash equilibrium points with the aid of a single conditional mean-field ABSDE for each player. However, it cannot handle the case where terminal cost function depends on average delay variable. For Theorem 4.3, the limitation of terminal cost function is removed by introducing three coupled conditional mean-field BSDEs as adjoint equations and imposing an assumption that the adjoint equation for point delay dynamics admits zero solution only. Therefore, we can choose between Theorems 4.2 and 4.3 in view of the form of terminal cost function in a specific problem. For example, in next section, Theorem 4.2 is applied in the advertising and dividend choice game problem in Section 5.2, and Theorem 4.3 is applied in the N-agent conditional mean–variance portfolio selection problem with capital adjustment in Section 5.1.

Remark 4.2. If we do not consider the delay factors in dynamics and cost functional, i.e., neither $Y_\cdot^v$ nor $Z_\cdot^v$ appears in (2) and (4), then Theorem 4.2 is identical to Theorem 4.3. So, both Theorems 4.2 and 4.3 apply to the application of inter-bank borrowing and lending with regime switching in Section 5.3.

5. Financial applications

5.1. N-Agent conditional mean–variance portfolio selection problem with capital adjustment

We consider an N-agent conditional mean–variance portfolio selection problem with capital adjustment. A new condition different with that in existing literature is proposed to guarantee that the problem is solvable. Under this condition, we obtain the explicit efficient portfolio and optimal cost for each agent.

Suppose that every agent invests between a common risk-less asset (bond) and an individual risk asset (stock). In fact, there are many factors which prompt the agents to focus on individual stocks, such as familiarity, informational frictions, learning cost reduction and so on [50]. The price of common bond evolves as

$$\begin{cases} dS_t = S_t \lambda(t, \alpha_t) dt, \\ S_0 = s, \end{cases}$$

where $\lambda(t, i) \ge 0$ denotes the interest rate of bond in market mode i ($i \in \mathcal{S}$). Agent k specializes in stock k ($k \in \mathcal{N}$) whose price is governed by

$$\begin{cases} dS_t^k = S_t^k \left[b^k(t, \alpha_t) dt + \sigma^k(t, \alpha_t) dW_t^k\right], \\ S_0^k = s^k, \end{cases}$$

where $b^k(t, i)$ and $\sigma^k(t, i) > 0$ denote the appreciate rate and volatility rate of kth ($k \in \mathcal{N}$) stock in market mode i ($i \in \mathcal{S}$), respectively. Assume that the coefficients $\lambda(t, i)$, $b^k(t, i)$, and $\sigma^k(t, i)$ ($k \in \mathcal{N}$) are measurable and uniformly bounded in t. It is a more realistic model that the key parameters are modulated by a random process switching among a finite number of states. For instance, the market modes can be generally classified as "bullish" and "bearish", under which some market parameters may be markedly different.

We suppose that the trading of stocks takes place continuously and no transaction cost is charged. If the initial wealth of the kth agent is x^k, then the instant wealth at time t for the kth agent's fund X_t^k satisfies

$$\begin{cases} dX_t^k = \left[\lambda(t, \alpha_t) X_t^k + v_t^k \mathbb{D}^k(t, \alpha_t)\right] dt + v_t^k \sigma^k(t, \alpha_t) dW_t^k, \\ X_0^k = x^k, \end{cases}$$

where $\mathbb{D}^k(t, \alpha_t) = b^k(t, \alpha_t) - \lambda(t, \alpha_t)$ and v_t^k is the total market value that agent k holds in stock at time t. We call $v_\cdot^k$ as a portfolio of agent k.

In reality, it is common that every agent injects or withdrawals capital dynamically on the basis of his investment performance, which can be characterized by

$$\begin{aligned} \vartheta_t^{k,1} &= X_t^k - \left[(1 - \epsilon^k)\check{Y}_t^k + \epsilon^k \check{Y}_t^{(N)}\right], \\ \vartheta_t^{k,2} &= X_t^k - \left[(1 - \epsilon^k) Z_t^k + \epsilon^k Z_t^{(N)}\right], \\ \vartheta_t^{k,3} &= X_t^k - \left[(1 - \epsilon^k) X_t^k + \epsilon^k X_t^{(N)}\right], \end{aligned} \tag{34}$$

where $k \in \mathcal{N}$, $\epsilon^k \in (0, 1)$ and

$$\begin{aligned} &Y_t^k = \int_{t-\delta}^t e^{-\rho(t-s)} X_s^k ds, \qquad \kappa(\rho, \delta) = \left(\int_{t-\delta}^t e^{-\rho(t-s)} ds\right)^{-1}, \qquad \check{Y}_t^k = \kappa(\rho, \delta) Y_t^k, \\ &Z_t^k = X_{t-\delta}^k, \qquad \Lambda_t^{(N)} = \frac{1}{N} \sum_{\ell=1}^N \Lambda_t^\ell, \quad \Lambda = X, Y, \check{Y}, Z. \end{aligned} \tag{35}$$

We see that $\vartheta_t^{k,\ell}$ $(k \in \mathcal{N}, \ell \in \{1,2,3\})$ consists of similar compositions. Here we take $\vartheta_t^{k,1}$ in (34) for detailed interpretation. $\check{Y}_t^k$ reflects the individual average performance of the kth agent's fund during $[t-\delta, t]$. $\check{Y}_t^{(N)}$ reflects the collective average performance of N agents' fund during $[t-\delta, t]$. Therefore, $\vartheta_t^{k,1}$ represents the gain/loss between the wealth of kth agent (i.e., X_t^k) and a comprehensive benchmark $(1-\epsilon^k)\check{Y}_t^k + \epsilon^k \check{Y}_t^{(N)}$ incorporating the individual average performance and collective average performance over a period. $\vartheta_t^{k,1} > 0$ (resp. $\vartheta_t^{k,1} < 0$) implies that the kth agent has a good (resp. bad) investment performance. Parameter ϵ^k in the benchmark reflects the degree of concern about collective fund performance. A higher ϵ^k means that the agent would pay more attention to other agents' investment performance while measuring his own. To make the problem tractable, we postulate that the instantaneous amount of capital outflow/inflow is governed by the following linear function:

$$f^k\left(t, \alpha_t, \vartheta_t^{k,1}, \vartheta_t^{k,2}, \vartheta_t^{k,3}\right) = \boldsymbol{\Phi}^k(t,\alpha_t)\vartheta_t^{k,1} + \beta^k \vartheta_t^{k,2} + \Pi^k \vartheta_t^{k,3}, \tag{36}$$

where $\boldsymbol{\Phi}^k(t,i), \beta^k, \Pi^k \in \mathbb{R}$ for $i \in \mathcal{S}, k \in \mathcal{N}$. If the capital adjustments are considered, then the modified wealth process X_t^k (without causing confusion, we still use X_t^k to denote the wealth process involving capital adjustments) becomes

$$\begin{cases} dX_t^k = \left[\lambda(t,\alpha_t)X_t^k + v_t^k \mathbb{D}^k(t,\alpha_t) - f^k\left(t,\alpha_t,\vartheta_t^{k,1},\vartheta_t^{k,2},\vartheta_t^{k,3}\right)\right]dt + v_t^k \sigma^k(t,\alpha_t)dW_t^k \\ \quad = \Big[\mu^k(t,\alpha_t)X_t^k + \Pi^k \epsilon^k X_t^{(N)} + \check{\boldsymbol{\Phi}}^k(t,\alpha_t)(1-\epsilon^k)Y_t^k + \check{\boldsymbol{\Phi}}^k(t,\alpha_t)\epsilon^k Y_t^{(N)} \\ \qquad + \beta^k(1-\epsilon^k)Z_t^k + \beta^k \epsilon^k Z_t^{(N)} + v_t^k \mathbb{D}^k(t,\alpha_t)\Big]dt + v_t^k \sigma^k(t,\alpha_t)dW_t^k, \quad t \in [0,T], \\ X_t^k = x^k, \quad t \in [-\delta, 0], \end{cases} \tag{37}$$

where $\mu^k(t,\alpha_t) = \lambda(t,\alpha_t) - \Pi^k\epsilon^k - \beta^k - \boldsymbol{\Phi}^k(t,\alpha_t)$ and $\check{\boldsymbol{\Phi}}^k(t,\alpha_t) = \boldsymbol{\Phi}^k(t,\alpha_t)\kappa(\rho,\delta)$. A reasonable interpretation about initial wealth $X_t^k = x^k$ for $t \in [-\delta, 0]$ is that the kth agent holds a reserve x^k during $[-\delta, 0]$ and starts to invest at time 0.

Remark 5.1. For illustration purposes, we assume that the agents in financial market are either long-term investors or short-term ones. The long-term investors tend to take part of their wealth for hedonic consumption when their investment performance is good and inject capital when their investment performance is bad. This situation can be described by letting $\boldsymbol{\Phi}^k(t,i), \beta^k, \Pi^k > 0$ in (37). On the contrary, the short-term investors will make a transfusion of cash to pursue higher profit if their investment performance is good and withdrawals capital quickly once their investment performance declines. This case corresponds to $\boldsymbol{\Phi}^k(t,i), \beta^k, \Pi^k < 0$ in (37).

Given that the random switching process $\alpha_\cdot$ is independent of the Brownian motion $W_\cdot^k$ $(k \in \mathcal{N})$, the underlying market is actually incomplete. Motivated by [23,51–53], we adopt the criterion about expectation of conditional mean–variance in cost functional for player k $(k \in \mathcal{N})$, which is

$$J^k[v_\cdot] = \mathbb{E}\left\{\frac{\eta^k(\alpha_T)}{2}\mathbb{VAR}\left[X_T^k + \theta^k\left((1-\epsilon^k)\check{Y}_T^k + \epsilon^k\check{Y}_T^{(N)}\right)\middle|\mathcal{F}_T^\alpha\right] - \mathbb{E}\left[X_T^k + \theta^k\left((1-\epsilon^k)\check{Y}_T^k + \epsilon^k\check{Y}_T^{(N)}\right)\middle|\mathcal{F}_T^\alpha\right]\right\}, \tag{38}$$

where $v_\cdot = (v_\cdot^1, \dots, v_\cdot^N)$ with $v_\cdot^k \in \mathcal{L}_{\mathcal{F}}^2([0,T];\mathbb{R})$. Here $\theta^k > 0$ $(k \in \mathcal{N})$ denotes the weight between terminal wealth X_T^k and $(1-\epsilon^k)\check{Y}_T^k + \epsilon^k\check{Y}_T^{(N)}$. The incorporation of $(1-\epsilon^k)\check{Y}_T^k + \epsilon^k\check{Y}_T^{(N)}$ in cost impels the agents to consider an overall performance involving all the agents over period $[T-\delta, T]$ rather than an absolute performance of individual wealth at T, which contributes to a certain sustainability of finance market. $\eta^k(\alpha_T) > 0$ $(k \in \mathcal{N})$ represents the kth agent's risk-aversion parameter and depends reasonably on the market mood since the risk tolerance is quite different in bull market and bear market.

Hereafter, for $k \in \mathcal{N}$, $\mathbb{I}^k = (0, \dots, 1, \dots, 0)^\top \in \mathbb{R}^N$ denotes the column vector which is formed by 1 in the kth component and zero elsewhere. Set $\check{\theta}^k = \theta^k\kappa(\rho,\delta)$ and $\mathbb{I}^{k,k} = \mathbb{I}^k(\mathbb{I}^k)^\top$ and introduce

$$\Gamma = \begin{pmatrix} \frac{\epsilon^1}{N} & \frac{\epsilon^1}{N} & \cdots & \frac{\epsilon^1}{N} \\ \frac{\epsilon^2}{N} & \frac{\epsilon^2}{N} & \cdots & \frac{\epsilon^2}{N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\epsilon^N}{N} & \frac{\epsilon^N}{N} & \cdots & \frac{\epsilon^N}{N} \end{pmatrix}, \quad x = \begin{pmatrix} x^1 \\ x^2 \\ \vdots \\ x^N \end{pmatrix}, \quad \Lambda_t = \begin{pmatrix} \Lambda_t^1 \\ \Lambda_t^2 \\ \vdots \\ \Lambda_t^N \end{pmatrix}, \quad \Lambda = X, Y, Z, \hat{X}, \hat{Y}, W,$$

$$\mathbb{A}(t,\alpha_t) = \sum_{\ell=1}^N \mathbb{I}^{\ell,\ell}\mu^\ell(t,\alpha_t) + \Gamma, \qquad \mathbb{B}(t,\alpha_t) = \left(\sum_{\ell=1}^N \mathbb{I}^{\ell,\ell}\check{\boldsymbol{\Phi}}^\ell(t,\alpha_t)\right)\left(\sum_{\ell=1}^N \mathbb{I}^{\ell,\ell}(1-\epsilon^\ell) + \Gamma\right),$$

$$\mathbb{F} = \sum_{\ell=1}^N \rho\mathbb{I}^{\ell,\ell}, \qquad \mathbb{C} = \left(\sum_{\ell=1}^N \mathbb{I}^{\ell,\ell}\beta^\ell\right)\left(\sum_{\ell=1}^N \mathbb{I}^{\ell,\ell}(1-\epsilon^\ell) + \Gamma\right),$$

$$\mathbb{G} = \sum_{\ell=1}^N e^{-\rho\delta}\mathbb{I}^{\ell,\ell}, \qquad \mathbb{M}^k = \check{\theta}^k(1-\epsilon^k)\mathbb{I}^k + \frac{1}{N}\sum_{\ell=1}^N \mathbb{I}^\ell\check{\theta}^\ell\epsilon^\ell.$$

Then it yields from (37) that X_t is governed by

$$\begin{cases} dX_t = \left[\mathbb{A}(t,\alpha_t)X_t + \mathbb{B}(t,\alpha_t)Y_t + \mathbb{C}Z_t + \sum_{\ell=1}^{N}\mathbb{I}^{\ell}v_t^{\ell}\mathbb{D}^{\ell}(t,\alpha_t)\right]dt + \sum_{\ell=1}^{N}\mathbb{I}^{\ell,\ell}\sigma^{\ell}(t,\alpha_t)v_t^{\ell}\,dW_t, \quad t\in[0,T], \\ X_t = x, \quad t\in[-\delta,0]. \end{cases} \tag{39}$$

Cost functional (38) is reformulated as

$$J^k[v_\cdot] = \mathbb{E}\left\{\frac{\eta^k(\alpha_T)}{2}\mathbb{VAR}\left[\left(\mathbb{I}^k\right)^{\top}X_T + \left(\mathbb{M}^k\right)^{\top}Y_T\middle|\mathcal{F}_T^{\alpha}\right] - \mathbb{E}\left[\left(\mathbb{I}^k\right)^{\top}X_T + \left(\mathbb{M}^k\right)^{\top}Y_T\middle|\mathcal{F}_T^{\alpha}\right]\right\}, \qquad k\in\mathcal{N}, \tag{40}$$

which implies that the terminal cost function g^k is

$$\begin{aligned} &g^k(i,x,y,\widehat{x},\widehat{y}) \\ &= \frac{\eta^k(i)}{2}\left[\left(x-\widehat{x}\right)^{\top}\mathbb{I}^{k,k}\left(x-\widehat{x}\right) + 2\left(x-\widehat{x}\right)^{\top}\mathbb{I}^{k}(\mathbb{M}^k)^{\top}\left(y-\widehat{y}\right) + \left(y-\widehat{y}\right)^{\top}\mathbb{M}^k\left(\mathbb{M}^k\right)^{\top}\left(y-\widehat{y}\right)\right] - \left[\left(\mathbb{I}^k\right)^{\top}x + \left(\mathbb{M}^k\right)^{\top}y\right]. \end{aligned} \tag{41}$$

For cost (40), finding portfolios in Nash equilibrium sense subject to (39) formulates an N-agent conditional mean–variance portfolio selection problem with capital adjustment. From (41), we know that cost functional (40) is covered by (4). Therefore, the N-agent conditional mean–variance portfolio selection problem, to our best knowledge, which cannot be solved by existing results, is a special case of Problem (CMFG).

According to (25), the Hamiltonian function for agent k ($k\in\mathcal{N}$) is

$$\begin{aligned} &\mathcal{H}^k(t,i,x,y,z,\widehat{x},\widehat{y},\widehat{z},v,p^k,q^k) \\ &= \left(\mathbb{A}(t,i)x + \mathbb{B}(t,i)y + \mathbb{C}z + \sum_{\ell=1}^{N}\mathbb{I}^{\ell}v^{\ell}\mathbb{D}^{\ell}(t,i)\right)^{\top}p^{k,1} + \operatorname{tr}\left[\left(\sum_{\ell=1}^{N}\sigma^{\ell}(t,i)v^{\ell}\mathbb{I}^{\ell,\ell}\right)q^{k,1}\right] + (x-\mathbb{F}y-\mathbb{G}z)^{\top}p^{k,2}, \end{aligned} \tag{42}$$

and the adjoint equations for agent k are

$$\begin{cases} dp_t^{k,1} = -\left[\mathbb{A}(t,\alpha_t)^{\top}p_t^{k,1} + p_t^{k,2}\right]dt + q_t^{k,1}dW_t + r_t^{k,1}\bullet dM_t, \\ p_T^{k,1} = \eta^k(\alpha_T)\left[\mathbb{I}^{k,k}\left(X_T-\widehat{X}_T\right) + \mathbb{I}^k\left(\mathbb{M}^k\right)^{\top}\left(Y_T-\widehat{Y}_T\right)\right] - \mathbb{I}^k, \end{cases} \tag{43}$$

$$\begin{cases} dp_t^{k,2} = -\left[\mathbb{B}(t,\alpha_t)^{\top}p_t^{k,1} - \mathbb{F}p_t^{k,2}\right]dt + q_t^{k,2}dW_t + r_t^{k,2}\bullet dM_t, \\ p_T^{k,2} = \eta^k(\alpha_T)\left[\mathbb{M}^k\left(\mathbb{I}^k\right)^{\top}\left(X_T-\widehat{X}_T\right) + \mathbb{M}^k\left(\mathbb{M}^k\right)^{\top}\left(Y_T-\widehat{Y}_T\right)\right] - \mathbb{M}^k, \end{cases} \tag{44}$$

and

$$\begin{cases} dp_t^{k,3} = -\left[\mathbb{C}^{\top}p_t^{k,1} - \mathbb{G}p_t^{k,2}\right]dt + q_t^{k,3}dW_t + r_t^{k,3}\bullet dM_t, \\ p_T^{k,3} = 0. \end{cases} \tag{45}$$

In order to guarantee (45) admits a zero solution merely, we impose the following structural restrictions about the coefficients:

$$\begin{aligned} \Pi^k &= -\beta^k e^{\rho\delta}, \\ \check{\theta}^k &= \beta^k e^{\rho\delta}, \\ \check{\Phi}^k(t,\alpha_t) &= \beta^k e^{\rho\delta}\left[\mu^k(t,\alpha_t) + \rho + (1-\epsilon^k)\beta^k e^{\rho\delta}\right]. \end{aligned} \tag{46}$$

Detailed computations yield that

$$\det(\mathbb{C}) = \left(1 + \sum_{\ell=1}^{N}\frac{\epsilon^{\ell}}{N(1-\epsilon^{\ell})}\right)\prod_{\ell=1}^{N}\beta^{\ell}(1-\epsilon^{\ell}),$$

where $\det(\mathbb{C})$ denotes the determinant of matrix $\mathbb{C}$. Therefore, $\mathbb{C}$ is invertible if $\beta^k\neq 0$ and $\epsilon^k\neq 1$ hold simultaneously for any $k\in\mathcal{N}$. Then we introduce a new triplet $(\tilde{p}_\cdot^{k,2},\tilde{q}_\cdot^{k,2},\tilde{r}_\cdot^{k,2})$ by

$$\begin{aligned} \widetilde{p}_t^{k,2} &= \left(\mathbb{C}^{\top}\right)^{-1}\mathbb{G}p_t^{k,2}, \\ \widetilde{q}_t^{k,2} &= \left(\mathbb{C}^{\top}\right)^{-1}\mathbb{G}q_t^{k,2}, \\ \left(\widetilde{r}_t^{k,2,\ell,i,j}\right)_{i,j=1}^{S\times S} &= \left(\mathbb{C}^{\top}\right)^{-1}\mathbb{G}\left(r_t^{k,2,\ell,i,j}\right)_{i,j=1}^{S\times S}, \qquad \ell\in\mathcal{N}, t\in[0,T]. \end{aligned} \tag{47}$$

It yields from (43), (44), (46), and (47) that $(\tilde{p}_\cdot^{k,2},\tilde{q}_\cdot^{k,2},\tilde{r}_\cdot^{k,2})$ and $(p_\cdot^{k,1},q_\cdot^{k,1},r_\cdot^{k,1})$ satisfy

$$\begin{cases} d\widetilde{p}_t^{k,2} = -\left[\left(\mathbb{C}^{\top}\right)^{-1}\mathbb{G}\mathbb{B}(t,\alpha_t)p_t^{k,1} - \mathbb{F}\widetilde{p}_t^{k,2}\right]dt + \widetilde{q}_t^{k,2}dW_t + \widetilde{r}_t^{k,2}\bullet dM_t, \\ \widetilde{p}_T^{k,2} = \eta^k(\alpha_T)\left[\mathbb{I}^{k,k}\left(X_T-\widehat{X}_T\right) + \mathbb{I}^k\left(\mathbb{M}^k\right)^{\top}\left(Y_T-\widehat{Y}_T\right)\right] - \mathbb{I}^k, \end{cases} \tag{48}$$

and

$$\begin{cases} dp_t^{k,1} = -\left[\mathbb{A}(t,\alpha_t)^\top p_t^{k,1} + \mathbb{G}^{-1}\mathbb{C}^\top \tilde{p}_t^{k,2}\right] dt + q_t^{k,1} dW_t + r_t^{k,1} \bullet dM_t, \\ p_T^{k,1} = \eta^k(\alpha_T)\left[\mathbb{I}^{k,k}\left(X_T - \widehat{X}_T\right) + \mathbb{I}^k\left(\mathbb{M}^k\right)^\top\left(Y_T - \widehat{Y}_T\right)\right] - \mathbb{I}^k, \end{cases} \tag{49}$$

respectively. In fact, based on (46), it can be derived that

$$\mathbb{F} + \mathbb{G}^{-1}\mathbb{C}^\top = (\mathbb{C}^\top)^{-1}\mathbb{G}\mathbb{B}(t,\alpha_t) - \mathbb{A}(t,\alpha_t)^\top. \tag{50}$$

If we set $\mathfrak{P}_\cdot^k = \tilde{p}_\cdot^{k,2} - p_\cdot^{k,1}$, $\mathfrak{Q}_\cdot^k = \tilde{q}_\cdot^{k,2} - q_\cdot^{k,1}$, and $\mathfrak{R}_\cdot^k = \tilde{r}_\cdot^{k,2} - r_\cdot^{k,1}$, then it follows from (48)–(50) that $(\mathfrak{P}_\cdot^k, \mathfrak{Q}_\cdot^k, \mathfrak{R}_\cdot^k)$ satisfies

$$\begin{cases} d\mathfrak{P}_t^k = \left(\mathbb{F} + \mathbb{G}^{-1}\mathbb{C}^\top\right)\mathfrak{P}_t^k dt + \mathfrak{Q}_t^k dW_t + \mathfrak{R}_t^k \bullet dM_t, \\ \mathfrak{P}_T^k = 0. \end{cases} \tag{51}$$

Note that (51) is a linear regime-switching BSDE with bounded coefficient, which has a unique solution $(\mathfrak{P}_\cdot^k, \mathfrak{Q}_\cdot^k, \mathfrak{R}_\cdot^k) = (0,0,0)$. It implies that, for any $t \in [0,T]$,

$$\mathfrak{P}_t^k = \tilde{p}_t^{k,2} - p_t^{k,1} = \left(\mathbb{C}^\top\right)^{-1}\mathbb{G}p_t^{k,2} - p_t^{k,1} = 0. \tag{52}$$

Combining (52) with adjoint equation (45), we obtain that $(p_\cdot^{k,3}, q_\cdot^{k,3}, r_\cdot^{k,3}) = (0,0,0)$ is the unique solution of (45). Further, recall that conditions (17) and (20) are equivalent since $\mathcal{H}^k$ is convex on υ_k. From the first order necessary condition for minimizing Hamiltonian function $\mathcal{H}^k$, we arrive at

$$\mathbb{D}^k(t,\alpha_t)(\mathbb{I}^k)^\top p_t^{k,1} + \sigma^k(t,\alpha_t)(\mathbb{I}^k)^\top q_t^{k,1}\mathbb{I}^k = 0. \tag{53}$$

It is plain to check that $\mathcal{H}^k$ is linear and thus convex on $(x,y,z,\hat{x},\hat{y},\hat{z},\upsilon^k)$, and g^k given by (41) is convex on $(x,\hat{x},y,\hat{y})$. Moreover, it has been shown that $(p_\cdot^{k,3}, q_\cdot^{k,3}, r_\cdot^{k,3}) = (0,0,0)$ is the unique solution of (45) under conditions (46). Therefore, we deduce from Theorem 4.3 that u^k satisfying (53) is an Nash equilibrium point. Next, we try to construct a feedback expression of $u_\cdot^k$.

Regarding adjoint processes $p_\cdot^{k,1} = (p_\cdot^{k,1,\ell}, \dots, p_\cdot^{k,1,N})^\top$, $q_\cdot^{k,1} = (q_\cdot^{k,1,\ell,m})_{\ell,m=1}^{N\times N}$, and $r_\cdot^{k,1} = ((r_\cdot^{k,1,1,i,j})_{i,j=1}^{S\times S}, \dots, (r_\cdot^{k,1,N,i,j})_{i,j=1}^{S\times S})$, from (46), (49), and (52), we obtain the following equation:

$$\begin{cases} dp_t^{k,1,\ell} = -\left[\mu^\ell(t,\alpha_t) + (1-\epsilon^\ell)\beta^\ell e^{\rho\delta}\right] p_t^{k,1,\ell} dt + \sum_{m=1}^N q_t^{k,1,\ell,m} dW_t^m + \sum_{i,j\in\mathcal{S}} r_t^{k,1,\ell,i,j} dM_t^{i,j}, \\ p_T^{k,1,\ell} = \eta^k(\alpha_T)\left[X_T^k - \widehat{X}_T^k + \breve{\theta}^k(1-\epsilon^k)\left(Y_T^k - \widehat{Y}_T^k\right) + \breve{\theta}^k\epsilon^k\left(Y_T^{(N)} - \widehat{Y}_T^{(N)}\right)\right]\delta_{k\ell} - \delta_{k\ell}. \end{cases} \tag{54}$$

Inspired by the terminal value $p_T^{k,1,\ell}$, we make the ansatz

$$p_t^{k,1,\ell} = \varphi^k(t,\alpha_t)\left[X_t^k - \widehat{X}_t^k + \breve{\theta}^k(1-\epsilon^k)\left(Y_t^k - \widehat{Y}_t^k\right) + \breve{\theta}^k\epsilon^k\left(Y_t^{(N)} - \widehat{Y}_t^{(N)}\right)\right]\delta_{k\ell} - \phi^k(t,\alpha_t)\delta_{k\ell}, \tag{55}$$

where $\varphi^k(t,i)$ and $\phi^k(t,i)$ are deterministic differentiable functions satisfying the terminal boundary condition:

$$\varphi^k(T,i) = \eta^k(i), \qquad \phi^k(T,i) = 1, \qquad \forall k \in \mathcal{N}, i \in \mathcal{S}.$$

For convenience, let $\dot{\varphi}^k(t,i)$ and $\dot{\phi}^k(t,i)$ denote the derivative of $\varphi^k(t,i)$ and $\phi^k(t,i)$ w.r.t. t. Applying Itô's formula for Markov-modulated process to the right side of (55), we have

$$\begin{aligned} dp_t^{k,1,k} =& \left(\dot{\varphi}^k(t,\alpha_t) + \sum_{j\in\mathcal{S}} \mathcal{Q}^{\alpha_t,j}\varphi^k(t,j)\right)\left[X_t^k - \widehat{X}_t^k + \breve{\theta}^k(1-\epsilon^k)\left(Y_t^k - \widehat{Y}_t^k\right) + \breve{\theta}^k\epsilon^k\left(Y_t^{(N)} - \widehat{Y}_t^{(N)}\right)\right] dt \\ &+ \sum_{i,j\in\mathcal{S}}\left(\varphi^k(t,j) - \varphi^k(t,i)\right)\left[X_t^k - \widehat{X}_t^k + \breve{\theta}^k(1-\epsilon^k)\left(Y_t^k - \widehat{Y}_t^k\right) + \breve{\theta}^k\epsilon^k\left(Y_t^{(N)} - \widehat{Y}_t^{(N)}\right)\right] dM_t^{i,j} \\ &+ \varphi^k(t,\alpha_t)\left[\mu^k(t,\alpha_t)(X_t^k - \widehat{X}_t^k) + \Pi^k\epsilon^k\left(X_t^{(N)} - \widehat{X}_t^{(N)}\right) + \breve{\Phi}^k(t,\alpha_t)(1-\epsilon^k)\left(Y_t^k - \widehat{Y}_t^k\right)\right. \\ &\left.+\breve{\Phi}^k(t,\alpha_t)\epsilon^k\left(Y_t^{(N)} - \widehat{Y}_t^{(N)}\right) + \beta^k(1-\epsilon^k)\left(Z_t^k - \widehat{Z}_t^k\right) + \beta^k\epsilon^k\left(Z_t^{(N)} - \widehat{Z}_t^{(N)}\right) + \left(u_t^k - \widehat{u}_t^k\right)\mathbb{D}^k(t,\alpha_t)\right] dt \\ &+ \varphi^k(t,\alpha_t)u_t^k\sigma^k(t,\alpha_t)dW_t^k + \varphi^k(t,\alpha_t)\breve{\theta}^k(1-\epsilon^k)\left[X_t^k - \widehat{X}_t^k - \rho\left(Y_t^k - \widehat{Y}_t^k\right) - e^{-\rho\delta}\left(Z_t^k - \widehat{Z}_t^k\right)\right] dt \\ &+ \varphi^k(t,\alpha_t)\breve{\theta}^k\epsilon^k\left[X_t^{(N)} - \widehat{X}_t^{(N)} - \rho\left(Y_t^{(N)} - \widehat{Y}_t^{(N)}\right) - e^{-\rho\delta}\left(Z_t^{(N)} - \widehat{Z}_t^{(N)}\right)\right] dt \\ &- \left(\dot{\phi}^k(t,\alpha_t) + \sum_{j\in\mathcal{S}}\mathcal{Q}^{\alpha_t,j}\phi^k(t,j)\right) dt - \sum_{i,j\in\mathcal{S}}\left(\phi^k(t,j) - \phi^k(t,i)\right) dM_t^{i,j}, \end{aligned} \tag{56}$$

where $\widehat{u}_t^k = \mathbb{E}\left[u_t^k | \mathcal{F}_t^\alpha\right]$.

Comparing (54) with (56), one gets

$$
\begin{aligned}
&-\left[\mu^k(t,\alpha_t)+(1-\epsilon^k)\beta^k e^{\rho\delta}\right]\left\{\varphi^k(t,\alpha_t)\left[X_t^k-\hat{X}_t^k+\check{\theta}^k(1-\epsilon^k)\left(Y_t^k-\hat{Y}_t^k\right)+\check{\theta}^k\epsilon^k\left(Y_t^{(N)}-\hat{Y}_t^{(N)}\right)\right]-\phi^k(t,\alpha_t)\right\}\\
&=\left(\dot{\varphi}^k(t,\alpha_t)+\sum_{j\in\mathcal{S}}\mathcal{Q}^{\alpha_t,j}\varphi^k(t,j)\right)\left[X_t^k-\hat{X}_t^k+\check{\theta}^k(1-\epsilon^k)\left(Y_t^k-\hat{Y}_t^k\right)+\check{\theta}^k\epsilon^k\left(Y_t^{(N)}-\hat{Y}_t^{(N)}\right)\right]\\
&+\varphi^k(t,\alpha_t)\left[\mu^k(t,\alpha_t)(X_t^k-\hat{X}_t^k)+\check{\boldsymbol{\Phi}}^k(t,\alpha_t)(1-\epsilon^k)\left(Y_t^k-\hat{Y}_t^k\right)+\check{\boldsymbol{\Phi}}^k(t,\alpha_t)\epsilon^k\left(Y_t^{(N)}-\hat{Y}_t^{(N)}\right)\right.\\
&\left.+\left(u_t^k-\hat{u}_t^k\right)\mathbb{D}^k(t,\alpha_t)\right]+\varphi^k(t,\alpha_t)\check{\theta}^k(1-\epsilon^k)\left[X_t^k-\hat{X}_t^k-\rho\left(Y_t^k-\hat{Y}_t^k\right)\right]\\
&-\varphi^k(t,\alpha_t)\check{\theta}^k\epsilon^k\rho\left(Y_t^{(N)}-\hat{Y}_t^{(N)}\right)-\dot{\phi}^k(t,\alpha_t)-\sum_{j\in\mathcal{S}}\mathcal{Q}^{\alpha_t,j}\phi^k(t,j),
\end{aligned}
\tag{57}
$$

$$
q_t^{k,1,k,m}=\begin{cases}\varphi^k(t,\alpha_t)u_t^k\sigma^k(t,\alpha_t), & m=k,\\ 0, & m\neq k,\end{cases}
\tag{58}
$$

and

$$
r_t^{k,1,k,i,j}=\left\{\left(\varphi^k(t,j)-\varphi^k(t,i)\right)\left[X_t^k-\hat{X}_t^k+\check{\theta}^k(1-\epsilon^k)\left(Y_t^k-\hat{Y}_t^k\right)+\check{\theta}^k\epsilon^k\left(Y_t^{(N)}-\hat{Y}_t^{(N)}\right)\right]-\left(\phi^k(t,j)-\phi^k(t,i)\right)\right\}.
$$

Set $\Theta^k(t,\alpha_t)=\left(\sigma^k(t,\alpha_t)\right)^2$. Substituting (55) and (58) into (53), we derive that the efficient portfolio u_t^k is

$$
u_t^k=-\frac{\mathbb{D}^k(t,\alpha_t)}{\Theta^k(t,\alpha_t)}\left\{X_t^k-\hat{X}_t^k+\check{\theta}^k(1-\epsilon^k)\left(Y_t^k-\hat{Y}_t^k\right)+\check{\theta}^k\epsilon^k\left(Y_t^{(N)}-\hat{Y}_t^{(N)}\right)-\frac{\phi^k(t,\alpha_t)}{\varphi^k(t,\alpha_t)}\right\}.
\tag{59}
$$

Taking $\mathbb{E}[\cdot|\mathcal{F}_t^\alpha]$ on both sides of (59) gives

$$
\hat{u}_t^k=\frac{\mathbb{D}^k(t,\alpha_t)\phi^k(t,\alpha_t)}{\Theta^k(t,\alpha_t)\varphi^k(t,\alpha_t)}.
\tag{60}
$$

Let $\varrho^k(t,\alpha_t)=\left(\mathbb{D}^k(t,\alpha_t)\right)^2/\Theta^k(t,\alpha_t)$. Inserting (59) and (60) into (57) and equalizing the coefficients of $X_t^k-\hat{X}_t^k$, $Y_t^k-\hat{Y}_t^k$, $Y_t^{(N)}-\hat{Y}_t^{(N)}$ as well as the non-homogeneous terms on both sides of (57), it follows that $\varphi^k(t,i)$ and $\phi^k(t,i)$ satisfy

$$
\begin{cases}\dot{\varphi}^k(t,i)=-\left\{2\left[\mu^k(t,i)+(1-\epsilon^k)\beta^k e^{\rho\delta}\right]-\varrho^k(t,i)\right\}\varphi^k(t,i)-\sum_{j\in\mathcal{S}}\mathcal{Q}^{i,j}\varphi^k(t,j),\\ \varphi^k(T,i)=\eta^k(i),\end{cases}
\tag{61}
$$

and

$$
\begin{cases}\dot{\phi}^k(t,i)=-\left[\mu^k(t,i)+(1-\epsilon^k)\beta^k e^{\rho\delta}\right]\phi^k(t,i)-\sum_{j\in\mathcal{S}}\mathcal{Q}^{i,j}\phi^k(t,j),\\ \phi^k(T,i)=1.\end{cases}
\tag{62}
$$

Both (61) and (62) admit a unique solution since they are first order linear ODEs. By employing the Feynman–Kac representation of the solutions to differential equations (see [54, Section 7, Chapter 2]), we have

$$
\varphi^k(t,i)=\mathbb{E}\left\{\eta^k(\alpha_T)\exp\left\{\int_t^T\left\{2\left[\mu^k(t,\alpha_s)+(1-\epsilon^k)\beta^k e^{\rho\delta}\right]-\varrho^k(t,\alpha_s)\right\}ds\right\}\middle|\alpha_t=i\right\},
\tag{63}
$$

and

$$
\phi^k(t,i)=\mathbb{E}\left\{\exp\left\{\int_t^T\left[\mu^k(t,\alpha_s)+(1-\epsilon^k)\beta^k e^{\rho\delta}\right]ds\right\}\middle|\alpha_t=i\right\}.
\tag{64}
$$

We are now in the position to compute the optimal cost for each agent. To ease the complexity of notations, we set $\Psi_t^k=X_t^k+\check{\theta}^k\left[(1-\epsilon^k)Y_t^k+\epsilon^k Y_t^{(N)}\right]$ and $\hat{\Psi}_t^k=\mathbb{E}\left[\Psi_t^k|\mathcal{F}_t^\alpha\right]$. Applying Itô's formula to $(\Psi_t^k-\hat{\Psi}_t^k)^2$ yields

$$
\begin{aligned}
d(\Psi_t^k-\hat{\Psi}_t^k)^2=&\left\{\left[2\left(\mu^k(t,\alpha_t)+(1-\epsilon^k)\beta^k e^{\rho\delta}\right)-\varrho^k(t,\alpha_t)\right](\Psi_t^k-\hat{\Psi}_t^k)^2-2\varrho^k(t,\alpha_t)\frac{\phi^k(t,\alpha_t)}{\varphi^k(t,\alpha_t)}(\Psi_t^k-\hat{\Psi}_t^k)\right.\\
&\left.+\varrho^k(t,\alpha_t)\left(\frac{\phi^k(t,\alpha_t)}{\varphi^k(t,\alpha_t)}\right)^2\right\}dt+2(\Psi_t^k-\hat{\Psi}_t^k)u_t^k\sigma^k(t,\alpha_t)dW_t^k.
\end{aligned}
\tag{65}
$$

Taking $\mathbb{E}\left[\cdot|\mathcal{F}_t^\alpha\right]$ on both sides of (65), we obtain

$$
d\mathbb{VAR}\left[\Psi_t^k|\mathcal{F}_t^\alpha\right]=\left\{\left[2\left(\mu^k(t,\alpha_t)+(1-\epsilon^k)\beta^k e^{\rho\delta}\right)-\varrho^k(t,\alpha_t)\right]\mathbb{VAR}\left[\Psi_t^k|\mathcal{F}_t^\alpha\right]+\varrho^k(t,\alpha_t)\left(\frac{\phi^k(t,\alpha_t)}{\varphi^k(t,\alpha_t)}\right)^2\right\}dt.
$$

Then applying Itô's formula to $\varphi^k(t,\alpha_t)\mathbb{VAR}\left[\Psi_t^k|\mathcal{F}_t^\alpha\right]$, it follows that

$$
d\left\{\varphi^k(t,\alpha_t)\mathbb{VAR}\left[\Psi_t^k|\mathcal{F}_t^\alpha\right]\right\}=\varrho^k(t,\alpha_t)\frac{(\phi^k(t,\alpha_t))^2}{\varphi^k(t,\alpha_t)}dt+\{\dots\}\bullet dM_t,
$$

which indicates

$$\mathbb{E}\left[\frac{\eta^k(\alpha_T)}{2}\mathbb{VAR}\left[\Psi_T^k|\mathcal{F}_T^\alpha\right]\right]=\frac{1}{2}\mathbb{E}\left[\int_0^T \varrho^k(t,\alpha_t)\frac{(\phi^k(t,\alpha_t))^2}{\varphi^k(t,\alpha_t)}dt\right]. \tag{66}$$

Similarly, we have

$$-\mathbb{E}\left[\Psi_T^k\right]=-\left[x^k(1+\theta^k)+\theta^k\epsilon^k\frac{1}{N}\sum_{\ell=1}^N(x^\ell-x^k)\right]\phi^k(0,i_0)-\mathbb{E}\left[\int_0^T \varrho^k(t,\alpha_t)\frac{(\phi^k(t,\alpha_t))^2}{\varphi^k(t,\alpha_t)}dt\right]. \tag{67}$$

Due to the fact that we assume the initial regime of Markov chain α_t is fixed at i_0, then it holds that

$$\mathbb{P}_i(t):=\mathbb{P}\{\alpha(t)=i\}=\mathbb{P}_{i_0,i}(t)=(\widetilde{\mathbb{I}}^{i_0})^\top e^{Qt}\widetilde{\mathbb{I}}^i, \tag{68}$$

where $\widetilde{\mathbb{I}}^i=(0,\dots,1,\dots,0)^\top\in\mathbb{R}^S$ is the column vector formed by 1 in the ith component and zero elsewhere.

Combining (66)–(68) gives

$$J^k[u_\cdot]=-\left[x^k(1+\theta^k)+\theta^k\epsilon^k\frac{1}{N}\sum_{\ell=1}^N(x^\ell-x^k)\right]\phi^k(0,i_0)-\frac{1}{2}\sum_{i\in S}\int_0^T \varrho^k(t,i)\frac{(\phi^k(t,i))^2}{\varphi^k(t,i)}(\widetilde{\mathbb{I}}^{i_0})^\top e^{Qt}\widetilde{\mathbb{I}}^i dt. \tag{69}$$

To summarize the above analysis, we obtain the result as follows.

Proposition 5.1. *Assume that identity (46) holds, $\varphi^k(t,i)$ and $\phi^k(t,i)$ satisfy (63) and (64), respectively. Then the strategy $u_\cdot=(u_\cdot^1,\dots,u_\cdot^N)$, where $u_\cdot^k$ $(k\in\mathcal{N})$ is given by (59), is an effective portfolio in Nash equilibrium sense of the N-agent conditional mean–variance portfolio selection problem with capital adjustment. Moreover, the optimal cost of agent k is (69).*

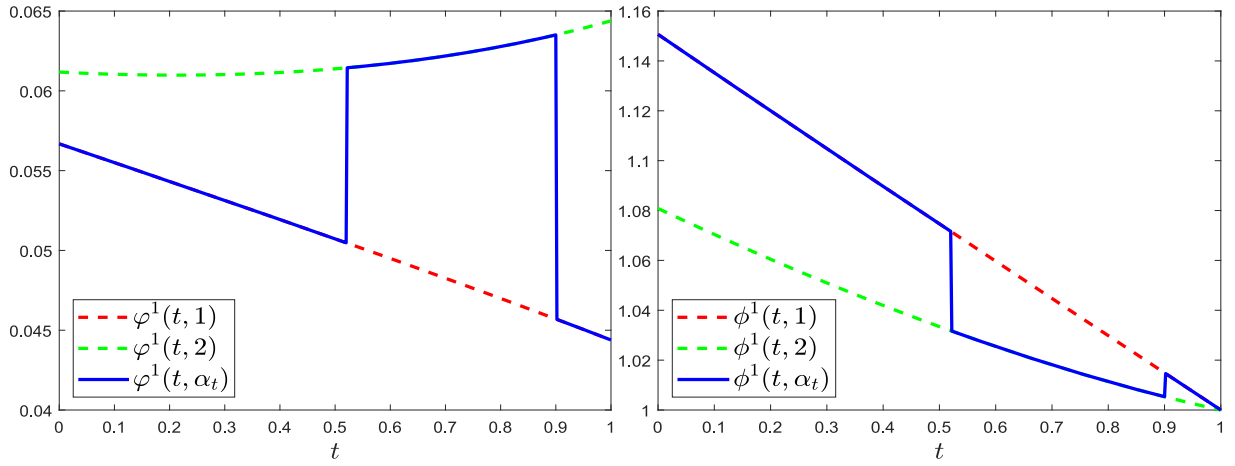
Remark 5.2. In general, the solutions for optimization problems of delayed systems such as (2) depend on initial datum $f_\cdot$, which belongs to an infinite-dimensional space $C([-\delta,0];\mathbb{R}^n)$. It is usually hard to obtain an explicit solution in infinite-dimensional space. However, if the controlled systems depend on delay term via $Y_\cdot^v$ and $Z_\cdot^v$ defined by (3) only, it is likely that the optimization problems are reduced to finite dimensional and become tractable under certain conditions. Essentially, it requires that the value function is dependent on the initial datum $f_\cdot$ exclusively through the following two functionals:

$$x=x(f_\cdot)=f_0,\qquad y=y(f_\cdot)=\int_{-\delta}^0 e^{\rho s}f_s\,ds.$$

From the view point of maximum principle, it is supposed that the adjoint equation for point delay process (28) admits zero solution only. Focusing on optimal control problem with delay, resurgent efforts have been devoted to make it solvable. Interested readers can refer to the discussions in [21,28,29,40] based on maximum principle and [38,39] on dynamic programming principle, respectively. In fact, all the works mentioned above proposed identical conditions (for example, formulas (3.32) in [29] and (35) in [38]), which are exactly the last two identities in (46). Actually, the N-agent conditional mean–variance portfolio selection problem with capital adjustment is an N-agent stochastic differential game for delayed system. The conditions to make the control problem tractable is not enough in our paper. Compared with those works, we initially introduce a new coefficient Π^k and impose the condition that $\Pi^k=-\beta^k e^{\rho\delta}$ (the first identity in (46)), which guarantees that the N-agent conditional mean–variance portfolio selection problem becomes finite dimensional and solvable. We emphasize that this condition is instructive to investigate the solvability for general stochastic differential games of delayed systems.

Remark 5.3. In Remark 5.1, we provide an economic interpretation of the cash flow adjustment parameter: positive values of $\Phi^k(t,i),\beta^k,\Pi^k$ correspond to long-term investment behavior, whereas negative values correspond to short-term behavior. The identity $\Pi^k=-\beta^k e^{\rho\delta}$ in condition (46) implies that parameters Π^k and β^k always have opposite signs. This reflects the fact that an investor optimizing a conditional mean–variance criterion necessarily incorporates both long-term and short-term considerations. Such a feature is intrinsic to the mean–variance framework and should not be viewed as contradictory. From a practical perspective, a long-term investor is conservative, typically buying stocks and holding them for a long period. He pays more attention to risk (i.e., variance) and expects that his fund grows steadily [55]. By contrast, a short-term investor is aggressive, frequently buying and selling. He seeks profit utilizing the temporary and volatile price fluctuations. Therefore, a short-term investor focuses primarily on return (i.e., expectation) instead of risk. Under the conditional mean–variance criterion, however, the objective of the investor is to maximize the return and minimize the risk at the same time [56]. This dual objective naturally requires the investor to select portfolios blending both the long-term and short-term investment style, which is reflected in the mandatory sign opposition for Π^k and β^k.

To intuitively illustrate the effectiveness of our results, we present some numerical examples for the N-agent conditional mean–variance portfolio selection problem with capital adjustment. Naturally, suppose that the market modes are divided into bullish and bearish, which are represented by regimes "1" and "2" of Markov chain α_t. Set $T=1$, $i_0=1$, $Q^{1,2}=0.25$, $Q^{2,1}=0.75$, $\mathcal{N}=\{1,\dots,100\}$, $x^k=100$, $\lambda(t,1)=0.15$, $\lambda(t,2)=0.05$, $b^k(t,1)=0.3$, $b^k(t,2)=0.1$, $\sigma^k(t,1)=0.45$, $\sigma^k(t,2)=0.4$, $\epsilon^k=0.8$, $\rho=0.5$, $\delta=0.01$, and $\theta^k=0.5$ for $k\in\mathcal{N}$. $\eta^k(1)$ is taken from $[0.04,0.06]$ randomly for different k. Usually, the agents would have a higher aversion about risk in bear market than bull market. Therefore, we set $\eta^k(2)=\eta^k(1)+0.02$. Then, according to (46), we derive that



(a) Trajectories of $\varphi^1(t,1)$, $\varphi^1(t,2)$, and $\varphi^1(t,\alpha_t)$.

(b) Trajectories of $\phi^1(t,1)$, $\phi^1(t,2)$, and $\phi^1(t,\alpha_t)$.

Fig. 1. Simulations for solutions to ODEs (61) and (62).

the remaining parameters are $\Phi^k(t,1)=0.0032$, $\Phi^k(t,2)=0.0027$, $\beta^k=0.4975$, and $\Pi^k=-0.5$. Using Euler–Maruyama method with step size $\Delta=2\times10^{-3}$, we obtain the numerical simulations in Figs. 1, 2, and 3. The detailed analyzes are given as follows.

Fig. 1 displays the trajectories of solutions to ODEs (61) and (62). Taking Fig. 1(a) for example, the red dotted line is the curve of $\varphi^1(t,1)$, the green dotted line is the curve of $\varphi^1(t,2)$, and the blue solid line is the curve of $\varphi^1(t,\alpha_t)$ switching between the lines $\varphi^1(t,1)$ and $\varphi^1(t,2)$.

Fig. 2 shows the trajectories of Markov chain α_t, optimal wealth processes X_t^k, effective portfolios u_t^k and the amount of cash flow adjustment $f^k(t,\alpha_t,\vartheta_t^{k,1},\vartheta_t^{k,2},\vartheta_t^{k,3})$ for $k\in\mathcal{N}$. From Fig. 2(a), we see that at the beginning, α_t is in regime 1, then it switches to regime 2, and finally goes back to regime 1, which means that the market is in bull mode and then comes into bear mode and finally returns to bull mode. It is observed from Fig. 2(b) that the agents' wealth grows rapidly in bull market and stays stagnant in bear market. From Fig. 2(c), we see that the agents hold certain amount of money in stock at the beginning when the market is in bull mode. Then, in order to avoid risk, the agents tend to sell out the risky asset as the market enters into bear mode. Finally, the agents increase the investment in risky asset to raise pay-back when the market reenters to bull mode, which is in agreement with the economic laws. Fig. 2(d) depicts the amount of cash flow adjustment for each agent, it demonstrates that the injection and withdrawal of capital in bullish time is larger than those in bearish time.

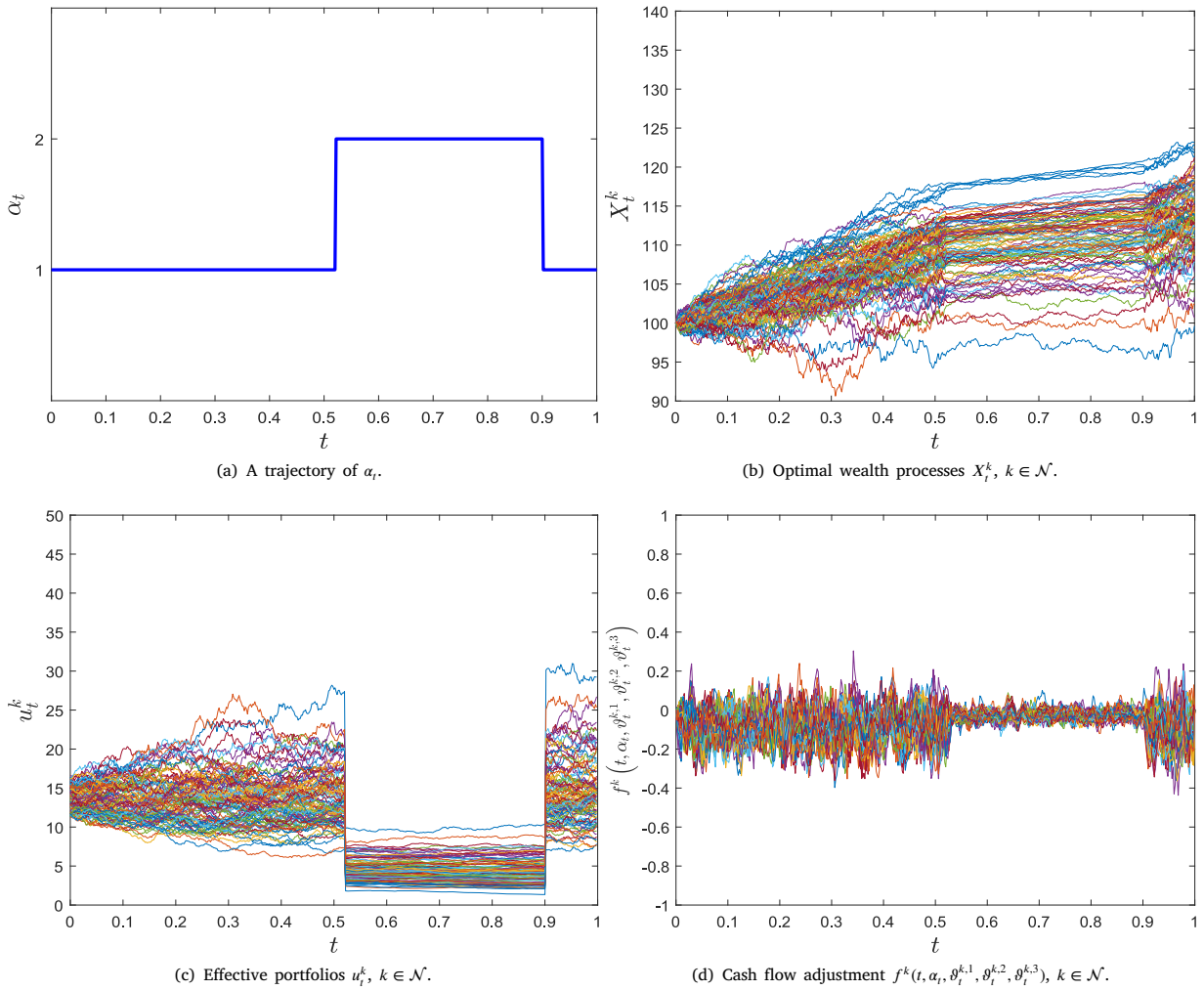
Fig. 3 illustrates the effect of delay parameter δ and initial regime i_0 on the optimal cost $J^1[u_\cdot]$. It is observed that the cost $J^1[u_\cdot]$ with $i_0=2$ is higher than that with $i_0=1$, which implies that an agent will undertake higher risk and obtain less return if he enters the market at bearish time than bullish time. Moreover, Fig. 3 reveals that the optimal cost increases as delay parameter δ. This means that it will lower gratification if an agent concerns his invest performance over a long period when adjusting his capital and proposing his utility function.

5.2. Advertising and dividend choice game problem

Optimal advertising is a classical problem in the field of mathematical finance, which originates from the seminal works of Nerlove and Arrow [57] and Vidale and Wolfe [58] with deterministic dynamics. A considerable amount of literature has extended the model to different frameworks. Specifically, stochastic dynamics for the stock of advertising goodwill with time lag structure were considered in [41–43,59] and so on. The main idea of those works is to firstly transform the original problem into the one driven by abstract evolution equation and then solve it via dynamic programming principle. However, only some existence results about optimal control were given in [41–43]. If point delay was considered in dynamics merely, [59] characterized the optimal strategy by value function. Inspired by those works, we study an advertising and dividend choice game problem, where average delay term in state is considered. Importantly, the explicit optimal strategies are derived.

Consider a firm preparing the launch of a new product at future time T. Usually, there exist two roles in a firm: investor (player 1) and manager (player 2). For the manager, he makes decision on the intensity of advertising spending $v_\cdot$ ($0\le v_\cdot\le\mathbf{M}$) to induce the stock of goodwill over the planning horizon $[0,T]$. Moreover, the firm is supposed to pay dividend $c_\cdot$ ($c_\cdot\ge0$) to the investor. Here we adopt the Nerlove–Arrow framework where the stock of goodwill X_t evolves as

$$\begin{cases} dX_t=\left[\mathbf{A}(t,\alpha_t)X_t+\displaystyle\int_{-\delta}^{0}\mathbf{B}_s X_{t+s}ds+\mathbf{D}(t,\alpha_t)v_t+\mathbf{F}(t,\alpha_t)c_t\right]dt+\sigma(t,\alpha_t,\displaystyle\int_{-\delta}^{0}\mathbf{B}_s X_{t+s}ds)dW_t, & t\in[0,T],\\ X_t=x, & t\in[-\delta,0]. \end{cases} \tag{70}$$



(a) A trajectory of α_t.

(b) Optimal wealth processes X_t^k, $k \in \mathcal{N}$.

(c) Effective portfolios u_t^k, $k \in \mathcal{N}$.

(d) Cash flow adjustment $f^k(t, \alpha_t, \vartheta_t^{k,1}, \vartheta_t^{k,2}, \vartheta_t^{k,3})$, $k \in \mathcal{N}$.

Fig. 2. Simulations for Markov chain α_t, optimal wealth processes X_t^k, effective portfolios u_t^k and cash flow adjustment $f^k(t, \alpha_t, \vartheta_t^{k,1}, \vartheta_t^{k,2}, \vartheta_t^{k,3})$ for $k \in \mathcal{N}$.

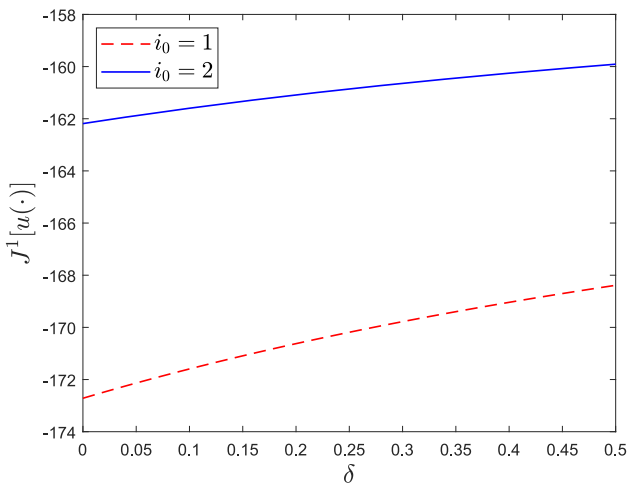


Fig. 3. The effect of delay parameter δ and initial regime i_0 on optimal cost $J^1[u_\cdot]$.

In (70), for each $i \in \mathcal{S}$, $\mathbf{A}(t,i) < 0$ is proportional rate of depreciation when advertising stops, $\mathbf{D}(t,i) > 0$ is the advertising effectiveness factor, and $\mathbf{F}(t,i) < 0$ is the dividend effectiveness factor. It should be noted that these coefficients are Markov-modulated processes that may vary quite a lot according to different states of economy. $x > 0$ is the history of the goodwill level. $\delta > 0$ represents the forgetting horizon. $\mathbf{B}_s$ is the distribution of forgetting time which is defined as $\mathbf{B}_s = e^{\rho s}\kappa(\rho,\delta)$ with constant $\kappa(\rho,\delta) = (\int_{-\delta}^{0} e^{\rho s} ds)^{-1}$. Using variable substitution, we rewrite the dynamics of stock of goodwill (70) as

$$\begin{cases} dX_t = \left[\mathbf{A}(t,\alpha_t)X_t + \kappa(\rho,\delta)Y_t + \mathbf{D}(t,\alpha_t)v_t + \mathbf{F}(t,\alpha_t)c_t\right]dt + \sigma(t,\alpha_t,\kappa(\rho,\delta)Y_t)dW_t, \quad t \in [0,T], \\ X_t = x, \quad t \in [-\delta, 0], \end{cases} \tag{71}$$

where $Y_t = \int_{t-\delta}^{t} e^{-\rho(t-s)} X_s ds$.

The objective of this investor is to maximize the expect utility of dividend c_t over the planning horizon $[0,T]$ and maximize the goodwill at T (i.e., X_T) since his payment is proportional to X_T. Here we use the concave logarithm utility function $\mathbf{U}(t,i,c) = \mathbf{G}(t,i)\ln(1+c)$ where $\mathbf{G}(t,i) > 0$ is bounded (more information about different utility functions can be found in [60]). Therefore, the cost functional for investor is formulated as

$$J^1[v_\cdot, c_\cdot] = \mathbb{E}\left[-\int_0^T \mathbf{G}(t,\alpha_t)\ln(1+c_t)dt - X_T\right].$$

For the manager, he aims to minimize the accumulative cost of advertising expenditure which is described by convex cost function $\mathbf{C}(t,i,v) = \frac{1}{2}\mathbf{S}(t,i)v^2$ where $\mathbf{S}(t,i) > 0$ is bounded. At the same time, he desires to maximize the stock of advertising goodwill at terminal time T. In view of this, we define the cost functional for manager as follows:

$$J^2[v_\cdot, c_\cdot] = \mathbb{E}\left[\frac{1}{2}\int_0^T \mathbf{S}(t,\alpha_t)(v_t)^2 dt - X_T\right].$$

To proceed, we derive that the Hamiltonian functions for the advertising and dividend choice game problem are

$$H^1(t,i,x,y,P^1,Q^1,v,c) = P^1\left(\mathbf{A}(t,i)x + \kappa(\rho,\delta)y + \mathbf{D}(t,i)v + \mathbf{F}(t,i)c\right) + Q^1\sigma(t,i,\kappa(\rho,\delta)y) - \mathbf{U}(t,i,c), \tag{72}$$

and

$$H^2(t,i,x,y,P^2,Q^2,v,c) = P^2\left(\mathbf{A}(t,i)x + \kappa(\rho,\delta)y + \mathbf{D}(t,i)v + \mathbf{F}(t,i)c\right) + Q^2\sigma(t,i,\kappa(\rho,\delta)y) + \mathbf{C}(t,i,v). \tag{73}$$

The adjoint processes $(P_\cdot^1, Q_\cdot^1, R_\cdot^1)$ and $(P_\cdot^2, Q_\cdot^2, R_\cdot^2)$ for the two players satisfy the identical equation, which is

$$\begin{cases} dP_t = -\left\{\mathbf{A}(t,\alpha_t)P_t + \kappa(\rho,\delta)\mathbb{E}\left[\int_t^{t+\delta}\left[P_s + \sigma_y(s,\alpha_s,\kappa(\rho,\delta)Y_s)Q_s\right]e^{\rho(t-s)}\mathbb{1}_{[0,T]}(s)ds\,\middle|\,\mathcal{F}_t\right]\right\}dt + Q_t dW_t + R_t \bullet dM_t, \\ P_T = -1. \end{cases} \tag{74}$$

Set $\mathbf{U}_c(t,i,c) = \frac{\partial \mathbf{U}(t,i,c)}{\partial c}$. Considering that $c_\cdot \geq 0$, $\mathbf{U}_c(t,i,0) = \mathbf{G}(t,i)$, and $\mathbf{U}_c(t,i,c)$ is strictly decreasing w.r.t. c, assume that

$$\frac{\mathbf{G}(t,\alpha_t)}{\mathbf{F}(t,\alpha_t)} \leq P_t < 0, \tag{75}$$

then applying the first order condition for minimizing Hamiltonian function H^1 gives

$$c_t^* = \frac{\mathbf{G}(t,\alpha_t)}{\mathbf{F}(t,\alpha_t)P_t} - 1. \tag{76}$$

Due to the fact that H^1 is convex on c, we obtain that condition (20) is satisfied for $c_\cdot^*$ given by (76). Moreover, it is obvious that H^1 is convex on (x,y,c) and $g^1(x) = -x$ is convex on x. Then, in view of Theorem 4.2, $c_\cdot^*$ is an optimal dividend.

Similarly, utilizing Theorem 4.1 and considering the requirement $0 \leq v_\cdot \leq \mathbf{M}$, we construct the following candidate optimal intensity of advertising spending:

$$u_t = \begin{cases} 0, & 0 \leq P_t, \\ -\frac{\mathbf{D}(t,\alpha_t)}{\mathbf{S}(t,\alpha_t)}P_t, & -\frac{\mathbf{S}(t,\alpha_t)}{\mathbf{D}(t,\alpha_t)}\mathbf{M} \leq P_t < 0, \\ \mathbf{M}, & P_t < -\frac{\mathbf{S}(t,\alpha_t)}{\mathbf{D}(t,\alpha_t)}\mathbf{M}. \end{cases}$$

Further, we check that all the conditions in Theorem 4.2 are fulfilled. Then $u_\cdot$ is an optimal strategy.

In addition, note that the terminal value P_T in (74) is deterministic. If $\mathbf{A}(t,\alpha_t) = \mathbf{A}_t$ is independent of switching process α_t, then we choose $Q_\cdot = R_\cdot = 0$. In this case, (74) becomes

$$\begin{cases} dP_t = -\left[\mathbf{A}_t P_t + \kappa(\rho,\delta)\int_t^{t+\delta} P_s e^{\rho(t-s)}\mathbb{1}_{[0,T]}(s)ds\right]dt, \\ P_T = -1, \end{cases} \tag{77}$$

which is a deterministic functional differential equation.

Summarizing above discussions, we have the result as follows.

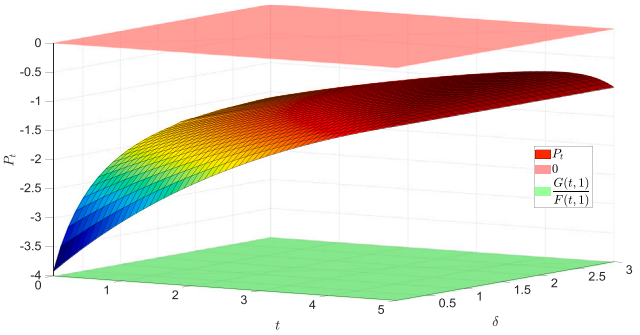
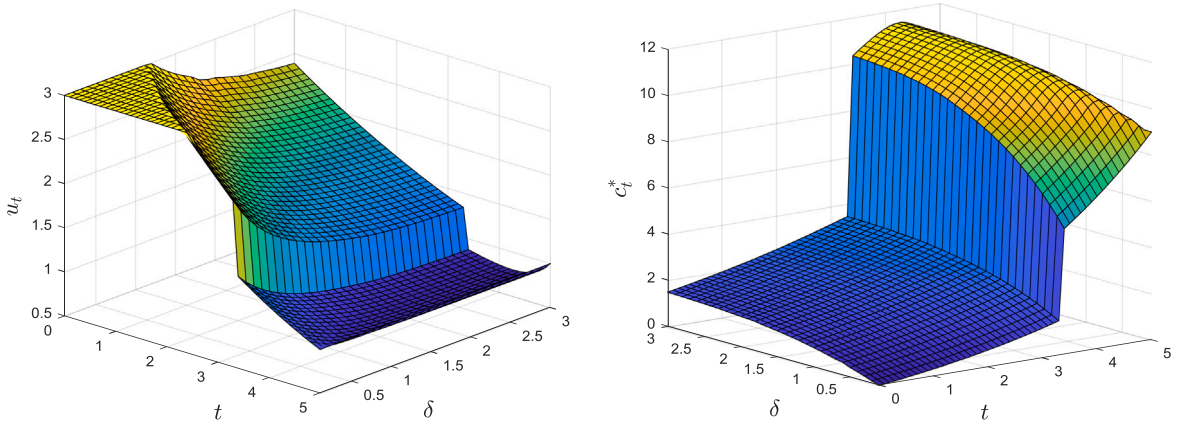


Fig. 4. The curves of P_t.



(a) Optimal intensity of advertising spending u_t.

(b) Optimal dividend c_t^*.

Fig. 5. The optimal intensity of advertising spending u_t and optimal dividend c_t^* corresponding to different delay parameters δ.

Proposition 5.2. *Suppose that $P_\cdot$ which is the solution of (74) satisfies identity (75), then $c_\cdot^*$ given by (76) is an optimal dividend, and $u_\cdot$ given by*

$$u_t = \begin{cases} -\frac{\mathbf{D}(t,\alpha_t)}{\mathbf{S}(t,\alpha_t)} P_t, & -\frac{\mathbf{S}(t,\alpha_t)}{\mathbf{D}(t,\alpha_t)}\mathbf{M} \le P_t < 0, \\ \mathbf{M}, & P_t < -\frac{\mathbf{S}(t,\alpha_t)}{\mathbf{D}(t,\alpha_t)}\mathbf{M} \end{cases} \tag{78}$$

is an optimal intensity of advertising spending of choice game problem. Specially, if the proportional rate of depreciation $\mathbf{A}(t,\alpha_t) = \mathbf{A}_t$ is independent of α_t, then process $P_\cdot$ in (76) and (78) is the solution of (77).

Next, we present some numerical examples for the advertising and dividend choice game problem. Suppose that the Markov chain α_t takes values in two regimes "1" and "2", which represent two different states of economy. Set $T = 5$, $i_0 = 1$, $\mathcal{Q}^{1,2} = 0.5$, $\mathcal{Q}^{2,1} = 1$, $\mathbf{A}_t = -0.8$, $\rho = 1.5$, $\mathbf{M} = 3$, $\mathbf{D}(t,1) = 1.5$, $\mathbf{D}(t,2) = 2$, $\mathbf{F}(t,1) = -1$, $\mathbf{F}(t,2) = -0.5$, $\mathbf{G}(t,1) = 4$, $\mathbf{G}(t,2) = 5$, $\mathbf{S}(t,1) = 1$, $\mathbf{S}(t,2) = 2$.

Note that we set $\mathbf{A}_t = -0.8$, which is independent of Markov chain α_t. Therefore, P_t satisfies Eq. (77). Using Euler–Maruyama method, Fig. 4 presents the numerical solutions of P_t. The pink and green planes represent 0 and $\frac{\mathbf{G}(t,1)}{\mathbf{F}(t,1)}$, respectively. The curve with gradient colors is the numerical solutions of P_t as δ varies from 0 to 3. It can be seen that $\frac{G(t,1)}{F(t,1)} < P_t < 0$ holds. Since in our setup $\frac{G(t,2)}{F(t,2)} < \frac{G(t,1)}{F(t,1)}$, then condition (75) is satisfied numerically.

Then we show the curves of optimal intensity of advertising spending u_t and optimal dividend c_t^* as δ varies from 0 to 3 in Fig. 5. In Fig. 5(a), it is shown that the optimal intensity of advertising u_t decreases as the forgetting horizon δ increases. This is because the longer the forgetting time δ is, the longer the residual advertising effect persists, which allows a reduction of advertising expenditure. Fig. 5(b) demonstrates that the optimal dividend payout exhibits an upward trend as δ increases. There exists a direct reason and an indirect one. The direct reason is that the forgetting structure of advertising effect defers the negative impact of dividend. The indirect reason is that the firm would be capable of providing more money to pay for the investor's dividend since the advertising expense retrenches.

5.3. N-Agent inter-bank borrowing and lending and systemic risk with regime switching

In this part, we consider an N-agent inter-bank borrowing and lending problem. Such kind of problems have been studied in [44], where a model with common Brownian noise was used, and in [61], where delayed control appears in dynamics.

Here we revisit the N-agent inter-bank borrowing and lending problem by letting the coefficients in both the dynamics and cost functional be Markov-modulated processes. Our motivations are fourfold. First, in the banking system, the regulator (i.e., central bank) adjusts monetary policies according prevailing market regimes. For example, in a stable and well-functioning market, the central bank may encourage higher inter-bank borrowing and lending rates while penalizing deviations from average capitalization in order to enhance liquidity and maintain systemic stability. In contrast, during a financial crisis, the central bank would restrict inter-bank borrowing and lending to prevent the amplification of systemic risk. To capture such regime-dependent behaviors, we introduce Markov-modulated parameters into both the dynamics and the cost functional. Second, the works [44,61] focus on deriving exact Nash equilibrium. In this paper, we are interested in searching for cooperative equilibrium, which is studied through a conditional mean-field type control problem (see, e.g., [62,63] for more details about cooperative equilibrium). Third, [44] reveals an intriguing phenomenon: when common Brownian noise is present, the probability of systemic risk does not vanish even as the number of banks grows large. We show that this phenomenon disappears when the common noise is instead modeled as a Markov switching process. In this case, the probability of systemic risk converges to zero as the number of banks tends to infinity (see Fig. 7 and the accompanying explanation). Finally, we find that, in a special case, the optimal control of the conditional mean-field type problem constitutes an ϵ-Nash equilibrium for the N-agent inter-bank borrowing and lending problem.

5.3.1. Problem formulation of N-agent inter-bank borrowing and lending with regime switching

In this section, we present the model of the N-agent inter-bank borrowing and lending problem with regime switching. For $k \in \mathcal{N}$, let $X_\cdot^k$ denote the log-monetary reserve of bank k, which evolves as

$$\begin{cases} dX_t^k = \left[\vartheta(t,\alpha_t)\left(X_t^{(N)} - X_t^k\right) + v_t^k\right]dt + \sigma(t,\alpha_t)dW_t^k, \\ X_0^k = x. \end{cases} \tag{79}$$

In (79), $X_\cdot^{(N)} = \frac{1}{N}\sum_{\ell=1}^N X_\cdot^\ell$ is the empirical mean of the monetary reserves of all banks, $W_\cdot^k$ ($k \in \mathcal{N}$) is a collection of independent standard Brownian motions (also independent of $\alpha_\cdot$) and denotes the idiosyncratic noise for bank k, $\vartheta(t,i)$ is the mean-reversion rate in market mode i. For bank k, $v_\cdot^k$ represents its borrowing and lending rates to central bank. $v_\cdot^k > 0$ (resp. $v_\cdot^k < 0$) means that bank k borrows from (resp. lends to) central bank.

The cost functional of bank k ($k \in \mathcal{N}$) is formulated as

$$J^k[v_\cdot] = \mathbb{E}\left\{\int_0^T \left[\frac{1}{2}(v_t^k)^2 - \eta(t,\alpha_t)v_t^k\left(X_t^{(N)} - X_t^k\right) + \frac{h(t,\alpha_t)}{2}\left(X_t^{(N)} - X_t^k\right)^2\right]dt + \frac{\psi(\alpha_T)}{2}\left(X_T^{(N)} - X_T^k\right)^2\right\}, \tag{80}$$

where $v_\cdot = (v_\cdot^1, \dots, v_\cdot^N)$. Suppose that $\eta(t,i) \geq 0$, $h(t,i) \geq 0$, $\psi(i) \geq 0$, and $\eta(t,i)^2 \leq h(t,i)$ for each $i \in \mathcal{S}$. $\eta(t,i)$ represents the incentive to borrowing or lending. In particular, bank k is encouraged to borrow if its monetary reserve X_t^k falls below the mean value $X_t^{(N)}$ and lend if its monetary reserve X_t^k rises above the mean value $X_t^{(N)}$. The quadratic terms $\frac{h(t,\alpha_t)}{2}(X_t^{(N)} - X_t^k)^2$ in running cost and $\frac{\psi(\alpha_T)}{2}(X_T^{(N)} - X_T^k)^2$ in terminal cost describe the costs incurred by the deviation from average. For any $i \in \mathcal{S}$, the coefficients $\vartheta(t,i)$, $\sigma(t,i)$, $\eta(t,i)$, and $h(t,i)$ are uniformly bounded for $t \in [0,T]$.

The inter-bank borrowing and lending with regime switching is actually an N-agent game problem with weakly coupled interaction among all the agents. For convenience, we state it as Problem (NG).

5.3.2. Conditional mean-field type control problem: An approximate problem of representative bank

In this part, we search for a cooperative equilibrium of Problem (NG). Suppose that the central bank chooses the feedback control $v_t^k = \pi(t,\alpha_t,X_t^k,X_t^{(N)})$ at time t, where π is a deterministic function. In this case, according to the laws of large numbers (see [20]), X_t^k ($k \in \mathcal{N}$) becomes independent and identically distributed given α_t as $N \to \infty$. Additionally, $X_t^{(N)}$ converges to the conditional expectation of X_t w.r.t. α_t, where X_t satisfies the following conditional mean-field SDE:

$$\begin{cases} dX_t = \left[\vartheta(t,\alpha_t)\left(\widehat{X}_t - X_t\right) + \pi(t,\alpha_t,X_t,\widehat{X}_t)\right]dt + \sigma(t,\alpha_t)dW_t, \\ X_0 = x. \end{cases} \tag{81}$$

In (81), $W_\cdot$ is a standard Brownian motion independent of $\alpha_\cdot$. Generally, here we do not consider the closed-loop control problem but investigate the open-loop control problem: the central bank seeks for a control $v_\cdot$ to minimize

$$J[v_\cdot] = \mathbb{E}\left\{\int_0^T \left[\frac{1}{2}(v_t)^2 - \eta(t,\alpha_t)v_t\left(\widehat{X}_t - X_t\right) + \frac{h(t,\alpha_t)}{2}\left(\widehat{X}_t - X_t\right)^2\right]dt + \frac{\psi(\alpha_T)}{2}\left(\widehat{X}_T - X_T\right)^2\right\}, \tag{82}$$

subject to

$$\begin{cases} dX_t = \left[\vartheta(t,\alpha_t)\left(\widehat{X}_t - X_t\right) + v_t\right]dt + \sigma(t,\alpha_t)dW_t, \\ X_0 = x. \end{cases} \tag{83}$$

In fact, limit equation (83) can be used to approximate the dynamics of log-monetary reserve of the representative bank. Problem (82)–(83) induced from the limiting dynamics and cost of Problem (NG) is a conditional mean-field type control problem, which will be referred to as Problem (CMFC).

In view of (15) and (16), the Hamiltonian function and adjoint equation of Problem (CMFC) are

$$H(t,i,x,\hat{x},v,P,Q) = P\left[\vartheta(t,i)\left(\hat{x}-x\right)+v\right] + Q\sigma(t,i) + \frac{1}{2}v^2 - \eta(t,i)v\left(\hat{x}-x\right) + \frac{h(t,i)}{2}\left(\hat{x}-x\right)^2, \tag{84}$$

and

$$\begin{cases} dP_t = \left[\vartheta(t,\alpha_t)(P_t-\widehat{P}_t) - h(t,\alpha_t)(X_t-\widehat{X}_t) - \eta(t,\alpha_t)(u_t-\widehat{u}_t)\right]dt + Q_t dW_t + R_t \bullet dM_t, \\ P_T = \psi(\alpha_T)(X_T-\widehat{X}_T), \end{cases} \tag{85}$$

where $\widehat{P}_t = \mathbb{E}[P_t|\mathcal{F}_t^{\alpha}]$ and $\widehat{u}_t = \mathbb{E}[u_t|\mathcal{F}_t^{\alpha}]$. Utilizing Theorem 4.1, minimizing Hamiltonian function H w.r.t v yields the following first order condition for optimal rates of borrowing and lending:

$$u_t = -P_t - \eta(t,\alpha_t)(X_t-\widehat{X}_t). \tag{86}$$

On the other hand, it is easily verified that terminal cost function $g(i,\hat{x},x) = \frac{\psi(i)(\hat{x}-x)^2}{2}$ is convex on $(\hat{x},x)$ since $\psi(i) > 0$ for any $i \in \mathcal{S}$, and Hamiltonian function H is convex on $(\hat{x},x,v)$ due to the fact that $h(t,i) \geq 0$ and $\eta(t,i)^2 \leq h(t,i)$ for any $t \in [0,T]$, $i \in \mathcal{S}$. Moreover, u_t defined in (86) minimizes $H(t,\alpha_t,x_t,\hat{x}_t,v,P_t,Q_t)$ for any $t \in [0,T]$, $v \in \mathbb{R}$. Therefore, we derive from Theorem 4.2 that $u_\cdot$ given by (86) is an optimal control of this problem. Next, we try to represent (86) in a feedback form w.r.t. X_t and $\widehat{X}_t$.

Assuming that

$$P_t = Y(t,\alpha_t)(X_t-\widehat{X}_t), \tag{87}$$

where $Y(t,i)$ is a deterministic differentiable function such that $Y(T,i) = \psi(i)$ for each $i \in \mathcal{S}$. Applying Itô's formula for Markov-modulated process to the right side of (87) follows

$$\begin{aligned} dP_t = &\left\{\dot{Y}(t,\alpha_t) - Y(t,\alpha_t)\left[\vartheta(t,\alpha_t)+Y(t,\alpha_t)+\eta(t,\alpha_t)\right] + \sum_{j\in\mathcal{S}} Q^{\alpha_t,j}Y(t,j)\right\}\left(X_t-\widehat{X}_t\right)dt \\ &+ Y(t,\alpha_t)\sigma(t,\alpha_t)dW_t + \sum_{i,j\in\mathcal{S}}\left(Y(t,j)-Y(t,i)\right)\left(X_t-\widehat{X}_t\right)dM_t^{i,j}, \end{aligned} \tag{88}$$

where $\dot{Y}(t,i)$ denotes the derivative of $Y(t,i)$ w.r.t. t. Inserting (86) and (87) into (85), we arrive at

$$dP_t = \left\{\left[\vartheta(t,\alpha_t)+\eta(t,\alpha_t)\right]Y(t,\alpha_t) - h(t,\alpha_t) + \eta(t,\alpha_t)^2\right\}\left(X_t-\widehat{X}_t\right)dt + Q_t dW_t + R_t \bullet dM_t. \tag{89}$$

Equalizing the coefficients of dt, dW_t, and dM_t in (88) and (89), it follows that

$$Q_t = Y(t,\alpha_t)\sigma(t,\alpha_t), \qquad R_t^{i,j} = \left(Y(t,j)-Y(t,i)\right)\left(X_t-\widehat{X}_t\right),$$

and

$$\begin{cases} \dot{Y}(t,i) = Y(t,i)^2 + 2\left[\vartheta(t,i)+\eta(t,i)\right]Y(t,i) - h(t,i) + \eta(t,i)^2 - \sum_{j\in\mathcal{S}} Q^{i,j}Y(t,j), \\ Y(T,i) = \psi(i). \end{cases} \tag{90}$$

Recall that $\psi(i) > 0$ and $\eta(t,i)^2 \leq h(t,i)$ hold for each $i \in \mathcal{S}$. It yields from Lemma 1 in [13] that (90) admits a unique solution $Y(t,i) > 0$. Then we obtain the following result.

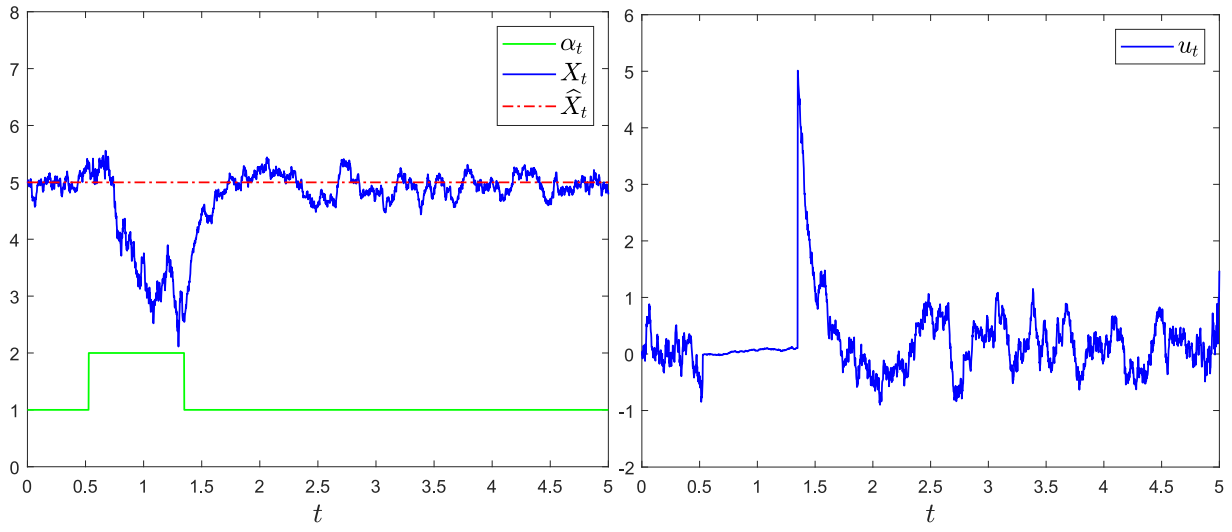
Proposition 5.3. *The control process*

$$u_t = -\left(Y(t,\alpha_t)+\eta(t,\alpha_t)\right)\left(X_t-\widehat{X}_t\right) \tag{91}$$

is an optimal borrowing and lending rates of the representative bank in Problem (CMFC), where $Y(t,i)$ satisfies (90), $\widehat{X}_t = x$, and X_t is governed by

$$\begin{cases} dX_t = -\left[\vartheta(t,\alpha_t)+Y(t,\alpha_t)+\eta(t,\alpha_t)\right]\left(X_t-\widehat{X}_t\right)dt + \sigma(t,\alpha_t)dW_t, \\ X_0 = x. \end{cases} \tag{92}$$

Next, we present some numerical examples. Suppose that the Markov chain α_t has two regimes: state 1 represents a well-functioning market, while state 2 corresponds to a financial crisis. The parameters are given by $T = 5$, $x = 5$, $i_0 = 1$, $Q^{1,2} = 0.25$, $Q^{2,1} = 0.75$, $\vartheta(t,1) = 10$, $\vartheta(t,2) = 0$, $\sigma(t,1) = 1$, $\sigma(t,2) = 2$, $\eta(t,1) = 2$, $\eta(t,2) = 0$, $h(t,1) = 5$, $h(t,2) = 0$, $\psi(1) = 5$, and $\psi(2) = 0$. It should be noted that we set $\vartheta(t,2) = \eta(t,2) = h(t,2) = \psi(2) = 0$ to describe the situation under which the borrowing and lending are limited and not encouraged when the financial crisis happens. Using Euler–Maruyama method with step size $\Delta = 2\times10^{-3}$, we obtain the trajectories of Markov chain α_t, optimal monetary reserve X_t, conditional expectation $\widehat{X}_t$, and optimal rates of borrowing and lending u_t shown in Fig. 6. We see that, at the beginning, the monetary reserve of bank is leveling out at around average reserve



(a) Trajectories of α_t, X_t, and $\widehat{X}_t$. (b) A trajectory of u_t.

Fig. 6. Simulations for α_t, X_t, $\widehat{X}_t$ and u_t.

5 through inter-bank borrowing and lending. Then the financial crisis happens, the monetary liquidity is limited which incurs that the reserve falls down rapidly. Finally, the financial crisis ends, the monetary reserve returns to average value quickly by borrowing a large amount of capitalization and then remains stable.

Finally, we investigate the probability of systemic event for the log-monetary reserves model (79) with regime switching in our paper when $v^k_\cdot = 0$ and compare it with the model in [44]. The log-monetary reserves model in [44] with $v^k_\cdot = 0$ is

$$\begin{cases} d\widetilde{X}^k_t = a\left(\widetilde{X}^{(N)}_t - \widetilde{X}^k_t\right)dt + \sigma\left(\rho d\widetilde{W}^0_t + \sqrt{1-\rho^2}d\widetilde{W}^k_t\right), \\ \widetilde{X}^k_0 = \widetilde{x}, \end{cases} \tag{93}$$

where the standard Brownian motion $\widetilde{W}^0_\cdot$, which denotes the common noise in the bank system, is independent of $\widetilde{W}^k_\cdot$ ($k \in \mathcal{N}$). In fact, the Markov chain $\alpha_\cdot$ in (79) acts as a common noise as well. We call it a systemic event when the ensemble averages $X^{(N)}_t$ in (79) and $\widetilde{X}^{(N)}_t$ in (93) reach a default level D. Here we define

$$\mathbb{P}^\alpha = \mathbb{P}\left\{\min_{0\le t\le T} X^{(N)}_t \le D\right\}, \qquad \text{and} \qquad \mathbb{P}^{\widetilde{W}^0} = \mathbb{P}\left\{\min_{0\le t\le T} \widetilde{X}^{(N)}_t \le D\right\} \tag{94}$$

to denote the probabilities of systemic events of models (79) and (93), where the superscript α and $\widetilde{W}^0$ highlight the fact that Markov chain $\alpha_\cdot$ and Brownian motion $\widetilde{W}^0_\cdot$ serve as common noises in (79) and (93) respectively. Set $\widetilde{x} = 5$, $a = 10$, $\sigma = 1$, $D = 4$, and $\rho = 0.5$. In this case, the initial log-monetary reserve, mean-reversion rate and volatility rate in (93) coincide with those designated in the above numerical example for model (83) when $i = 1$. Using Monte Carlo method with 10^4 simulations, we obtain the values of $\mathbb{P}^\alpha$ and $\mathbb{P}^{\widetilde{W}^0}$ as N increases from 1 to 200. The result is demonstrated in Fig. 7. As pointed out in [44] and shown in Fig. 7, when the common noise is characterized by Brownian motion which adds the same shocks on the monetary reserve of all the banks simultaneously, the possibility that systemic event happens will not disappear even if N becomes large. However, we check from Fig. 7 that, when the common noise is a Markov chain that brings a random switching of economic conditions, the probability of systemic event will vanish as N increases.

5.3.3. ϵ-Nash equilibrium of problem (NG)

This section is devoted to showing that, in a special case, the optimal control of Problem (CMFC) provides an ϵ-Nash Equilibrium for Problem (NG). In game theory, the definition of ϵ-Nash Equilibrium of Problem (NG) is given as follows.

Definition 5.1. An admissible control $u_\cdot = (u^1_\cdot, \dots, u^N_\cdot)$ is called an ϵ-Nash Equilibrium, if there exists $\epsilon = \epsilon(N) > 0$ with $\lim_{N\to\infty} \epsilon(N) = 0$ such that, for any $k \in \mathcal{N}$, the following identity holds:

$$J^k[u^{-k}_\cdot, u^k_\cdot] \le J^k[u^{-k}_\cdot, v^k_\cdot] + \epsilon, \tag{95}$$

where $v^k_\cdot$ is an arbitrary admissible control.

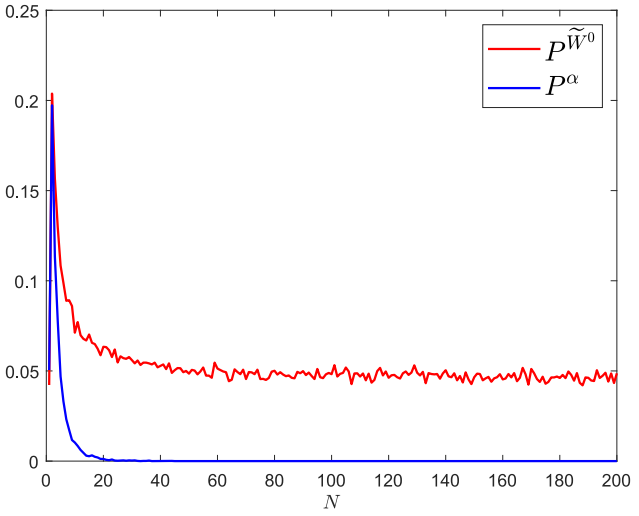


Fig. 7. The probabilities of systemic event in two models.

Clearly, if we take $\epsilon = 0$ in (95), then it reduces to (5). This means that an ϵ-Nash Equilibrium can achieve approximate optimality with acceptable error.

An effective method to derive the ϵ-Nash equilibrium of the N-player game problem is the fixed point approach (see, e.g., [64]), which mainly includes the following procedures. First, it freezes the coupling term with a fixed process (to be determined later), which leads to a decoupled control problem from the original N-player game problem with weak interaction. Second, employing maximum principle or dynamics programming principle to solve the control problem. Finally, find a solution such that the fixed point condition (also referred to as consistency condition) is satisfied and verify that the solution is an ϵ-Nash equilibrium of the N-player game problem. Despite strong similarities, there is in general no equivalence between the solution of a mean-field type control problem and that of the fixed point problem [62]. In fact, without further restriction, it cannot be shown that the optimal control of Problem (CMFC) is equivalent to the solution of the fixed point problem associated with Problem (NG). In what follows, we restrict our attention to cases in which these two solutions coincide. For this purpose, the following assumption is introduced and will be in force throughout this part.

Assumption (A6). $\vartheta(t,i) = \eta(t,i) = 0$ for any $t \in [0,T]$ and $i \in S$.

Under Assumption (A6), dynamics (83) and cost (82) in Problem (CMFC) reduces to

$$\begin{cases} dX_t = v_t dt + \sigma(t,\alpha_t) dW_t, \\ X_0 = x, \end{cases} \tag{96}$$

and

$$J[v_\cdot] = \mathbb{E}\left\{ \int_0^T \left[\frac{1}{2}(v_t)^2 + \frac{h(t,\alpha_t)}{2} \left(\widehat{X}_t - X_t \right)^2 \right] dt + \frac{\psi(\alpha_T)}{2} \left(\widehat{X}_T - X_T \right)^2 \right\}, \tag{97}$$

respectively.

We now provide a further analysis of Problem (CMFC).

Proposition 5.4. *Suppose Assumption (A6) holds, the following statements are valid.*

(a) Cost functional (97) is uniformly convex and has a unique optimal control $\bar{u}_\cdot$.

(b) $\bar{u}_\cdot = -\bar{p}_\cdot$ and $\bar{X}_\cdot$ are the optimal control and optimal state of Problem (CMFC) if and only if $(\bar{X}_\cdot, \bar{p}_\cdot, \bar{q}_\cdot, \bar{r}_\cdot)$ solves the following Hamiltonian system

$$\begin{cases} d\bar{X}_t = \bar{u}_t dt + \sigma(t,\alpha_t) dW_t, \\ d\bar{p}_t = h(t,\alpha_t)(\widehat{\bar{X}}_t - \bar{X}_t)\mathrm{d}t + \bar{q}_t dW_t + \bar{r}_t \bullet dM_t, \\ \bar{X}_0 = x, \qquad \bar{p}_T = \psi(\alpha_T)(\bar{X}_T - \widehat{\bar{X}}_T), \qquad \bar{u}_t = -\bar{p}_t. \end{cases} \tag{98}$$

(c) Hamiltonian system (98) admits a unique solution $(\bar{X}_\cdot, \bar{p}_\cdot, \bar{q}_\cdot, \bar{r}_\cdot) \in \mathcal{S}^2_{\mathcal{F}}([0,T];\mathbb{R}) \times \mathcal{S}^2_{\mathcal{F}}([0,T];\mathbb{R}) \times \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}) \times \mathcal{K}^2_{\mathcal{F}}([0,T];\mathbb{R})$.

Proof. Conclusion (a) follows from an argument similar to that in [63, Lemma 2.2]. The sufficiency part of (b) is obtained by Theorem 4.2, while the necessity follows from Theorem 4.1. Finally, conclusion (c) is a direct consequence of (a) and (b). □

Next, we derive the ϵ-Nash Equilibrium of Problem (NG) using the fixed point approach. To this end, we choose an $\mathcal{F}_t^{\theta}$-adapted process $\xi_\cdot$ to replace $X_\cdot^{(N)}$ in Problem (NG) and construct the following auxiliary control problem: minimize

$$\mathbb{J}^k[v_\cdot^{k,\dagger}] = \mathbb{E}\left\{\int_0^T \left[\frac{1}{2}(v_t^{k,\dagger})^2 + \frac{h(t,\alpha_t)}{2}\left(\xi_t - X_t^{k,\dagger}\right)^2\right]dt + \frac{\psi(\alpha_T)}{2}\left(\xi_T - X_T^{k,\dagger}\right)^2\right\},$$

subject to

$$\begin{cases} dX_t^{k,\dagger} = v_t^{k,\dagger}dt + \sigma(t,\alpha_t)dW_t^k, \\ X_0^{k,\dagger} = x. \end{cases}$$

By Theorems 4.1 and 4.2, the optimal control of the auxiliary problem is

$$\bar{u}_t^{k,\dagger} = -\bar{p}_t^{k,\dagger},$$

and the related Hamiltonian system is

$$\begin{cases} d\bar{X}_t^{k,\dagger} = -\bar{u}_t^{k,\dagger}dt + \sigma(t,\alpha_t)dW_t^k, \\ d\bar{p}_t^{k,\dagger} = h(t,\alpha_t)(\xi_t - \bar{X}_t^{k,\dagger})\mathrm{d}t + \bar{q}_t^{k,\dagger}dW_t^k + \bar{r}_t^{k,\dagger} \bullet dM_t, \\ \bar{X}_0^{k,\dagger} = x, \quad \bar{p}_T^{k,\dagger} = \psi(\alpha_T)(\bar{X}_T^{k,\dagger} - \xi_T), \quad \bar{u}_t^{k,\dagger} = -\bar{p}_t^{k,\dagger}. \end{cases} \tag{99}$$

Since we approximate X_t^k in Problem (NG) by $X_t^{k,\dagger}$, and replace $X_t^{(N)}$ with ξ_t, then ξ_t in (99) should satisfy the consistency condition, and it follows from the law of large numbers [20] that

$$\xi_t = \lim_{N\to\infty}\frac{1}{N}\sum_{k=1}^N \bar{X}_t^{k,\dagger} = \hat{\bar{X}}_t^{k,\dagger}, \qquad \mathbb{P}-a.s.$$

Substituting ξ_t with $\hat{\bar{X}}_t^{k,\dagger}$ in (99), we arrive at the consistency condition system

$$\begin{cases} d\bar{X}_t^{k,\dagger} = -\bar{u}_t^{k,\dagger}dt + \sigma(t,\alpha_t)dW_t^k, \\ d\bar{p}_t^{k,\dagger} = h(t,\alpha_t)(\hat{\bar{X}}_t^{k,\dagger} - \bar{X}_t^{k,\dagger})\mathrm{d}t + \bar{q}_t^{k,\dagger}dW_t^k + \bar{r}_t^{k,\dagger} \bullet dM_t, \\ \bar{X}_0^{k,\dagger} = x, \quad \bar{p}_T^{k,\dagger} = \psi(\alpha_T)(\bar{X}_T^{k,\dagger} - \hat{\bar{X}}_T^{k,\dagger}), \quad \bar{u}_t^{k,\dagger} = -\bar{p}_t^{k,\dagger}. \end{cases} \tag{100}$$

By Proposition 5.4, we obtain the following result concerning the above fixed point problem.

Proposition 5.5.

(a) Consistency condition system (100) admits a unique solution $(\bar{X}_\cdot^{k,\dagger}, \bar{p}_\cdot^{k,\dagger}, \bar{q}_\cdot^{k,\dagger}, \bar{r}_\cdot^{k,\dagger}) \in \mathcal{S}^2_{\mathcal{F}}([0,T];\mathbb{R}) \times \mathcal{S}^2_{\mathcal{F}}([0,T];\mathbb{R}) \times \mathcal{L}^2_{\mathcal{F}}([0,T];\mathbb{R}) \times \mathcal{K}^2_{\mathcal{F}}([0,T];\mathbb{R})$.

(b) Suppose that Assumption (A6) holds and replace $W_\cdot$ *with* $W_\cdot^k$ *in (96), then the solution of the fixed point problem coincides with the optimal control of Problem (CMFC).*

Proof. The proposition follows by noting that consistency condition system (100) coincides with (98). □

We now present the following result.

Proposition 5.6. *Under Assumption (A6),* $\bar{u}_\cdot^{\dagger} = (\bar{u}_\cdot^{1,\dagger}, \ldots, \bar{u}_\cdot^{N,\dagger})$ *constitutes an ϵ-Nash equilibrium of Problem (NG).*

Proof. This proof is divided into three steps.

Step 1: We first show that, for any $k \in \mathcal{N}$,

$$\left|J^k[\bar{u}_\cdot^{-k,\dagger}, \bar{u}_\cdot^{k,\dagger}] - \mathbb{J}^k[\bar{u}_\cdot^{k,\dagger}]\right| = O\left(\frac{1}{\sqrt{N}}\right). \tag{101}$$

Under Assumption (A6), the optimal state $\bar{X}_\cdot^{k,\dagger}$ in the fixed point problem is also the solution of (79) corresponding to $v_\cdot^k = \bar{u}_\cdot^{k,\dagger}$ in Problem (NG). Taking $\mathbb{E}[\cdot|\mathcal{F}_t^{\alpha}]$ on the forward and backward components in (100), we obtain that $\hat{\bar{X}}_t^{k,\dagger} = x$ and $\hat{\bar{u}}_t^{k,\dagger} = -\hat{\bar{p}}_t^{k,\dagger} = 0$. Define $\Xi_\cdot^k = \bar{X}_\cdot^{(N),\dagger} - \hat{\bar{X}}_\cdot^{k,\dagger}$. Then $\Xi_\cdot^k$ satisfies

$$\begin{cases} d\Xi_t^k = \dfrac{1}{N}\displaystyle\sum_{\ell=1}^N \bar{u}_t^{\ell,\dagger}dt + \frac{1}{N}\sum_{\ell=1}^N \sigma(t,\alpha_t)dW_t^{\ell}, \\ \Xi_0^k = 0. \end{cases} \tag{102}$$

Noting that $\bar{u}_t^{k,\dagger}$ $(k \in \mathcal{N})$ is identically distributed and independent given $\mathcal{F}_t^{\alpha}$ and $\hat{\bar{u}}_t^{k,\dagger} = 0$ for $t \in [0,T]$, it follows

$$\mathbb{E}\left|\frac{1}{N}\sum_{\ell=1}^{N}\bar{u}_r^{\ell,\dagger}\right|^2 = \frac{1}{N^2}\mathbb{E}\sum_{\ell=1}^{N}\left|\bar{u}_r^{\ell,\dagger}\right|^2 + \frac{1}{N^2}\mathbb{E}\sum_{\ell\neq j}^{N}\bar{u}_r^{\ell,\dagger}\bar{u}_r^{j,\dagger} = \frac{1}{N}\mathbb{E}\left|\bar{u}_r^{k,\dagger}\right|^2. \tag{103}$$

From (102), (103), Cauchy–Schwartz inequality, and B-D-G inequality, there exists a constant C (depending only on T, possibly changing from line to line) such that

$$\begin{aligned}\mathbb{E}\sup_{0\le s\le t}\left|\Xi_s^k\right|^2 &\le 2\mathbb{E}\sup_{0\le s\le t}\left|\int_0^s \frac{1}{N}\sum_{\ell=1}^{N}\bar{u}_r^{\ell,\dagger}dr\right|^2 + 2\mathbb{E}\sup_{0\le s\le t}\left|\int_0^s \frac{1}{N}\sum_{\ell=1}^{N}\sigma(r,\alpha_r)dW_r^{\ell}\right|^2\\ &\le \frac{C}{N}\mathbb{E}\int_0^t\left|\bar{u}_s^{k,\dagger}\right|^2 ds + \frac{C}{N}\mathbb{E}\int_0^t|\sigma(s,\alpha_s)|^2 ds\\ &= O\left(\frac{1}{N}\right),\end{aligned} \tag{104}$$

since $\bar{u}_\cdot^{k,\dagger} \in S_{\mathcal{F}}^2([0,T];\mathbb{R})$ and $\sigma(t,i)$ is uniformly bounded on $[0,T]$ for any $i \in \mathcal{S}$.

Recall that

$$J^k[\bar{u}_\cdot^{-k,\dagger},\bar{u}_\cdot^{k,\dagger}] = \mathbb{E}\left\{\int_0^T\left[\frac{1}{2}\left(\bar{u}_t^{k,\dagger}\right)^2 + \frac{h(t,\alpha_t)}{2}\left(\bar{X}_t^{(N),\dagger} - \bar{X}_t^{k,\dagger}\right)^2\right]dt + \frac{\psi(\alpha_T)}{2}\left(\bar{X}_T^{(N),\dagger} - \bar{X}_T^{k,\dagger}\right)^2\right\}, \tag{105}$$

and

$$\mathbb{J}^k[\bar{u}_\cdot^{k,\dagger}] = \mathbb{E}\left\{\int_0^T\left[\frac{1}{2}\left(\bar{u}_t^{k,\dagger}\right)^2 + \frac{h(t,\alpha_t)}{2}\left(\hat{\bar{X}}_t^{k,\dagger} - \bar{X}_t^{k,\dagger}\right)^2\right]dt + \frac{\psi(\alpha_T)}{2}\left(\hat{\bar{X}}_T^{k,\dagger} - \bar{X}_T^{k,\dagger}\right)^2\right\}. \tag{106}$$

Exploiting (104) and the estimate $\mathbb{E}\sup_{0\le s\le t}|\bar{X}_s^{k,\dagger}|^2 < \infty$, we obtain

$$\begin{aligned}&\left|\mathbb{E}\int_0^T\left[\frac{h(t,\alpha_t)}{2}\left(\bar{X}_t^{(N),\dagger} - \bar{X}_t^{k,\dagger}\right)^2 - \frac{h(t,\alpha_t)}{2}\left(\hat{\bar{X}}_t^{k,\dagger} - \bar{X}_t^{k,\dagger}\right)^2\right]dt\right|\\ &= \left|\mathbb{E}\int_0^T\left[\frac{h(t,\alpha_t)}{2}\left(\bar{X}_t^{(N),\dagger} - \hat{\bar{X}}_t^{k,\dagger}\right)^2 + h(t,\alpha_t)\left(\bar{X}_t^{(N),\dagger} - \hat{\bar{X}}_t^{k,\dagger}\right)\left(\hat{\bar{X}}_t^{k,\dagger} - \bar{X}_t^{k,\dagger}\right)\right]dt\right|\\ &\le C\mathbb{E}\int_0^T\left[\left(\bar{X}_t^{(N),\dagger} - \hat{\bar{X}}_t^{k,\dagger}\right)^2 + \left|\left(\bar{X}_t^{(N),\dagger} - \hat{\bar{X}}_t^{k,\dagger}\right)\left(\hat{\bar{X}}_t^{k,\dagger} - \bar{X}_t^{k,\dagger}\right)\right|\right]dt\\ &\le C\int_0^T\left\{\mathbb{E}\left(\bar{X}_t^{(N),\dagger} - \hat{\bar{X}}_t^{k,\dagger}\right)^2 + \left[\mathbb{E}\left(\bar{X}_t^{(N),\dagger} - \hat{\bar{X}}_t^{k,\dagger}\right)^2\mathbb{E}\left(\hat{\bar{X}}_t^{k,\dagger} - \bar{X}_t^{k,\dagger}\right)^2\right]^{\frac{1}{2}}\right\}dt\\ &= O\left(\frac{1}{\sqrt{N}}\right).\end{aligned} \tag{107}$$

Similarly, one has

$$\left|\mathbb{E}\left[\frac{\psi(\alpha_T)}{2}\left(\bar{X}_T^{(N),\dagger} - \bar{X}_T^{k,\dagger}\right)^2 - \frac{\psi(\alpha_T)}{2}\left(\hat{\bar{X}}_T^{k,\dagger} - \bar{X}_T^{k,\dagger}\right)^2\right]\right| = O\left(\frac{1}{\sqrt{N}}\right). \tag{108}$$

Combining (105)–(108), we conclude that estimate (101) holds.

Step 2: We then show that, for any perturbation control $v_\cdot^k$ and $k \in \mathcal{N}$,

$$\left|J^k[\bar{u}_\cdot^{-k,\dagger},v_\cdot^k] - \mathbb{J}^k[v_\cdot^k]\right| = O\left(\frac{1}{\sqrt{N}}\right). \tag{109}$$

According to Definition 5.1, it suffices to verify that

$$\inf_{v_\cdot^k\in\mathcal{U}^k} J^k[\bar{u}_\cdot^{-k,\dagger},v_\cdot^k] \ge J^k[\bar{u}_\cdot^{-k,\dagger},\bar{u}_\cdot^{k,\dagger}] - \epsilon. \tag{110}$$

Then it is enough to consider the perturbation control $v_\cdot^k$ satisfying $J^k[\bar{u}_\cdot^{-k,\dagger},v_\cdot^k] \le J^k[\bar{u}_\cdot^{-k,\dagger},\bar{u}_\cdot^{k,\dagger}]$. Otherwise, identity (110) holds trivially. With this in mind, we have

$$\frac{1}{2}\mathbb{E}\int_0^T\left|v_t^k\right|^2 dt \le J^k[\bar{u}_\cdot^{k,\dagger},v_\cdot^k] \le J^k[\bar{u}_\cdot^{-k,\dagger},\bar{u}_\cdot^{k,\dagger}] \le \mathbb{J}^k[\bar{u}_\cdot^{k,\dagger}] + O\left(\frac{1}{\sqrt{N}}\right),$$

which implies

$$\mathbb{E}\int_0^T\left|v_t^k\right|^2 dt \le C. \tag{111}$$

Define

$$\Pi_\cdot^k = \frac{1}{N}\sum_{\ell\neq k}^{N} \bar{X}_\cdot^{\ell,\dagger} + \frac{1}{N}X_\cdot^k - \hat{X}_\cdot^{k,\dagger}.$$

Then Π_t^k satisfies

$$\begin{cases} d\Pi_t^k = \left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{u}_t^{\ell,\dagger} + \frac{1}{N}v_t^k\right)dt + \frac{1}{N}\sum_{\ell=1}^{N}\sigma(t,\alpha_t)dW_t^\ell, \\ \Pi_0^k = 0. \end{cases}$$

Similar to (103), we have

$$\mathbb{E}\left|\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{u}_t^{\ell,\dagger}\right|^2 = \frac{N-1}{N^2}\mathbb{E}\left|\bar{u}_t^{k,\dagger}\right|^2. \tag{112}$$

Together with (111), (112), B-D-G inequality, and the boundedness of $\sigma(t,i)$, we derive

$$\begin{aligned} \mathbb{E}\sup_{0\le s\le t}\left|\Pi_s^k\right|^2 &\le C\int_0^t \mathbb{E}\left|\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{u}_s^{\ell,\dagger} + \frac{1}{N}v_s^k\right|^2 ds + C\mathbb{E}\sup_{0\le s\le t}\left|\int_0^s \frac{1}{N}\sum_{\ell=1}^{N}\sigma(r,\alpha_r)dW_r^\ell\right|^2 \\ &\le C\frac{N-1}{N^2}\mathbb{E}\int_0^t\left|\bar{u}_s^{k,\dagger}\right|^2 ds + C\frac{1}{N^2}\mathbb{E}\int_0^t\left|v_s^k\right|^2 ds + C\frac{1}{N}\mathbb{E}\int_0^t\left|\sigma(s,\alpha_s)\right|^2 ds \\ &= O\left(\frac{1}{N}\right). \end{aligned} \tag{113}$$

For the perturbed control $v_\cdot^k$,

$$J^k[\bar{u}_\cdot^{-k,\dagger}, v_\cdot^k] = \mathbb{E}\left\{\int_0^T\left[\frac{1}{2}\left(v_t^k\right)^2 + \frac{h(t,\alpha_t)}{2}\left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{X}_t^{\ell,\dagger} + \frac{1}{N}X_t^k - X_t^k\right)^2\right]dt + \frac{\psi(\alpha_T)}{2}\left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{X}_T^{\ell,\dagger} + \frac{1}{N}X_T^k - X_T^k\right)^2\right\}, \tag{114}$$

and

$$\mathbb{J}^k[v_\cdot^k] = \mathbb{E}\left\{\int_0^T\left[\frac{1}{2}\left(v_t^k\right)^2 + \frac{h(t,\alpha_t)}{2}\left(\hat{\bar{X}}_t^{k,\dagger} - X_t^k\right)^2\right]dt + \frac{\psi(\alpha_T)}{2}\left(\hat{\bar{X}}_T^{k,\dagger} - X_T^k\right)^2\right\}. \tag{115}$$

From (113) and the estimate $\mathbb{E}\sup_{0\le s\le t}\left|X_s^k\right|^2 < \infty$, we have

$$\begin{aligned} &\left|\mathbb{E}\int_0^T\left[\frac{h(t,\alpha_t)}{2}\left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{X}_t^{\ell,\dagger} + \frac{1}{N}X_t^k - X_t^k\right)^2 - \frac{h(t,\alpha_t)}{2}\left(\hat{\bar{X}}_t^{k,\dagger} - X_t^k\right)^2\right]dt\right| \\ &= \left|\mathbb{E}\int_0^T\left[\frac{h(t,\alpha_t)}{2}\left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{X}_t^{\ell,\dagger} + \frac{1}{N}X_t^k - \hat{\bar{X}}_t^{k,\dagger}\right)^2 + h(t,\alpha_t)\left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{X}_t^{\ell,\dagger} + \frac{1}{N}X_t^k - \hat{\bar{X}}_t^{k,\dagger}\right)\left(\hat{\bar{X}}_t^{k,\dagger} - X_t^k\right)\right]dt\right| \\ &\le C\mathbb{E}\int_0^T\left[\left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{X}_t^{\ell,\dagger} + \frac{1}{N}X_t^k - \hat{\bar{X}}_t^{k,\dagger}\right)^2 + \left|\left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{X}_t^{\ell,\dagger} + \frac{1}{N}X_t^k - \hat{\bar{X}}_t^{k,\dagger}\right)\left(\hat{\bar{X}}_t^{k,\dagger} - X_t^k\right)\right|\right]dt \\ &\le C\int_0^T\left\{\mathbb{E}\left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{X}_t^{\ell,\dagger} + \frac{1}{N}X_t^k - \hat{\bar{X}}_t^{k,\dagger}\right)^2 + \left[\mathbb{E}\left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{X}_t^{\ell,\dagger} + \frac{1}{N}X_t^k - \hat{\bar{X}}_t^{k,\dagger}\right)^2\mathbb{E}\left(\hat{\bar{X}}_t^{k,\dagger} - X_t^k\right)^2\right]^{\frac{1}{2}}\right\}dt \\ &= O\left(\frac{1}{\sqrt{N}}\right). \end{aligned} \tag{116}$$

Similarly,

$$\left|\mathbb{E}\left[\frac{\psi(\alpha_T)}{2}\left(\frac{1}{N}\sum_{\ell\neq k}^{N}\bar{X}_T^{\ell,\dagger} + \frac{1}{N}X_T^k - X_T^k\right)^2 - \frac{\psi(\alpha_T)}{2}\left(\hat{\bar{X}}_T^{k,\dagger} - X_T^k\right)^2\right]\right| = O\left(\frac{1}{\sqrt{N}}\right). \tag{117}$$

Thus, combining (114)–(117), we obtain error estimate (109).

Step 3: We now establish the ϵ-Nash equilibrium property of $\bar{u}_\cdot^\dagger = (\bar{u}_\cdot^{1,\dagger}, \dots, \bar{u}_\cdot^{N,\dagger})$.

By (101) and (109), we obtain

$$J^k[\bar{u}_\cdot^{-k,\dagger}, \bar{u}_\cdot^{k,\dagger}] \le \mathbb{J}^k[\bar{u}_\cdot^{k,\dagger}] + O\left(\frac{1}{\sqrt{N}}\right) \le \mathbb{J}^k[v_\cdot^k] + O\left(\frac{1}{\sqrt{N}}\right) \le J^k[\bar{u}_\cdot^{-k,\dagger}, v_\cdot^k] + O\left(\frac{1}{\sqrt{N}}\right),$$

which implies that $\bar{u}^{\dagger}_{\cdot} = (\bar{u}^{1,\dagger}_{\cdot}, \dots, \bar{u}^{N,\dagger}_{\cdot})$ is an ϵ-Nash equilibrium of Problem (NG).

The proof is completed. □

From Propositions 5.5 and 5.6, we establish the following relationship between the optimal control of Problem (CMFC) and the ϵ-Nash equilibrium of Problem (NG).

Proposition 5.7. *Suppose Assumption (A6) holds and replace $W_{\cdot}$ with $W^k_{\cdot}$ in the dynamics of Problem (CMFC), then the optimal control (91) of representative bank in Problem (CMFC) also constitutes an ϵ-Nash equilibrium of Problem (NG).*

6. Conclusion

This paper studies a conditional mean-field type game of nonlinear hybrid switching diffusions system with delay. As a prerequisite, an existence and uniqueness result is firstly obtained for a nonlinear conditional mean-field ABSDE with regime switching under mild conditions. Then, using established results on conditional mean-field ABSDE, one necessary condition in the form of Pontryagin's type maximum principle for open-loop Nash equilibrium point is established. Two verification theorems under different assumptions are also provided which give sufficient conditions for Nash equilibrium. Finally, we present three financial applications. By virtue of the theoretical result derived, explicit optimal strategies for all financial applications are obtained. Some numerical examples are provided with detailed interpretations, which verify the effectiveness of our conclusions.

For interacting particles systems with a common noise driving by Markov chain $\alpha_{\cdot}$, the convergence from delayed empirical mean $\frac{1}{N}\sum_{n=1}^{N}\int_{t-\delta}^{t} e^{-\rho(t-s)}X^n_s ds$ and $\frac{1}{N}\sum_{n=1}^{N} X^n_{t-\delta}$ towards delayed conditional expectation $\int_{t-\delta}^{t} e^{-\rho(t-s)}\mathbb{E}\left[X_s|\mathcal{F}^{\alpha}_s\right] ds$ and $\mathbb{E}[X_{t-\delta}|\mathcal{F}^{\alpha}_{t-\delta}]$ as $N \to \infty$ are at a heuristic level now. The formal conclusion regarding this point should be based on rigorous arguments. We will investigate this issue carefully in the near future.

CRediT authorship contribution statement

Zhongbin Guo: Writing – review & editing, Writing – original draft, Conceptualization. **Qingshuo Song:** Writing – review & editing, Supervision. **Guangchen Wang:** Writing – review & editing, Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Proof of Lemma 4.1

Proof. With Assumption (A1), all the coefficients in (14) are bounded. Therefore, Eq. (14) admits a unique solution $\mathbb{X}^k_{\cdot}$ satisfying

$$\mathbb{E}\sup_{0\le s\le t}|\mathbb{X}^k_s|^2 < \infty. \tag{A.1}$$

Define

$$\mathscr{X}^k_t = \frac{X^{(u^{-k},u^{k,\varepsilon})}_t - X_t}{\varepsilon} - \mathbb{X}^k_t, \qquad \mathscr{Y}^k_t = \int_{t-\delta}^{t} e^{-\rho(t-s)}\mathscr{X}^k_s ds, \qquad \mathscr{Z}^k_t = \mathscr{X}^k_{t-\delta},$$

and

$$\widehat{\mathscr{X}^k_t} = \mathbb{E}[\mathscr{X}^k_t|\mathcal{F}^{\alpha}_t], \qquad \widehat{\mathscr{Y}^k_t} = \mathbb{E}[\mathscr{Y}^k_t|\mathcal{F}^{\alpha}_t], \qquad \widehat{\mathscr{Z}^k_t} = \mathbb{E}[\mathscr{X}^k_{t-\delta}|\mathcal{F}^{\alpha}_{t-\delta}] = \widehat{\mathscr{X}^k_{t-\delta}}.$$

Then, from (2) and (14), we derive that $\mathscr{X}^k_{\cdot}$ satisfies

$$\begin{cases} d\mathscr{X}^k_t = \left[\check{b}_x(t)\mathscr{X}^k_t + \check{b}_y(t)\mathscr{Y}^k_t + \check{b}_z(t)\mathscr{Z}^k_t + \check{b}_{\hat{x}}(t)\widehat{\mathscr{X}^k_t} + \check{b}_{\hat{y}}(t)\widehat{\mathscr{Y}^k_t} + \check{b}_{\hat{z}}(t)\widehat{\mathscr{Z}^k_t} + A(t)\right]dt, \\ \qquad\quad + \sum_{\ell=1}^{d}\left[\check{\sigma}^{\ell}_x(t)\mathscr{X}^k_t + \check{\sigma}^{\ell}_y(t)\mathscr{Y}^k_t + \check{\sigma}^{\ell}_z(t)\mathscr{Z}^k_{t-\delta} + \check{\sigma}^{\ell}_{\hat{x}}(t)\widehat{\mathscr{X}^k_t} + \check{\sigma}^{\ell}_{\hat{y}}(t)\widehat{\mathscr{Y}^k_t} + \check{\sigma}^{\ell}_{\hat{z}}(t)\widehat{\mathscr{Z}^k_t} + B^{\ell}(t)\right]dW^{\ell}_t, \qquad t\in[0,T], \\ \mathscr{X}^k_t = 0, \qquad t\in[-\delta,0]. \end{cases} \tag{A.2}$$

Here, for functions $\phi = b, \sigma^{\ell}$ and variables $\jmath = x, y, z, \hat{x}, \hat{y}, \hat{z}, v^k$, we denote

$$\begin{aligned} \check{\phi}_{\jmath}(t) &= \int_0^1 \phi^{k,\lambda,\varepsilon}_{\jmath}(t)d\lambda, \\ \phi^{k,\lambda,\varepsilon}(t) &= \phi\left(t, X_t + \lambda\left(X^{(u^{-k},u^{k,\varepsilon})}_t - X_t\right), Y_t + \lambda\left(Y^{(u^{-k},u^{k,\varepsilon})}_t - Y_t\right), Z_t + \lambda\left(Z^{(u^{-k},u^{k,\varepsilon})}_t - Z_t\right),\right. \\ &\qquad \left.\hat{X}_t + \lambda\left(\hat{X}^{(u^{-k},u^{k,\varepsilon})}_t - \hat{X}_t\right), \hat{Y}_t + \lambda\left(\hat{Y}^{(u^{-k},u^{k,\varepsilon})}_t - \hat{Y}_t\right), \hat{Z}_t + \lambda\left(\hat{Z}^{(u^{-k},u^{k,\varepsilon})}_t - \hat{Z}_t\right), u^{-k}, u^k + \lambda\Delta v^k\right), \\ A(t) &= \left(\check{b}_x(t) - b_x(t)\right)\mathbb{X}^k_t + \left(\check{b}_y(t) - b_y(t)\right)\mathbb{Y}^k_t + \left(\check{b}_z(t) - b_z(t)\right)\mathbb{Z}^k_t \\ &\quad + \left(\check{b}_{\hat{x}}(t) - b_{\hat{x}}(t)\right)\widehat{\mathbb{X}}^k_t + \left(\check{b}_{\hat{y}}(t) - b_{\hat{y}}(t)\right)\widehat{\mathbb{Y}}^k_t + \left(\check{b}_{\hat{z}}(t) - b_{\hat{z}}(t)\right)\widehat{\mathbb{Z}}^k_t + \left(\check{b}_{v^k}(t) - b_{v^k}(t)\right)\Delta v^k, \end{aligned}$$

and $B^{\ell}(t)$ is defined analogously by replacing b with σ^{ℓ} in $A(t)$.

By (A.1) and Cauchy–Schwartz inequality, it follows that

$$\begin{aligned}\mathbb{E}\int_0^t \left|\mathbb{Y}_s^k\right|^2 ds &= \mathbb{E}\int_0^t \left|\int_{s-\delta}^s e^{-\rho(s-\tau)}\mathbb{X}_\tau^k d\tau\right|^2 ds \le \mathbb{E}\int_0^t \int_{s-\delta}^s e^{-2\rho(s-\tau)} d\tau \int_{s-\delta}^s \left|\mathbb{X}_\tau^k\right|^2 d\tau ds \\ &\le \frac{1-e^{-2\rho\delta}}{2\rho}\mathbb{E}\int_0^t \int_{-\delta}^t \left|\mathbb{X}_\tau^k\right|^2 d\tau ds = \frac{1-e^{-2\rho\delta}}{2\rho}\mathbb{E}\int_0^t \int_0^t \left|\mathbb{X}_\tau^k\right|^2 d\tau ds \\ &\le \frac{1-e^{-2\rho\delta}}{2\rho} T\mathbb{E}\int_0^t \left|\mathbb{X}_s^k\right|^2 ds \\ &< \infty,\end{aligned} \tag{A.3}$$

and

$$\mathbb{E}\int_0^t \left|\mathbb{Z}_s^k\right|^2 ds = \mathbb{E}\int_0^t \left|\mathbb{X}_{s-\delta}^k\right|^2 ds = \mathbb{E}\int_{-\delta}^{t-\delta} \left|\mathbb{X}_s^k\right|^2 ds \le \mathbb{E}\int_0^t \left|\mathbb{X}_s^k\right|^2 ds < \infty. \tag{A.4}$$

Therefore, under Assumption (A1), by (A.1), (A.3), (A.4), and the square-integrability of admissible controls, we have

$$\lim_{\varepsilon\to 0^+} \mathbb{E}\int_0^t |A(s)|^2 ds = 0, \qquad \lim_{\varepsilon\to 0^+} \mathbb{E}\int_0^t \left|B^{\ell}(s)\right|^2 ds = 0, \quad \forall \ell \in \{1,\dots,d\}. \tag{A.5}$$

From (A.2), exploiting B-D-G inequality and following similar arguments as in (A.3) and (A.4), we derive

$$\begin{aligned}\mathbb{E}\sup_{0\le s\le t}\left|\mathscr{X}_s^k\right|^2 &\le C\mathbb{E}\int_0^t \left(\left|\mathscr{X}_s^k\right|^2 + \left|\mathscr{Y}_s^k\right|^2 + \left|\mathscr{Z}_s^k\right|^2 + |A(s)|^2 + \sum_{\ell}^{d}\left|B^{\ell}(s)\right|^2\right) ds \\ &\le C\mathbb{E}\int_0^t \left|\mathscr{X}_s^k\right|^2 ds + C\mathbb{E}\int_0^t \left(|A(s)|^2 + \sum_{\ell}^{d}\left|B^{\ell}(s)\right|^2\right) ds,\end{aligned} \tag{A.6}$$

where C denotes a generic constant depending on ρ, δ, T (and can be different from line to line).

Applying Gronwall's inequality to identity (A.6) yields

$$\mathbb{E}\sup_{0\le s\le t}\left|\mathscr{X}_s^k\right|^2 \le C\mathbb{E}\int_0^t \left(|A(s)|^2 + \sum_{\ell}^{d}\left|B^{\ell}(s)\right|^2\right) ds,$$

which, in conjunction with (A.5), implies

$$\lim_{\varepsilon\to 0^+} \mathbb{E}\sup_{0\le s\le t}\left|\mathscr{X}_s^k\right|^2 = 0.$$

This completes the proof. □

Data availability

Data will be made available on request.

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